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ENGINEERING

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SHEET	DATE
	SCALE AS SHOWN
	JOB NUMBER

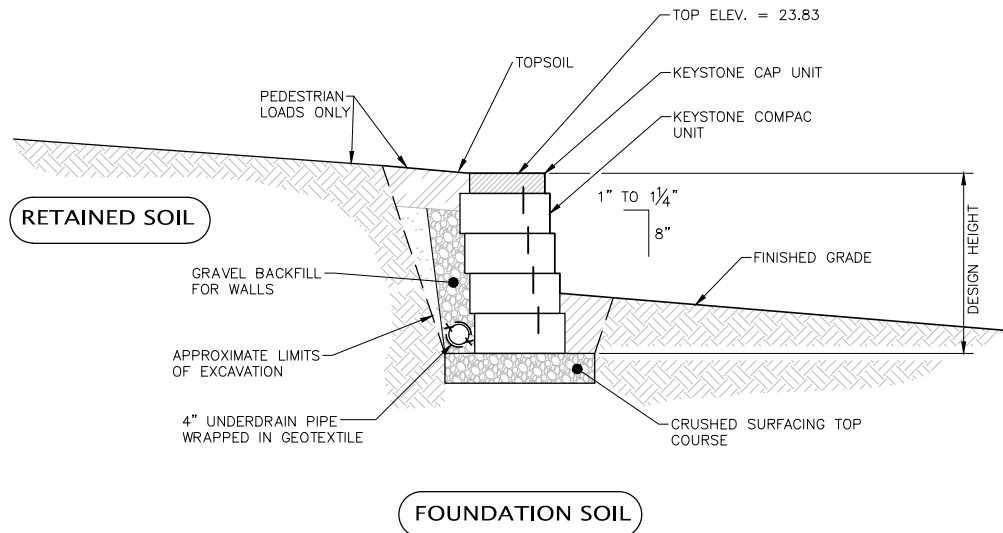
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OF

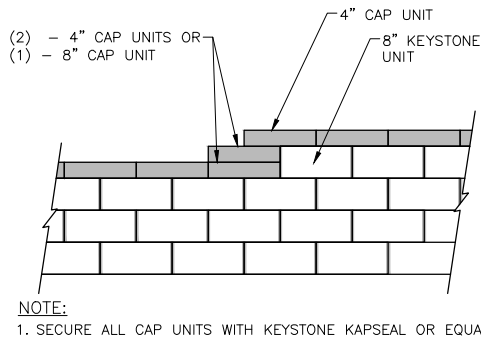
96

CIVIL DETAILS

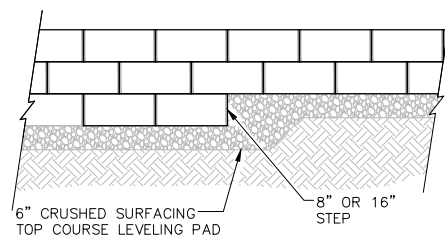
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 W:\2020\2020-070 EASTSOUND WWTP DESIGN\DWG\2020-070 C7.1-C7.8 CIVIL DETAILS.DWG - 11/2/2022 3:26 PM - Jeff Smith



TYPICAL GRAVITY WALL SECTION
 STANDARD UNIT - 1" SETBACK

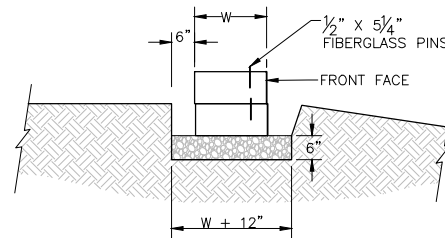


TOP OF WALL STEPS

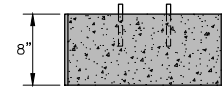


ELEVATION

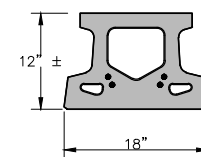
NOTE:
 1. THE LEVELING PAD IS TO BE CONSTRUCTED OF CRUSHED CRUSHED SURFACING TOP COURSE.



SECTION
 LEVELING PAD DETAIL

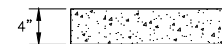


COMPAC III ELEVATION

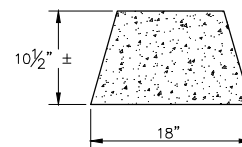


COMPAC III PLAN

COMPAC III UNIT
 *
 by Region

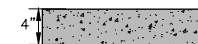


CAP UNIT ELEVATION

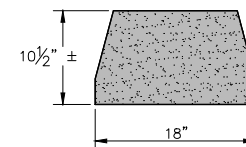


CAP UNIT PLAN

CAP UNIT OPTION



CAP UNIT ELEVATION

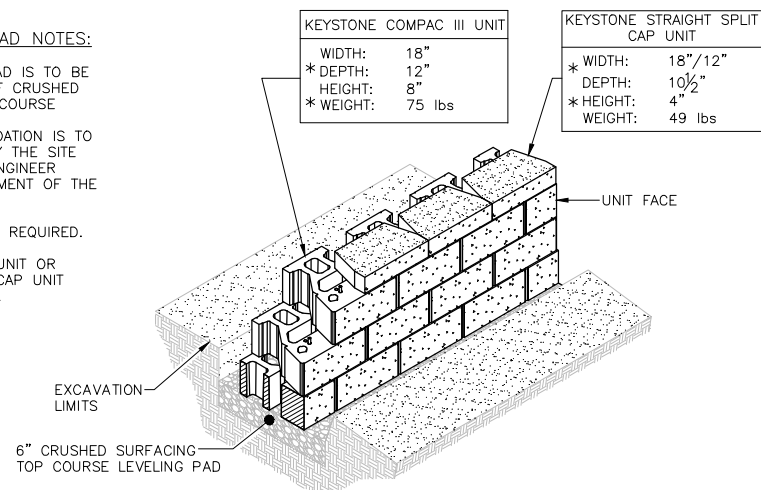


CAP UNIT PLAN

STRAIGHT SPLIT
 CAP UNIT OPTION

BASE LEVELING PAD NOTES:

1. THE LEVELING PAD IS TO BE CONSTRUCTED OF CRUSHED SURFACING TOP COURSE
2. THE BASE FOUNDATION IS TO BE APPROVED BY THE SITE GEOTECHNICAL ENGINEER PRIOR TO PLACEMENT OF THE LEVELING PAD.
3. FIBERGLASS PINS REQUIRED.
4. UNIVERSAL CAP UNIT OR STRAIGHT SPLIT CAP UNIT OPTION ALLOWED.



COMPAC III UNIT/BASE PAD ISOMETRIC SECTION VIEW
 * DIMENSIONS & WEIGHT MAY VARY BY REGION

KEYSTONE RETAINING WALL SYSTEM
 COMPAC III UNIT - STRAIGHT FACE DETAILS
 NOT TO SCALE

NO.	REVISIONS	BY	DATE



BID SET

11-02-2022

WILSON ENGINEERING

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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

CIVIL DETAILS

DATE	SCALE	AS SHOWN	JOB NUMBER

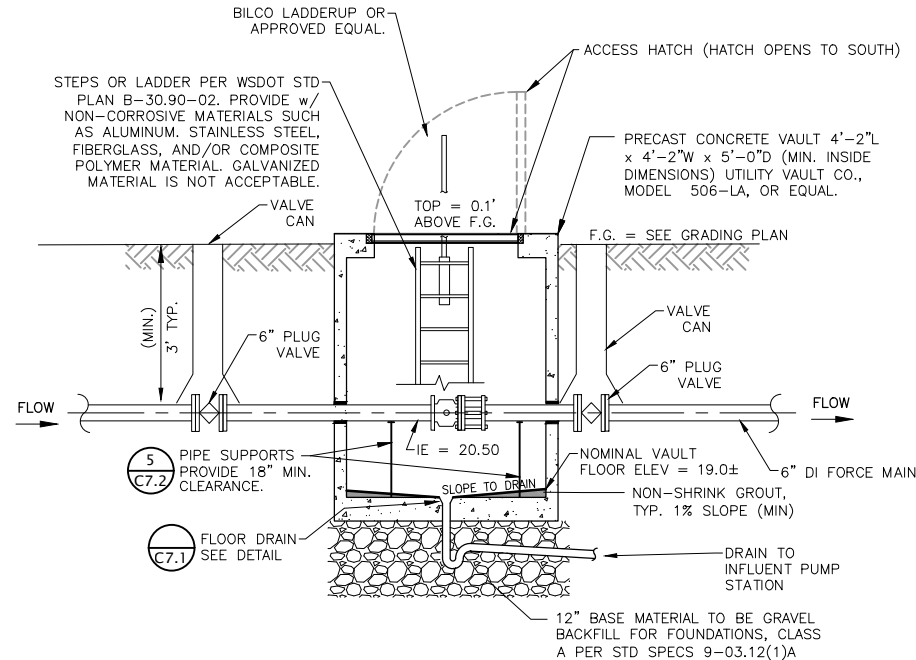
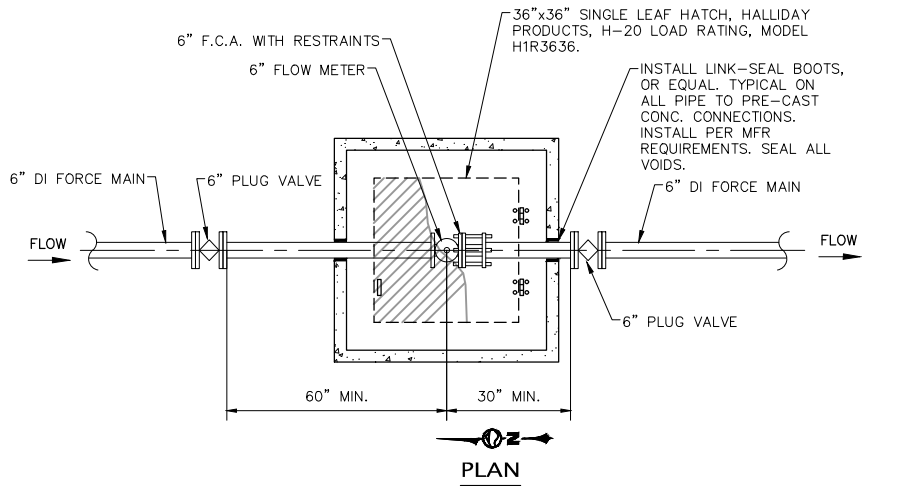
SHEET

50 OF 96

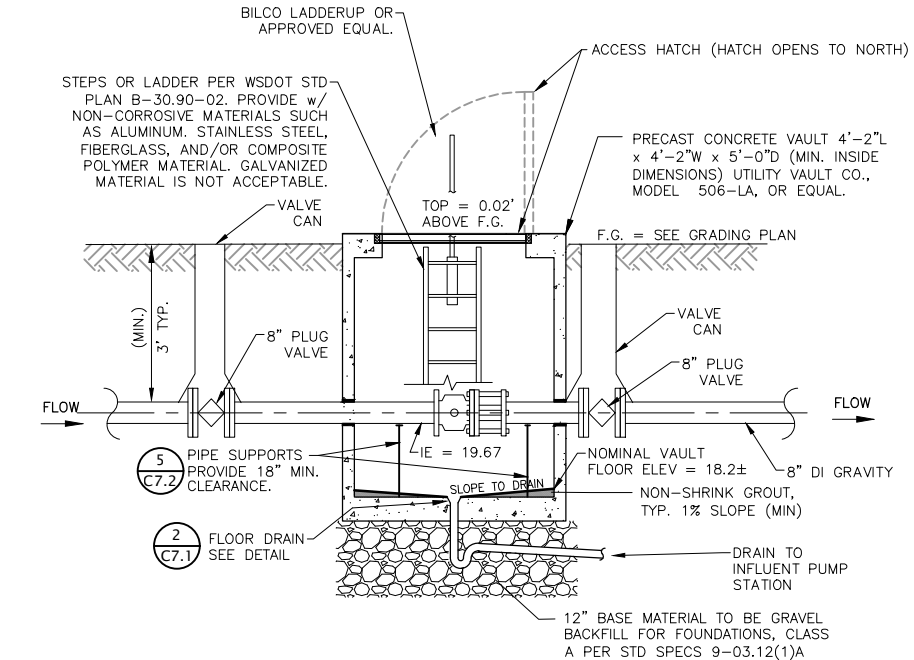
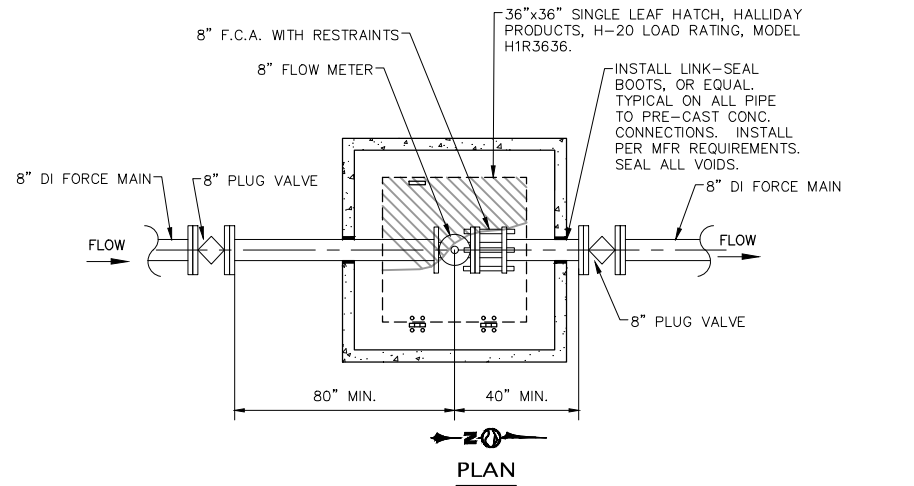
THIS DRAWING, TITLED, COMPAC III UNIT - STRAIGHT FACE DETAILS, WAS DOWNLOADED FROM HTTP://WWW.KEYSTONEWALLS.COM ON 9/1/2021. THE STANDARD DETAILS WERE MODIFIED TO SUIT THE PROJECT. THE ORIGINAL DRAWING WAS COPYRIGHTED IN 2010 BY KEYSTONE RETAINING WALL SYSTEMS.

NOV 2 2022
 11:54 AM
 May 2022

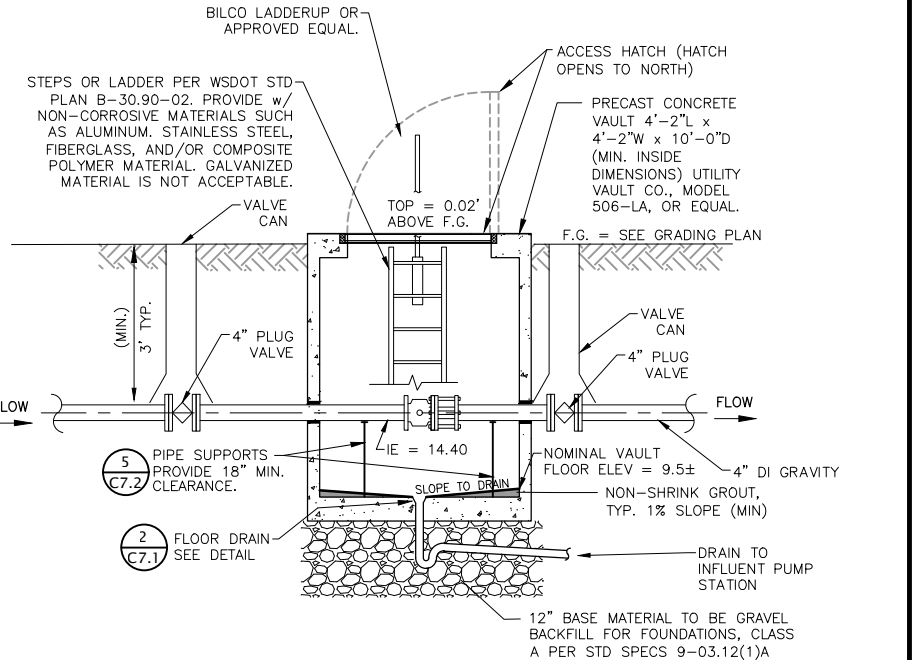
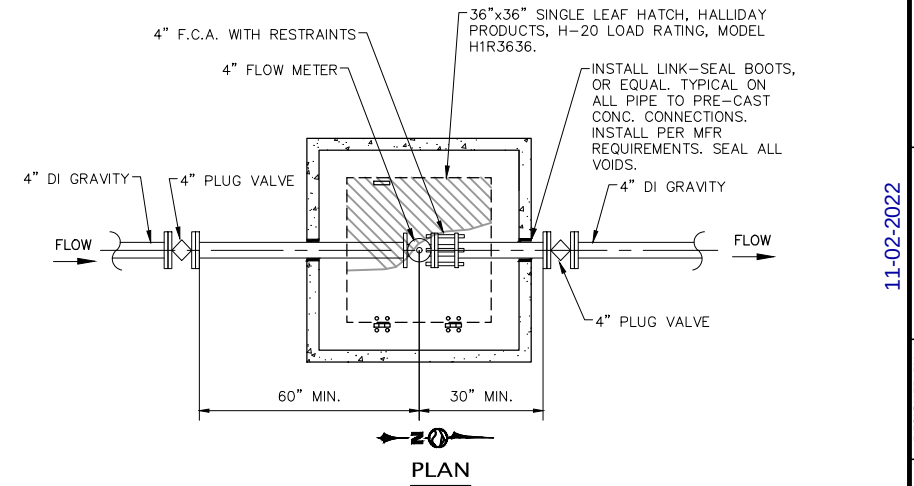
PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 C7.9-C7.10 FLOW METER & VALVE VAULT DETAILS.DWG - 11/2/2022 3:27 PM - Jeff Smith



INFLUENT FLOW METER VAULT, PLAN
 NOT TO SCALE



EFFLUENT FLOW METER VAULT, PLAN
 NOT TO SCALE



PLANT DRAIN FLOW METER VAULT, PLAN
 NOT TO SCALE

Call 811
 two business days
 before you dig

BID SET

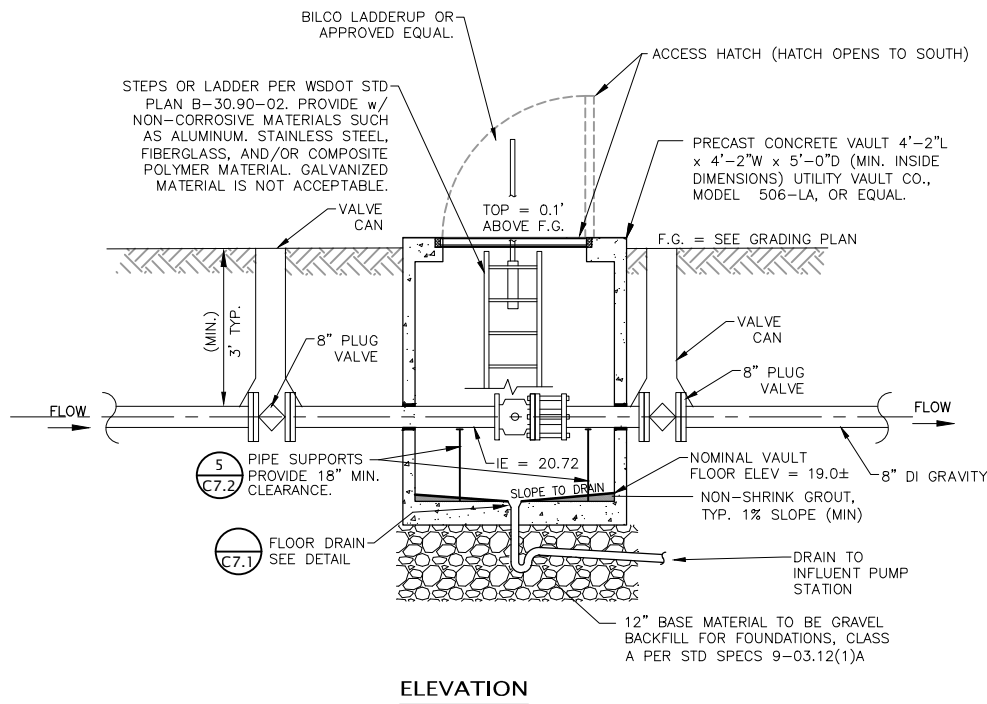
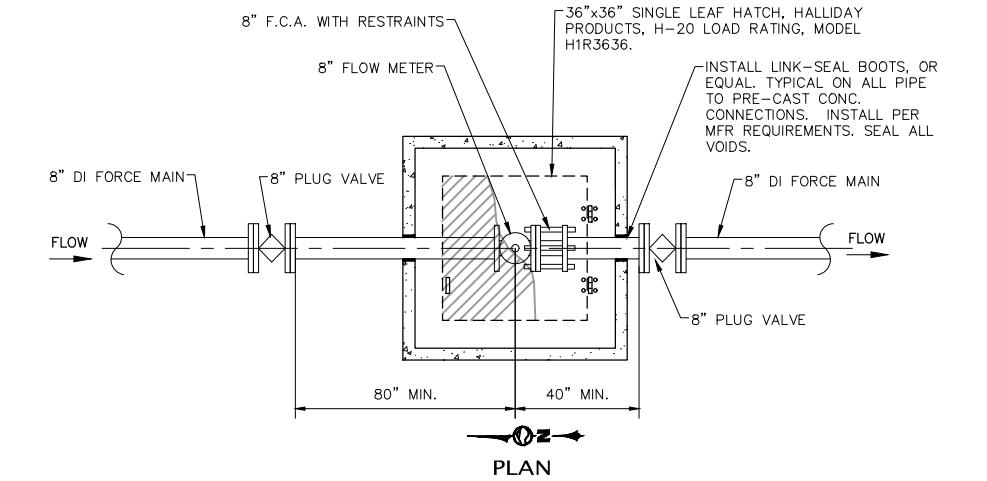
NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

11-02-2022

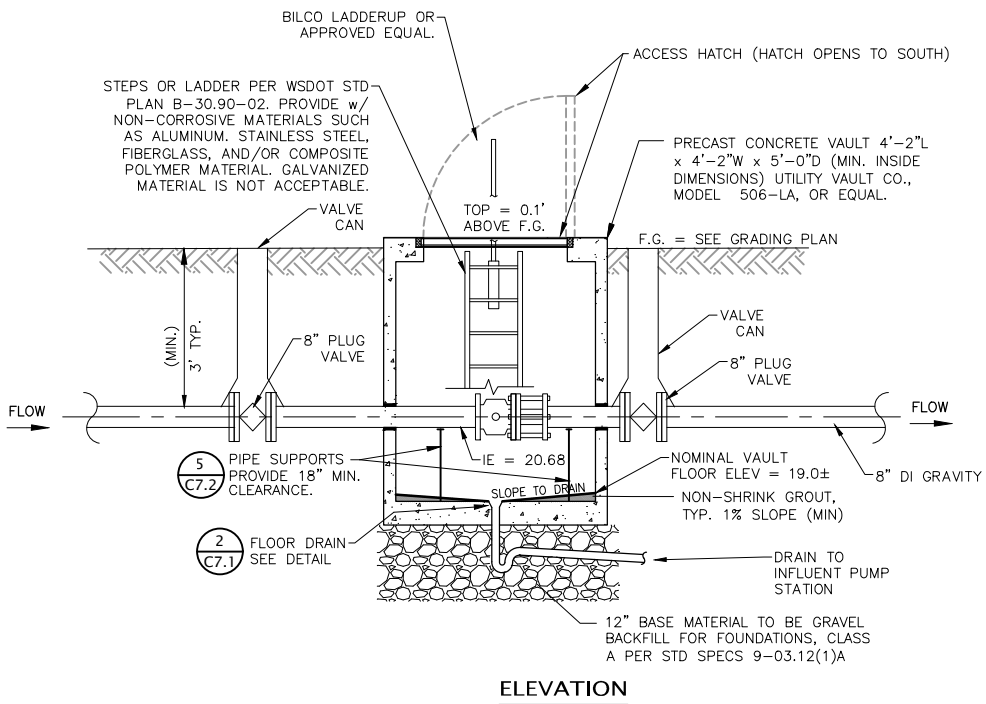
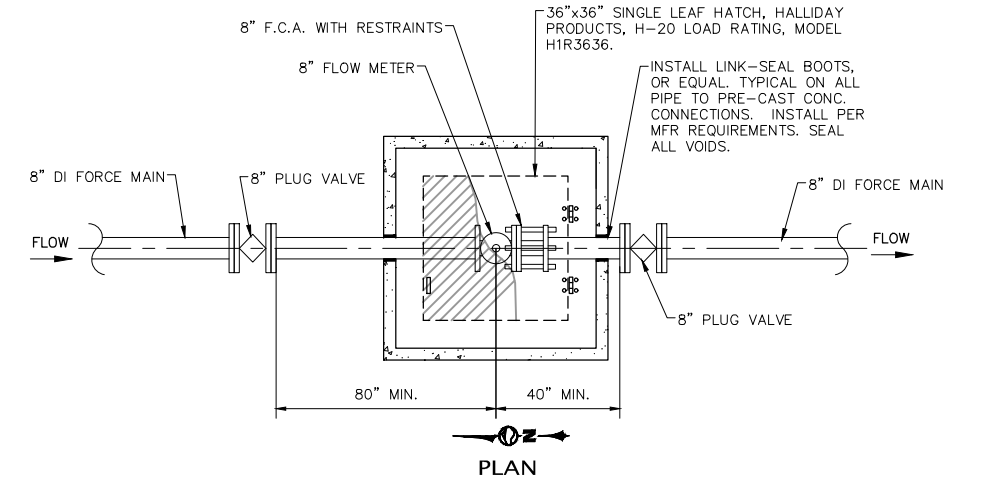
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EASTSOUND SEWER AND WATER DISTRICT					
WASHINGTON					
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I					
FLOW METER & VALVE VAULT DETAILS					
DATE	SCALE	AS SHOWN		JOB NUMBER	
				51 OF 96	

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020_2020-070 C7.9-C7.10 FLOW METER & VALVE VAULT DETAILS.DWG - 11/2/2022 3:27 PM - Jeff Smith



TRAIN 2 FLOW METER VAULT, PLAN
 NOT TO SCALE



TRAIN 3 FLOW METER VAULT, PLAN
 NOT TO SCALE

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

11-02-2022

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DRAWN BY: JGS

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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY, WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

FLOW METER & VALVE VAULT DETAILS

DATE: _____

SCALE: AS SHOWN

JOB NUMBER: _____

SHEET: _____

52 OF 96



BID SET

NOV 2, 2022

PLOT SETTINGS: Adobe PDF1.pc3, ANSI full bleed B (17.00 x 11.00 inches), Portrait, 1:2, WE APWA_UNSCREENED.ctb
 W:\2020\2020-070 EASTSOUND WWP DESIGN\DWG\2020-070 A1.1-A1.3 BLOWER BUILDING.DWG - 11/2/2022 3:27 PM - Jeff Smith

NO.	REVISIONS	BY	DATE
6	ECOLOGY REVIEW COMMENTS	JGC	8/9/22

GENERAL NOTES:

1. PROCESS AIR PIPING AND 2W PIPING NOT SHOWN FOR CLARITY, SEE CIVIL SHEETS C1.1, C6.4, AND C6.5 FOR PIPING INFORMATION.

FINISH SCHEDULE

ROOM NO.	FLOOR		BASE	WALL		CEILING		
	MAT'L	FINISH		MAT'L	FINISH	MAT'L	FINISH	HEIGHT
BLOWER ROOM	CONCRETE	SEALER	6" R.B.	GWB	PAINT	EXP STR	PAINT	VARIES

ABBREVIATIONS
 R.B. = RUBBER BASE
 EXP STR = EXPOSED STRUCTURE
 GWB = GYPSUM WALL BOARD
 AC = ACOUSTICAL WALL BOARD
 FCS = FIBER CEMENT SIDING
 TILE = SUSPENDED CEILING TILE
 PLY = 1/2" PLYWOOD
 ROOM NAME

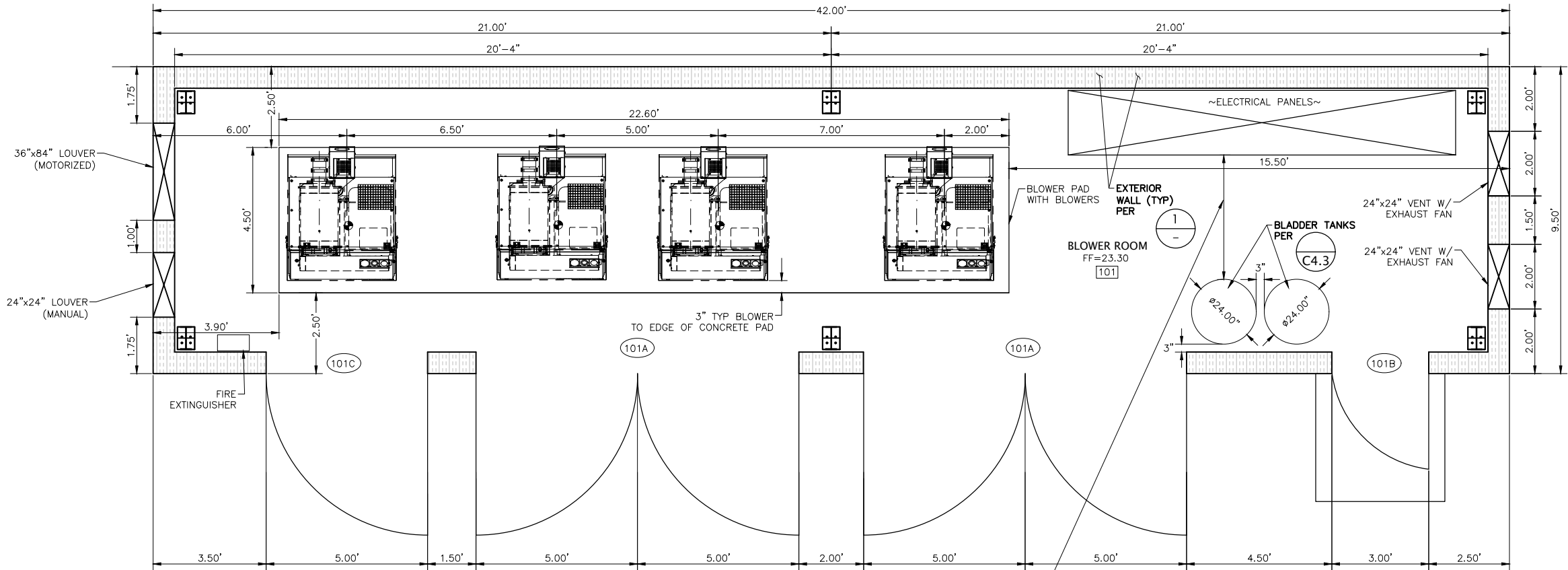
DOOR SCHEDULE

DOOR NO.	DOOR SIZE WxHxT	TYPE	MATERIAL	FINISH	HARDWARE GROUP #
101A	10'-0"x8'-0"x1 3/4"	B	INS HM	PT	-
101B	3'-0"x6'-8"x1 3/4"	A	INS HM	PT	-
101C	5'-0"x8'-0"x1 3/4"	A	INS HM	PT	-

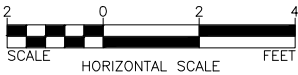
FLUSH PANEL
 TYPE A

TYPE B
 FLUSH PANEL

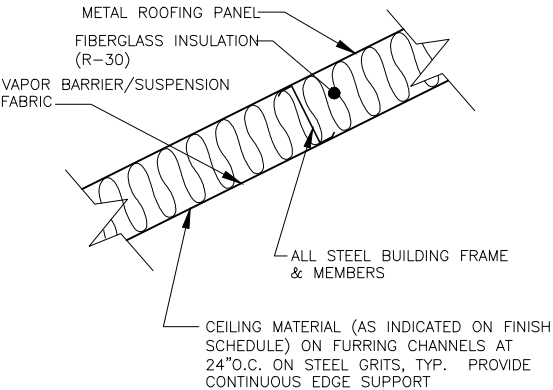
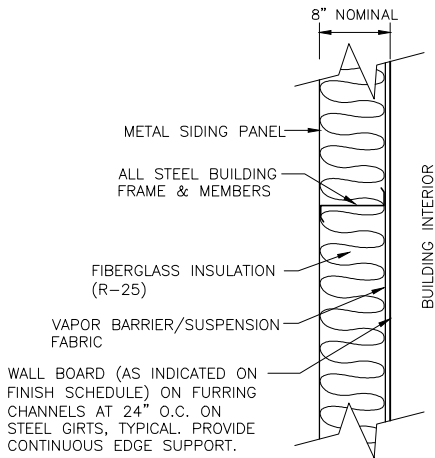
ABBREVIATIONS
 INS HM = INSULATED HOLLOW METAL
 PT = PAINT



BLOWER BUILDING – PLAN



CONTRACTOR TO COORDINATE AND ENSURE ALL ELECTRICAL PANELS ARE CONFIGURED WITH 3.5 FOOT CLEARANCE, AS NEEDED, PER ELECTRICAL REQUIREMENTS.



BID SET

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DRAWN BY: SNR

CHECKED BY: AWL

EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

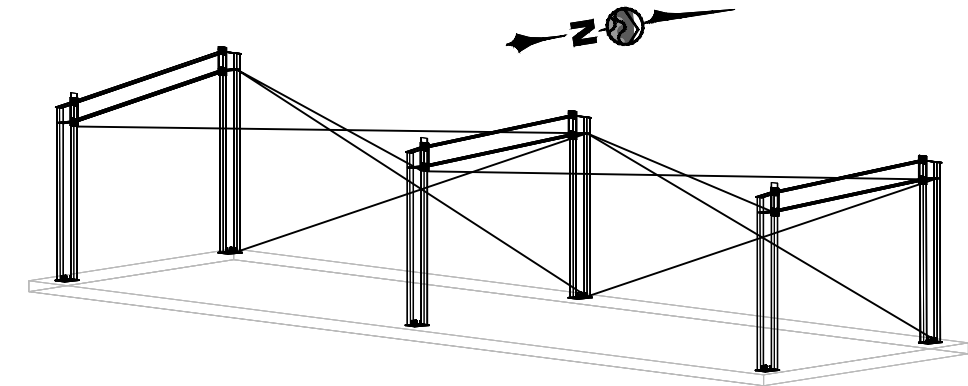
BLOWER BUILDING PLAN

DATE: 11-02-2022

SCALE: AS SHOWN

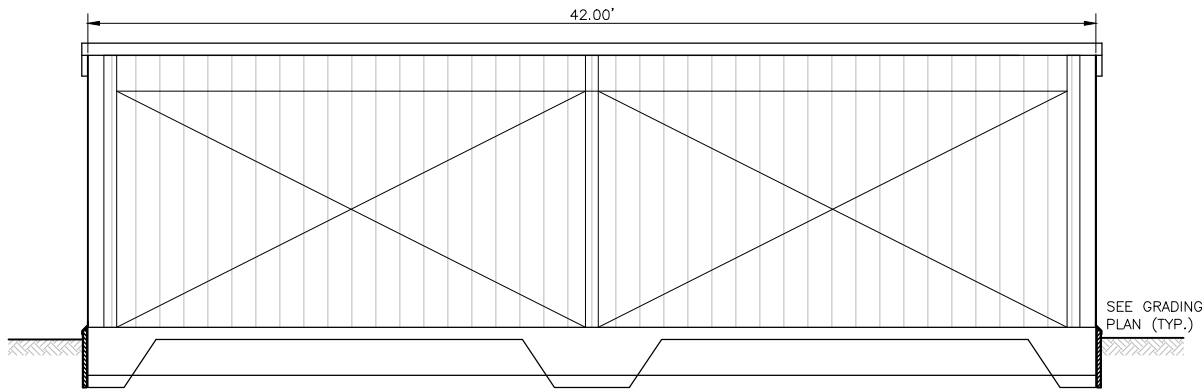
JOB NUMBER: 53 OF 96

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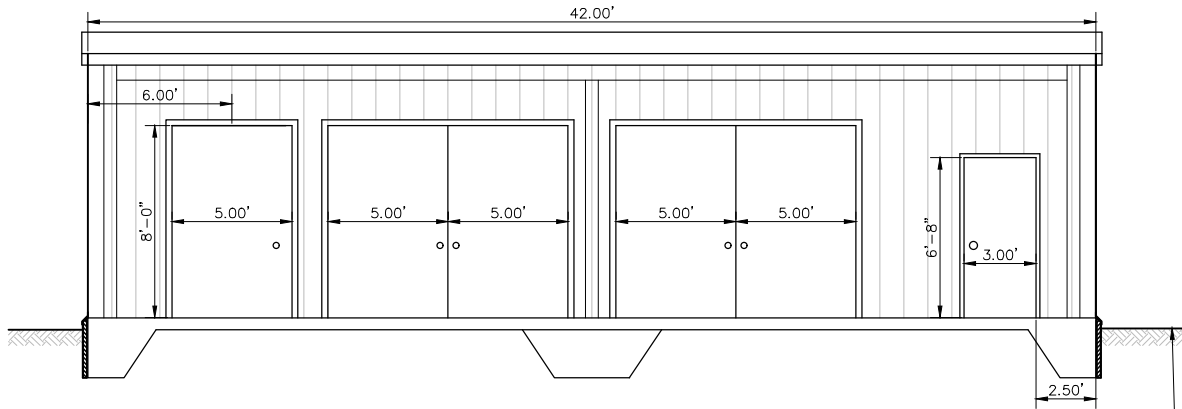
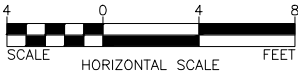


CONCEPTUAL FRAMING
NOT TO SCALE

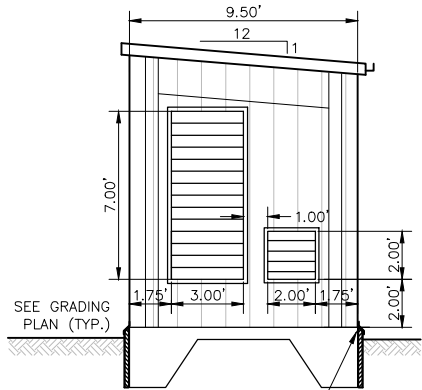
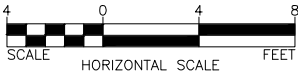
NOTE: CONTRACTOR SHALL PROVIDE
PRE-ENGINEERED METAL BUILDING WITH
SIMILAR COLUMN AND BRACING LAYOUT.



NORTH ELEVATION

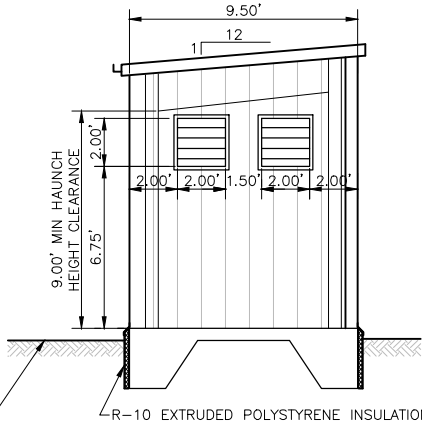
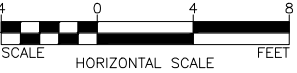


SOUTH ELEVATION

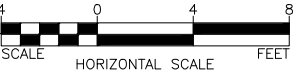


FLASHING TO COVER CHAMFERED
EDGE OF INSULATION BOARD.
MATCH METAL BUILDING TRIM
COLOR. LAP UNDER SIDING AND
OVER INSULATION BOARD A
MINIMUM OF 3-INCHES (TYP.).

WEST ELEVATION



EAST ELEVATION



BID SET



11-02-2022

DESIGNED BY	JCC
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CHECKED BY	AWL

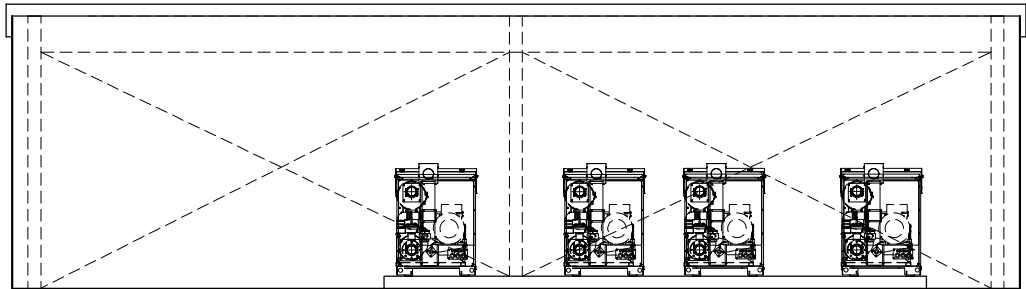
EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

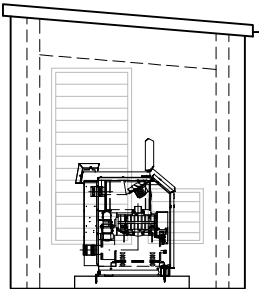
BLOWER BUILDING EXTERIOR ELEVATIONS 1

DATE	SCALE	JOB NUMBER
	AS SHOWN	
SHEET	54	96

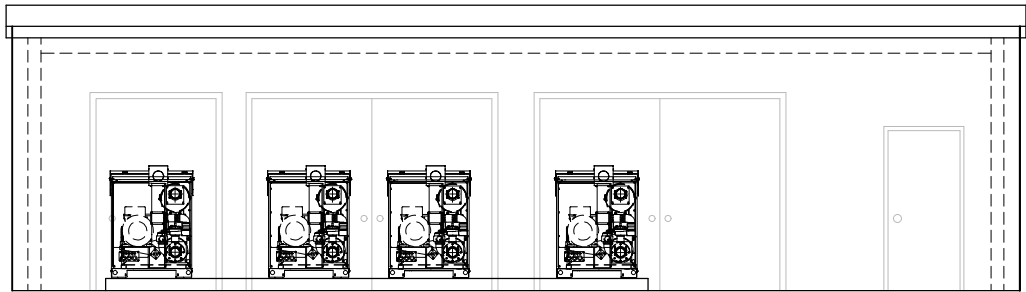
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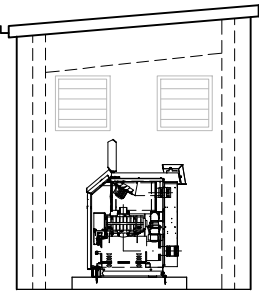
NORTH ELEVATION



WEST ELEVATION

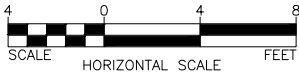


SOUTH ELEVATION

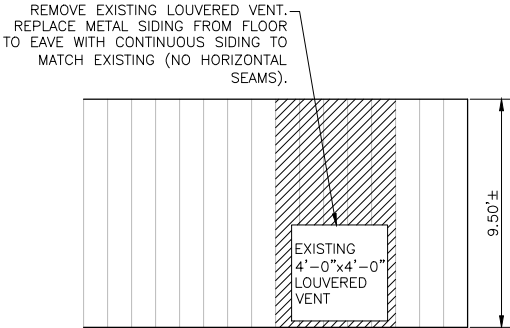


EAST ELEVATION

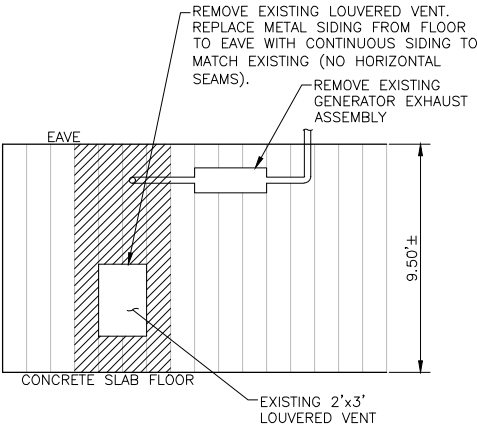
PROPOSED BLOWER BUILDING INTERIOR ELEVATIONS



GENERAL NOTES:
 1) PROCESS AIR PIPING AND 2W PIPING NOT SHOWN FOR CLARITY.
 SEE CIVIL SHEETS C1.1, C6.4, AND C6.5 FOR PIPING INFORMATION

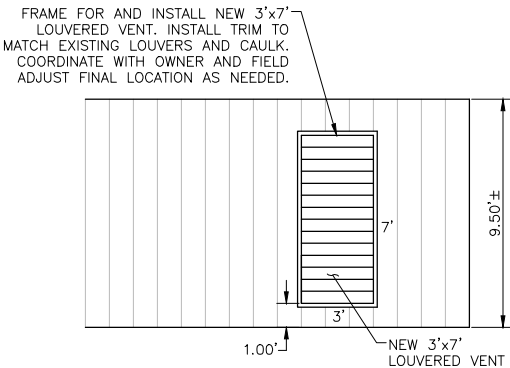
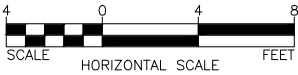


EXISTING OFFICE BUILDING
SOUTH ELEVATION @ 4'x4' LOUVER

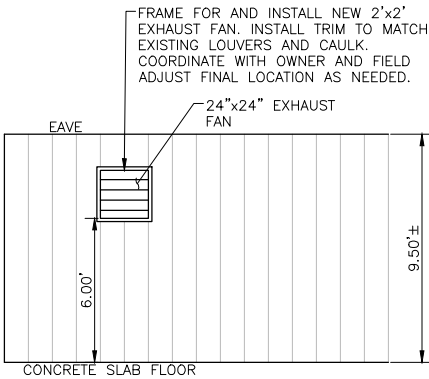


EXISTING OFFICE BUILDING
EAST ELEVATION @ 2'x3' LOUVER

EXISTING OFFICE
BUILDING ELEVATIONS

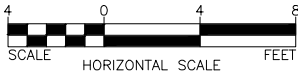


NEW LOUVER FOR
EXISTING WWTP BUILDING
(TYPICAL FOR ALL THREE LOCATIONS)



NEW EXHAUST FAN
FOR EXISTING BUILDING
(TYPICAL FOR ALL THREE LOCATIONS)

EXISTING WWTP
BUILDING ELEVATIONS



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EASTSOUND SEWER AND WATER DISTRICT
 SAN JUAN COUNTY
 WASHINGTON
 WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
 BUILDING DETAILS

DATE	SCALE	JOB NUMBER
	AS SHOWN	96

SHEET
 55 OF 96

-- GENERAL NOTES --

- 1DESIGN BASIS: Designed in accordance with the 2018 International Building Code (IBC).
- 2RISK CATEGORY: III (per IBC Table 1604.5)
- 3DESIGN DEAD / LIVE LOADS
- Floor Live Load: 100 PSF Typical
- 4DESIGN SNOW LOADS (ASCE 7-16)
- Ground Snow Load: Pg = 25 PSF
- Flat Roof Snow Load: Pf = 25 PSF
- Minimum Uniform Roof Snow Load = 25 PSF (unreducible)
- Thermal Factor: Ct = 1.1 / Exposure Factor: Ce = 1.1 / Snow Load Importance Factor: Is = 1.1
- 5DESIGN WIND LOADS (ASCE 7-16)
- V(ult) = 105 MPH / V(asd) = 81 MPH
- Exposure: C / Internal Pressure Coefficient: +/- 0.18
- Components & Cladding: Components and cladding-wind pressures to be used for the design of exterior components and cladding (by others) shall be determined in accordance with ASCE 7 and the IBC.
- 6DESIGN SEISMIC LOADS (ASCE 7-16)
- Site Class = D (default)
- Seismic Design Category = D / Importance Ie = 1.25
- Ss = 1.13g / S1 = 0.40g
- Sds = 0.79g / Sd1 = 0.51g
- 7QUALITY: Contractor shall ensure high standards of workmanship throughout, with strict adherence to the contract documents and all governing codes and standards.
- 8DESIGN RESPONSIBILITY: Kingworks is responsible only for the design of the primary structural system as shown in the contract documents. Design of all secondary structure or non-structural elements are by others.
- 9DISCREPANCIES: Notify Wilson Engineering immediately of any discrepancies between these notes, the contract drawings, the specification, or the governing code. Wilson Engineering shall reply in writing. Any related work performed by the Contractor prior to receiving a reply from Wilson Engineering is at the Contractor's sole risk. For purposes of bidding, the most stringent of the conflicting documents shall apply.
- 10VERIFICATIONS: Verify all existing conditions; verify all dimensions in the field; verify architectural, mechanical and electrical openings for size, location and number; notify Wilson Engineering of any discrepancies, substandard existing conditions, or conditions not included in or contrary to the Contract Documents prior to shop drawing submittal or construction.
- 11DRAWING COORDINATION: Coordinate the structural drawings with drawings from all other disciplines (including but not limited to Architectural, Civil, Mechanical, and Electrical).
- 12COMPLETED FORM: The structure shown in these drawings is designed to be stable and to resist the loads above only in a fully completed form. Contractor shall ensure that the structure is adequately braced and shored during construction for all temporary loads until all elements are in place, and shall ensure that temporary loadings do not exceed the allowable capacity of any structural elements both before and after these elements are in place.
- 13MEANS AND METHODS: Contractor is solely responsible for site safety, coordination, procedures, construction methodology, shoring, bracing, sequencing, and all other "means and methods" of construction except where specifically shown in the Contract Documents.
- 14PROTECTION AND BRACING: Contractor is solely responsible for the protection of existing buildings, utilities, streets, equipment, etc. during construction. Provide temporary bracing and protection as required.
- 15SCALING: Do not scale drawings. See civil drawings for dimensions, and notify Wilson Engineering of any discrepancies.
- 16ALTERATIONS: Any holes or other alterations to the structure which are not specifically detailed on the Contract Drawings shall be submitted to the engineer for approval.
- 17LOAD COORDINATION: The design of all pre-engineered components shall include provisions for precise locations and weights of all mechanical units and other concentrated loads. Concentrated loads are not necessarily indicated on the Structural Drawings, and must be coordinated by the General Contractor, who shall also inform Wilson Engineering of any significant loads not shown in the contract documents. The Contractor shall assume full responsibility for coordination of weights, locations, hanger spacings, methods of attachment and seismic bracing of all mechanical units, sprinklers, pipes, ductwork, and other miscellaneous MEP assemblies.
- 18MEP CONNECTIONS TO STRUCTURE: Unless specifically shown otherwise in the Contract Documents, all anchorage, support, and seismic bracing of mechanical and electrical equipment, piping, ceilings, fixtures and other non-structural components shall be designed by a Washington State Licensed Professional Engineer and installed by the Contractor. The engineer shall be retained by the Contractor, and the Contractor is responsible for all associated engineering, component, and installation costs. Design shall be per ASCE 7 Chapter 13.
- 19DELIVERY, STORAGE AND HANDLING: All products shall be delivered, stored, and handled according to the Manufacturer's recommendations and installation instructions. Protect all items from damage, moisture, corrosion, or other deterioration before, during and after installation.
- 20COPYRIGHT: These drawings, and all designs shown within these drawings, are copyrighted by Kingworks Structural Engineers. Duplication is not permitted without written permission. The designs shown herein are intended for this project only and may not be used on any other project or for any other purpose.

-- SUBMITTALS --

- 1GENERAL: Provide PDF of all submittals to Wilson Engineering. Allow two weeks for review. Submittals will be reviewed for general conformance to the contract documents. Responsibility for adherence to the contract documents lies solely with the Contractor, including but not limited to dimensions, sizes, connections, and quantities.
- 2CONTRACTOR REVIEW: Contractor shall review, mark, and stamp all submittals before submittal to Wilson Engineering. Unreviewed or unstamped submittals will be returned to the Contractor without review.
- 3RESUBMITTALS: Resubmittals shall have all revisions clearly identified with "drawing clouds" and revision dates. KW shall not be responsible for review of any unmarked revisions.

-- SUBMITTALS --

- 4SHOP DRAWINGS: To include typical and unique conditions and all connections, shall be submitted to the Structural Engineer of Record for the following products prior to fabrication. Shop drawings shall clearly demonstrate the Contractor's understanding of the contract documents. The following shall be considered minimum structural submittals for this project:
- Concrete Reinforcing Steel (incl. wall elevations)
- Concrete Mix Designs (confirm f'c prior to construction)
- Slab-on-Grade Jointing Plan
- Structural Steel
- 5DEFERRED SUBMITTALS: Shall be designed by a Washington State Licensed Professional Engineer (Structural Engineer where required by jurisdiction). Submittals shall clearly indicate all loads and reactions to be imposed on the primary structure. Submittals shall include all components and connections required to perform the work. Where the Contract Documents indicate member or connection qualities and/or quantities, these shall be considered a minimum and remain the design responsibility of the component engineer to increase as required to satisfy load and deflection requirements. Deferred submittals shall include calculation set. Both the calculation set and all related shop drawings of the deferred submittal shall be sealed prior to submission. Wilson Engineering shall forward deferred submittals to the Building Official where required. The following shall be considered Deferred Submittals on this project:
- Metal Stairs & Landings
- Pre-engineered Metal Building for Blower Building (See Specifications)
- 6SUBMITTAL REVIEW COMMENTS: Engineer marks and comments on shop drawings and other submittals are a normal and expected part of the submittal process, and are not to be used as a basis for change orders except in cases where these marks result in or derive from substantial changes to the Contract Drawings. Time required to revise and resubmit any submittal shall be considered inherent to the submittal review process and shall not be deemed a change order. If discrepancies are discovered between the submittals and the Contract Documents (either before, during, or after submittal review), the Contract Documents shall govern and be implemented unless specifically directed otherwise.

-- SPECIAL INSPECTIONS, TESTING, AND STRUCTURAL OBSERVATION --

- 1GENERAL: A special inspection agency shall be retained by the owner to perform inspections according to IBC Chapter 17. The following scheduled structural special inspection and testing regimen shall be cross-referenced with the IBC and its referenced standards for more specific requirements and exceptions. Special inspection and test reports shall be submitted to the Building Official, Civil Engineer, and Structural Engineer in accordance with IBC 1704.2.4.
- 2NONSTRUCTURALCOMPONENTS: Nonstructural components are also subject to special inspection for conformance to the Seismic Design Requirements of ASCE 7-10 Chapter 13. The support and seismic bracing of nonstructural components shall be designed by a Washington State Licensed Engineer, who shall be retained by the Contractor. Nonstructural components subject to seismic design and periodic special inspection requirements of ASCE 7-10 and IBC Chapter 17 include (but are not limited to) the following:
- Architectural components (IBC 1705.12.5)
- Nonstructural components (IBC 1705.13.2)
- MEP Components (IBC 1705.12.6)
- Reference the architectural, electrical, and mechanical drawings and/or specifications for additional special inspection and testing requirements.
- 3STRUCTURAL OBSERVATION: Kingworks will perform Structural Observations in accordance with IBC Section 1704.6 if/as required. These observations provide intermittent checks of general conformance to the design intent and are in addition to (not replacing) the third-party special inspection regimen. It shall be the Contractor's responsibility to keep the Structural Engineer apprised of the general schedule of construction, such that observations may be made at appropriate stages before significant structural components (such as reinforcing bars, framing members, or wall holdowns) are obscured.

-- STRUCTURAL SPECIAL INSPECTION SCHEDULE --

	STRUCTURAL ITEM	FREQUENCY (C=continuous, P=periodic)	REFERENCE (2018 IBC Section, uon)
1	STEEL		1705.2, 1705.12.1
	- Structural Steel - All	Varies	AISC 360-16 Chap N
	- Steel Deck	P	1705.2.2
	- Cold-Formed Steel	P	1705.11.2, 1705.12.3, AWS D1.3
	- SLRS Steel	Varies	AISC 360-16 Chap N, AISC 341-16 Appx Q, 1705.13
2	CONCRETE		1705.3, ACI 318-14 26.13
	- Reinf Placement	P (and prior to all pours)	Table 1705.3, ACI 318-14 26.6.2
	- Anchor Placement	P (and prior to all pours)	Table 1705.3
	- Concrete Placement	C	Table 1705.3
	- Concrete Testing	Per ACI 318-14 26.12	Table 1705.3
	- Formwork & Curing	P	Table 1705.3
3	POST-INSTALLED ANCHORS		Table 1705.3
	- Epoxy or Adhesive Anchors Used in Horizontal or Overhead Position	C	See ICC-ES report
	- All Other Anchors Installed in Hardened Concrete or Masonry	P (except where C req'd by ICC-ES report)	See ICC-ES report
4	SOILS & FOUNDATIONS		1705.6, Table 1705.6, Geotech Report
	- Subgrade Adequacy	P (beneath fill and/or foundations)	Table 1705.6
	- Excavation Depth	P	Table 1705.6
	- Fill Materials	P	Table 1705.6
	- Fill Placement & Compaction	C	Table 1705.6

-- FOUNDATIONS & SUBGRADE --

- 1SOIL ALLOWABLE BEARING PRESSURE: 2500 PSF (x 1.33 when subjected to wind or seismic loadings).
- 2GEOTECHNICAL REPORT: Reference GeoEngineers geotechnical report dated September 7, 2021 for subgrade preparation, backfill, compaction, wet weather work, and other recommendations. All provisions of the geotechnical report shall be enacted by the Contractor.
- 3GEOTECHNICAL VERIFICATION: The Owner shall retain a geotechnical inspection agency to inspect the excavations and confirm that the provisions of the Geotechnical Report are followed. All recommendations of the geotechnical inspector shall be followed and communicated with the design team via inspection report. Contractor shall coordinate this inspection effort with the Owner.
- 4UTILITIES: Utilities are not to pass through or beneath footings, stemwalls, and other concrete work on grade except as shown in specific details.
- 5MISCELLANEOUS VERIFICATIONS: Verify sizes, slopes and locations of tunnels, electrical cells, pits, pipes, floor drains, trenches and floor recesses with architectural, mechanical and electrical contractors.
- 6ALIGNMENT: All footings shall be centered below columns and walls, unless dimensioned otherwise.
- 7BACKFILL: Do not backfill against below-grade walls until strength has been achieved and floor framing is in place.
- 8FROST PROTECTION: Maintain minimum 1'-6" soil cover, measured from finished grade to the bottom of the footing, for perimeter wall foundations and isolated exterior foundations.
- 9EXCAVATION SLOPE: Excavation slope shall not exceed that permitted by local regulation, except as specifically approved by the geotechnical engineer.

-- REINFORCED CONCRETE --

- 1REFERENCED STANDARDS: Reinforced concrete shall conform to the requirements of the following documents, except as modified in these drawings.
- ACI 301-16 "Standards Specifications for Structural Concrete"
- ACI 318-14 "Building Code Requirements for Structural Concrete"
- 2MATERIALS
- Conform to the following concrete material schedule requirements:

-- CONCRETE MATERIAL SCHEDULE --

	f'c (PSI)	W/CM (MAX)	COARSE AGG (MAX)	AIR CONTENT	MIN FLY ASH AND/OR SLAG
BLOWER BUILDING SOG & GRADE BEAMS, MISC EQUIPMENT PADS	4500	0.45	3/4"	6%	30%
HYDRAULIC STRUCTURES (SEE ADDTL REQ BELOW)	4500	0.45	1"	SEE NOTES	20%
DRAIN PUMP STATION LID	4500	0.45	3/4"	5%	20%

- Required concrete strength "f'c" shall be evaluated at 28 days (56 days OK for fly ash or slag concrete).
- Cementitious Materials: Provide Type II Blended Hydraulic Cement per ASTM C595 for hydraulic structures. Provide Type I or II Cement per ASTM C150 (Type III OK for Precast) for all else. Cementitious materials for use in concrete exposed to soil or weather must meet or exceed the S1 exposure class per ACI 318-14 Table 19.3.2.1.
- Fly Ash: Class F per ASTM C618.
- Slag: GGBF Slag per ASTM C989, Grade 100 minimum.
- Air Entrainment: Provide air entrainment per concrete mix table +/- 1.5%, all other locations exposed to weather shall have air entrainment of 6% (+/- 1.5%).
- Typical Reinforcing Steel: ASTM A615, Grade 60 Typ, Grade 75 for Train #3 (H) bars
- Weldable Reinforcing Steel: ASTM A706, Grade 60
- Welded Wire Fabric: ASTM A185
- Mixes for hydraulic structures shall also satisfy the following:
A. Shrinkage reducing admixture per ASTM C-494 shall be added so that test mixes shall demonstrate shrinkage of no more than 400 microstrain at 7 days.
B. Air percentage shall be per the manufacturer of the shrinkage reducing admixture recommendation.
C. Hydraulic structure concrete shall contain a minimum of 500 pounds of cementitious materials per cubic yard.
D. For hydraulic structures, the requirement for stone aggregate shall not be waived.
- 3SPICES: All reinforcing steel lap splices are to be per typical schedule unless noted otherwise. All welded wire reinforcement lap slices shall be the greater of one space plus 2 inches or 6 inches, unless otherwise noted.
- 4CONTINUITY: Horizontal reinforcing steel in walls and wall footings shall be continuous around corners, same size and spacing. At intersections of walls or footings, extend all bars as far as possible into continuous element and terminate with standard hook.
- 5CLEAR COVER: Provide clear cover from outermost reinforcing to surface of concrete in accordance with the following:
- Elements cast against and permanently exposed to earth: 3" dr
- Elements formed and exposed to earth or weather: 2" dr (#6-#18), 1 1/2" dr (#3-#5)
- 6INTEGRAL COLUMN & WALL FOOTINGS: Where column footings are shown integral with wall strip footings, transverse wall footing bars need not continue through width of column footing. Longitudinal wall footing bars shall extend continuous through column footing, in addition to column footing bars per typical detail.
- 7ACCESSORIES: Provide all accessories, chairs, spacer bars and supports necessary to secure steel in accordance with ACI Code of Standard Practice.
- 8CHAMFER: Chamfer all exposed corners and edges above grade per Civil.
- 9FORM STRIPPING: Do not strip forms until concrete has reached adequate strength.
- 10SLEEVES / OPENINGS: Furnish and place all sleeves and openings as shown on the drawings or as specified.

-- REINFORCED CONCRETE --

- 1REINFORCING FABRICATION: All reinforcing shall be shop fabricated. Exception: #3 or #4 bars may be field bent one time in any location, do not rebend or restraighthen.
- 2HOOKS: All hooked bars shall be a standard shop fabricated hook with bend radii and length per ACI 318, UON.
- 3TESTING: Test cylinders shall be taken by qualified personnel according to ACI 318-14 Section 26.12.
- 4MEMBRANE: Provide vapor barrier (per arch/spec, 10-mil min where not otherwise specified) directly below slab-on-grade (above compacted fill), reinforce all seams and repair tears as necessary.
- 5WEATHER PROVISIONS: Observe all ACI recommendations for hot or cold weather concreting cure slabs using an approved curing compound or wet cure system per ACI recommendations, with special consideration for slag and fly ash concrete as appropriate.
- 6CONSTRUCTION JOINTS: All construction joint locations shall be submitted to Wilson Engineering for approval. Roughen joints to 1/4-inch minimum amplitude, remove all laitance. Soak joint continuously for 2 hours minimum, then remove all standing water, immediately prior to second pour. Unless noted otherwise, all reinforcing shall be spliced/continuous across the construction joints.
- 7RIGID FOAM: Where rigid foam is indicated on plans to be placed below concrete slabs, it shall consist of cellular polystyrene conforming to ASTM D6817. Polystyrene shall have a minimum compressive resistance of 3.6 psi at 1 percent deformation unless otherwise noted. Secure foam in place per the manufacturer's recommendations. Offset joints between foam blocks 24" minimum between adjacent layers. At contractors option it is acceptable to substitute polystyrene conforming to ASTM C578 Type XIV rated for 40 psi compressive resistance at 10 percent deformation with a minimum thickness of 2" per layer.

-- ANCHORAGE TO CONCRETE --

- 1MATERIALS (unless noted otherwise in the drawings)
- Concrete or grout must cure for a minimum of 21 days prior to drilling any holes or placing post-installed anchors.
- Anchor type shall be according to the drawings. All post-installed anchors installed in concrete shall have ICC-ES reports demonstrating IBC compliance for use in cracked concrete and for seismic loading. Substitutions not permitted without written permission by KW.
- Pre-approved Epoxy for post-installed threaded rod or reinforcing in concrete base material: HILTI HIT-RE 500 V3 or Simpson SET-3G.
- Pre-approved "Expansion Anchors" in concrete base material: HILTI Kwik-Bolt TZ or Simpson Strong-Bolt 2, or DEWALT/Powers Power-Stud+ SD2.
- Pre-approved "Screw Anchors" in concrete base material: HILTI HUS-EZ or Simpson Titen HD or DEWALT/Powers Screw-Bolt+.
- Post-installed or Cast-in-Place Threaded Rod (Anchor): ASTM A36
- Post-installed Reinforcing: ASTM A615 Grade 60
- 2EMBEDMENT: Anchor embedment in base material shall be per the drawings, which shall govern over the typical values shown below. Where not otherwise indicated, provide embedment as follows:
- Epoxy to Concrete: Min Embed = 12 x (Rod/Bar Ø) for 5/8"Ø (#5) or smaller, 14 x (Rod/Bar Ø) for 3/4"Ø (#6) or larger
- Expansion Anchor to Concrete: Min Embed = 8 x (Anchor Ø)
- Screw Anchor to Concrete: Min Embed = 9 x (Anchor Ø)
- Cast-In-Place Anchor: Min Embed = 7" to top of embedded washer or hook
- 3INSTALLATION: Post-installed anchor hole diameter, drilling depth, cleaning and installation procedure shall be in accordance with the current Manufacturer's Printed Installation Instructions (MPII) provided in the ICC/ES report. Holes shall be drilled with rotohammer equipment. Core-drilled holes are not permitted unless specifically noted otherwise.
- 4COLD-WEATHER INSTALLATION: Do not use epoxy or adhesive anchors outside of their rated temperature range. Contact the Structural Engineer for alternate if the base material temperature may be less than 40 degrees during installation or curing.
- 5CAST-IN-PLACE ANCHORS: Cast-in-place anchors shall have nut and washer at embedded end, UON. Anchors shall be affixed to the form to prevent movement during pouring, vibration, or set-up and shall not be "stabbed" into wet concrete or grout. Verify adequate length of exposed thread to fully engage all attached work.
- 6FINISHES: All anchors used at exterior, or where subject to moisture, or where in contact with pressure treated wood, shall be hot-dip galvanized per ASTM A153 or stainless steel, including matching washers and nuts.
- 7REINFORCEMENT LOCATIONS: All post-installed anchors shall be located to avoid drilling into reinforcement, unless specifically approved by the Engineer. Reinforcement shall be placed with consideration for locations of post-installed anchors. Do not damage reinforcing during drilling operations.

-- STRUCTURAL STEEL --

- 1REFERENCED STANDARDS: Fabrication, erection and dimensional tolerances of structural steel members shall be in accordance with AISC 303-16 "Code of Standard Practice for Steel Buildings and Bridges" and ASTM A6.
- 2MATERIALS (unless otherwise noted on drawings)
- Wide Flange and WT shapes: ASTM A992, fy=50 ksi.
- Hollow Structural Sections (HSS): A500 Grade B, Rectangular fy=46 ksi, Round fy=42 ksi.
- Pipe Sections: A53 Grade B, fy=35 ksi.
- Bolts: A325(-X or -N or -SC per details) Type 1 high strength bolts with A563 nuts and F436 washers unless noted otherwise.
- Welding Electrodes: E70XX
- Anchor Rods: ASTM F1554 Grade 36
- Angles / Channels / Plates / Rods / Miscellaneous: ASTM A36
- 3WELDING: All welding to be performed by qualified welders and shall conform to A.W.S. D1.1.
- 4GALVANIZING: All steel elements and fasteners exposed to the exterior shall be hot-dip galvanized in accordance with ASTM A153/A123.
- 5GALVANIZING REPAIR: Repair all damage to hot-dip galvanized coating by hot-stick or high-zinc paint method in accordance with ASTM A780. Where welding of galvanized steel is required, all galvanizing on the area of steel to be welded, plus 3/4-inch each side of the welded area, shall be ground off prior to welding; affected area shall be repaired per ASTM A780.

NO.	REVISIONS	BY	DATE
2	Bid Set	TH	9/6/22



-- STRUCTURAL STEEL --

- 6ERECTION: Contractor shall comply with all current OSHA/WISHA requirements concerning steel erection. This may require additional bolts, shims, guys, bracing, and other elements that are not detailed in the Contract Documents but remain the Contractor's responsibility to implement.
- 7BOLT SPECIFICATION: All bolted connections shall conform to the current "RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts". "-SC" bolts shall be approved twist-off tension-control type per ASTM F1852 or shall utilize compressible-washer-type direct tension indicators per ASTM F959.
- 8BOLT INSTALLATION: Bolts shall be snug-tightened (-ST) unless noted as "-SC". "-SC" bolts shall be pretensioned with Class A faying surfaces per the RCSC Spec.
- 9BOLT HOLES: Bolt holes in steel shall be 1/16" larger than bolt diameter, except where specifically detailed otherwise.
- 10GROUT: Grout for baseplates, sleeves, and embedded steel shall be an approved nonshrink cementitious grout containing natural aggregates delivered to the job site in factory prepackaged containers requiring only the addition of water. The minimum 26-day compressive strength shall be at least 1000 psi higher than the supporting concrete strength, UON. Grout shall be mixed, applied, and cured strictly in accordance with the manufacturer's printed instructions. For grouting under base plates, grout shall be proportioned as a flowable mix. When a flowable mix does not provide the required strength, an epoxy grout shall be used. Grout thickness shall be per the typical details.



Verify Scale:

Enclosing box measures 1/2 inch tall x 2 inch wide when drawings are printed at full scale

DESIGNED BY	THaidersdorf	DRAWN BY	CKinney	CHECKED BY	THaidersdorf
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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON

SAN JUAN COUNTY

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

STRUCTURAL NOTES

SHEET	DATE	SCALE	JOB NUMBER	PAGE
S1.1	10-15-2021	(on 22"x34")	Wilson: 2020-070 KW: 211.31	56 OF 96



BID SET

ELEVATION

Diagram illustrating the reinforcement layout for a circular foundation, showing concentric rings and radial bars.

Key components and labels:

- EDGE OF PEDESTAL ABOVE, SEE 2/S2.1**: Points to the outermost boundary of the foundation.
- SMALLEST RING = 12'-0"**: Points to the innermost ring of reinforcement.
- ADDITIONAL #5 x 8'-0" TOP BAR**: Points to a bar in the top ring.
- ADDITIONAL #5 x 8'-0" BOTTOM BAR**: Points to a bar in the bottom ring.
- LAP PER TYP DTS**: Points to a lap joint in a bar.
- RADIAL BARS PER 1/S2.1**: Points to the radial bars extending from the center to the outer edge.
- TERMINATE TYP AT 3'-0" RING, TERMINATE EVERY FIFTH BAR AT 1'-0" RING**: Points to the termination of radial bars.

3' 0" DIA.

12" DIA. PEDESTAL

3" CLR

3" CLR

CONT #5 AA @ 12" OC

#5 RADIAL BAR @ 16" OC AT OUTER EDGE OF PEDESTAL, 2" CLR FROM TOP.

- NOTES**
1. CENTER OF PEDESTAL = CENTER OF TRAIN #3 SLAB, REFER TO 1/S2.1
 2. TOP BARS SHOWN, BOTTOM BARS SIM (3" CLR FROM BOT).

1
S4.1

#5 (V) @ 16" OC AA,
TYP LAP AT TOP,
STD HOOK AT BOT.

1'-0" AA

CL

GUARDRAIL, MECHANICAL
EQUIPMENT, AND LADDERS
PER CIVIL

(2) #4 (H) AT TOP

CONCRETE WALL PER PLAN

#5 (H) @ 12" OC EF,
GRADE 75 BARS,
STAGGER LAPS PER 5/S2.1

#5 (V) @ 12" OC EF, 3" CLR
EF AND FROM BOT

GROUT SURFACE
PER SPEC

PEDESTAL REINFORCING
PER 2/S2.1

HYDROPHILIC WATER STOP AA,
SUBMIT FOR APPROVAL

MAT FOUNDATION &
REINFORCING PER PLAN

#5 (V) @ 16" OC AA,
STD HOOK T&B

SUBGRADE PREP PER
STRUCTURAL NOTES
& GEOTECH. REPORT

cl1

1'-0" Typ AA

12" CONCRETE WALL AA

12" MAT SLAB FOUNDATION

PERIMETER OF RAISED PEDESTAL, SEE 2/S2.1

3 S2.1

clA

4 S2.1

clB

CONT #5 AA @ 12" OC

#5 RADIAL BAR @ 12" OC AT OUTER EDGE OF SLAB, 2" CLR FROM TOP, TERMINATE EVERY OTHER BAR W/ STD HOOK AGAINST (V) BAR AT PEDESTAL PERIMETER

5 S2.1

cl1

NOTE
TOP BARS SHOWN, BOTTOM BARS SIM (3" CLR FROM BOT).

TRUE NORTH

NOTE
TOP BARS SHOWN, BOTTOM
BARS SIM (3" CLR FROM BOT).



NO.	REVISIONS	BY	DATE
2	Bid Set	TH	9/6/22



THIS ELECTRONIC SEAL AND SIGNATURE IS APPLIED WITHIN
AN ELECTRONIC DOCUMENT. THE DIGITAL SIGNATURE HAS
BEEN REMOVED FOR DOCUMENT COMPILED. A STORED
COPY IS AVAILABLE ON REQUEST.



WILSON
ENGINEERING

WILSONEN INEERIN.COM



**KINGWORKS
STRUCTURAL
ENGINEERS**

00 Dupont St, Suite B
Bellingham, WA 98225

Verify Scale:

Enclosing box measures 1/2 inch tall x 2 inch wide when drawings are printed at full scale.

DESIGNED BY
THAIDORSDORF

THEORETICAL	DRAWN BY

DRAWN BY
C. Kinnear

SKINNEY	CHECKED BY
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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY

WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

TRAIN 3 PLANS & SECTION

DATE 10-15-2010

1.2

As indicated

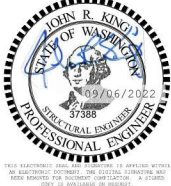
FOR NUMBER (on 22" x 34")

OF 96

811 Call 811
two business days
before you dig

BID SET

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22



Verify Scale:
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when drawings are printed at full scale.

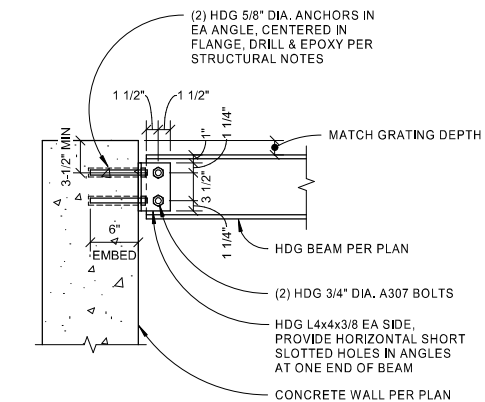
DESIGNED BY	THEIDERSDORF	DRAWN BY	CKINNEY	CHECKED BY	THEIDERSDORF
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1
UV DISINFECTION CHAMBER & DIGESTER PLAN & SECTIONS

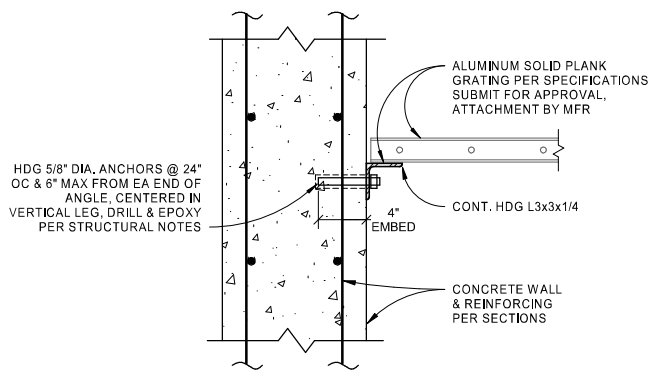
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SHEET	S2.2	PAGE	58 OF 96



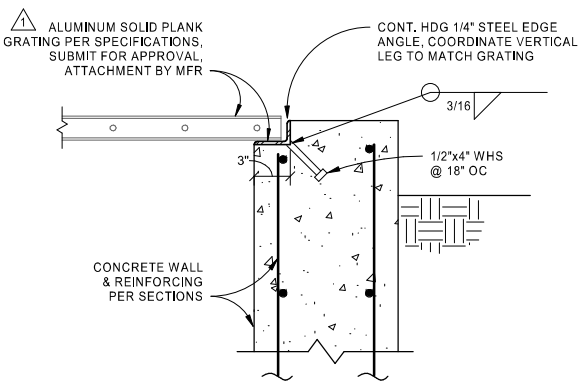
BID SET



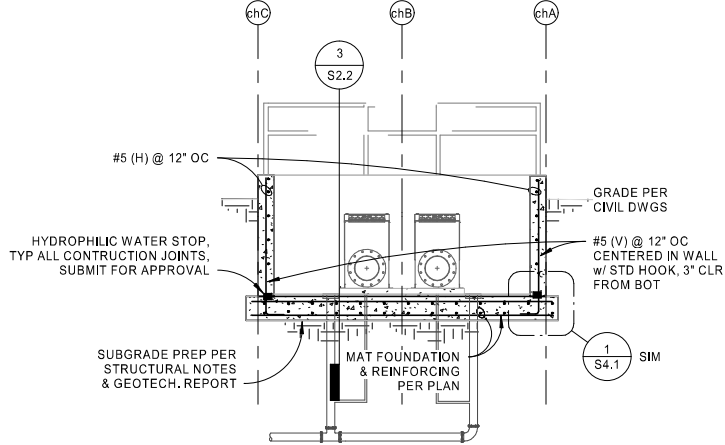
7 STEEL BEAM TO CONCRETE WALL
1" = 1'-0"



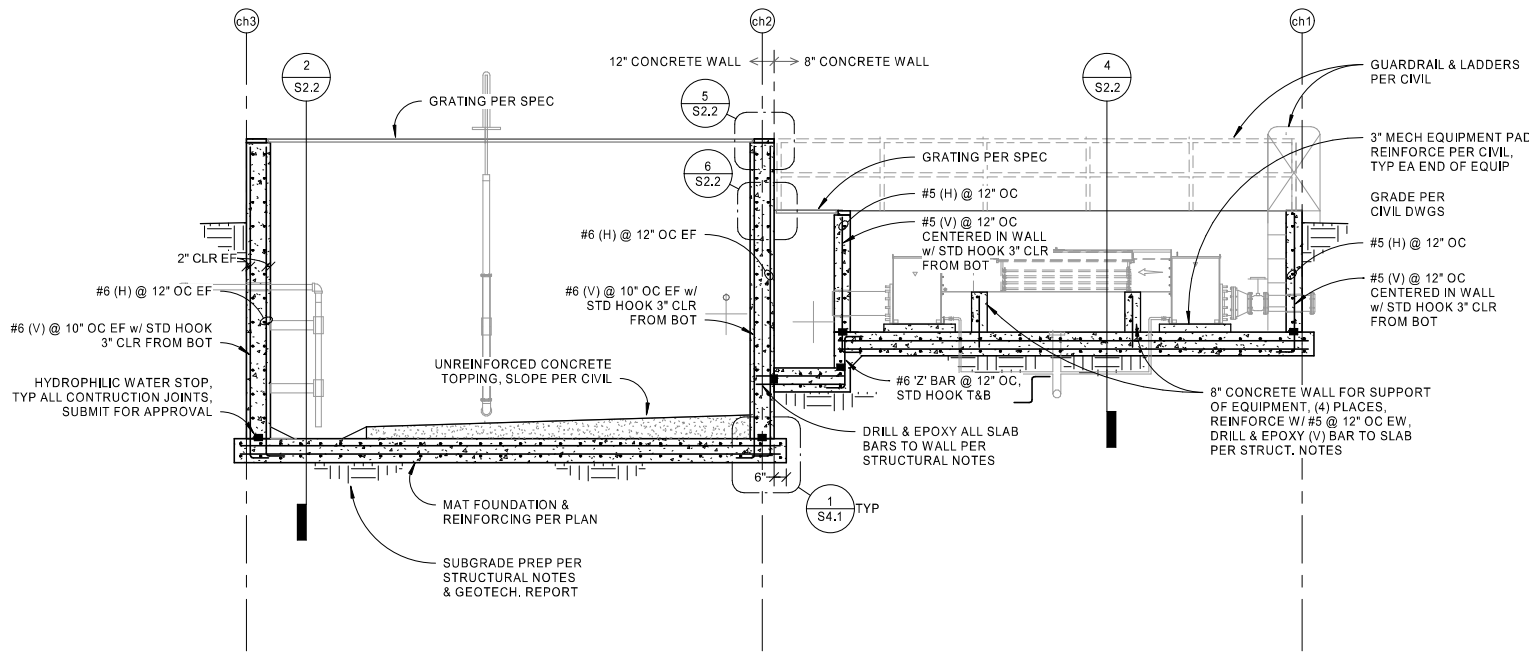
6 TYP GRATING SUPPORT AT SIDE OF CONCRETE WALL
1 1/2" = 1'-0"



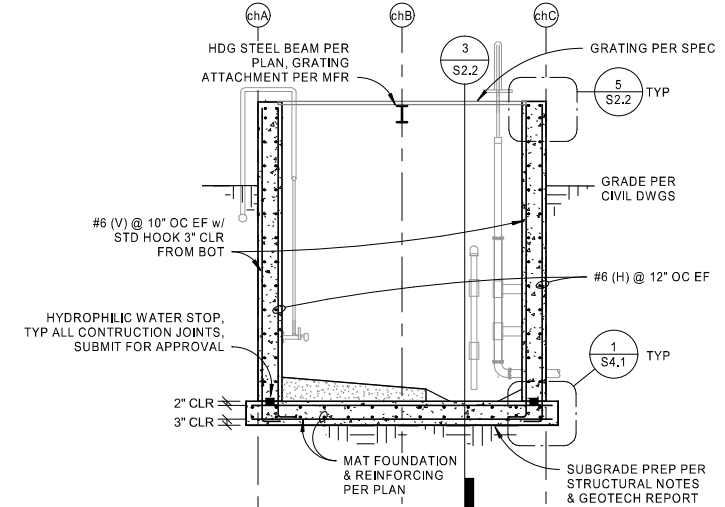
5 TYP GRATING SUPPORT AT TOP OF CONCRETE WALL
1 1/2" = 1'-0"



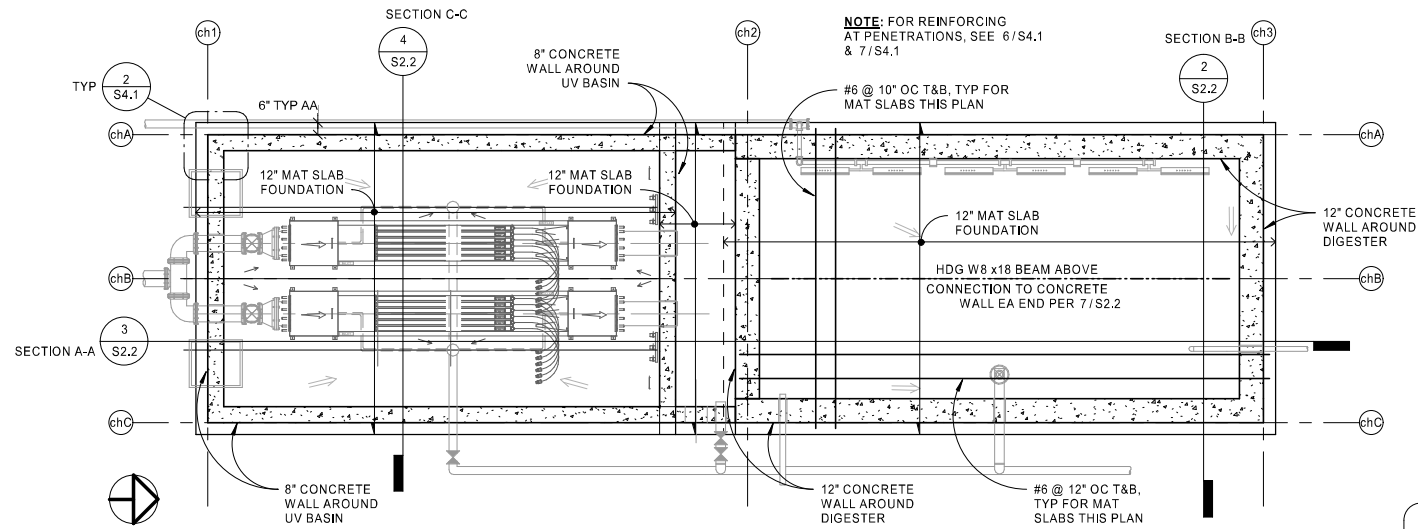
4 UV DISINFECTION CHAMBER & DIGESTER SECTION C-C
1/4" = 1'-0"



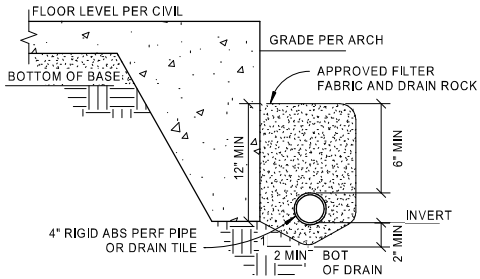
3 UV DISINFECTION CHAMBER & DIGESTER SECTION A-A
1/4" = 1'-0"



2 UV DISINFECTION CHAMBER & DIGESTER SECTION B-B
1/4" = 1'-0"



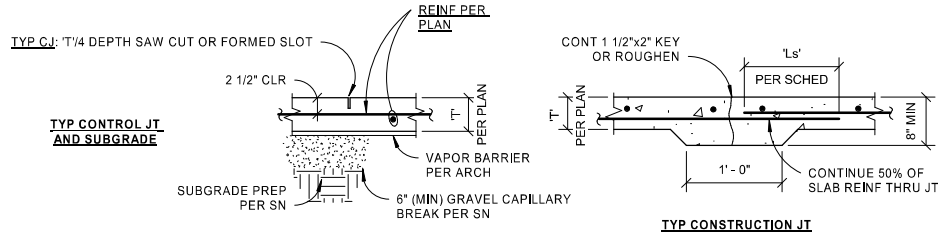
1 UV DISINFECTION CHAMBER & DIGESTER FOUNDATION PLAN
1/4" = 1'-0"



PERIMETER FOUNDATION DRAINS
- INVERT MUST BE BELOW FLOOR LEVEL.
- BOTTOM OF DRAIN MUST BE BELOW BOT OF BASE.

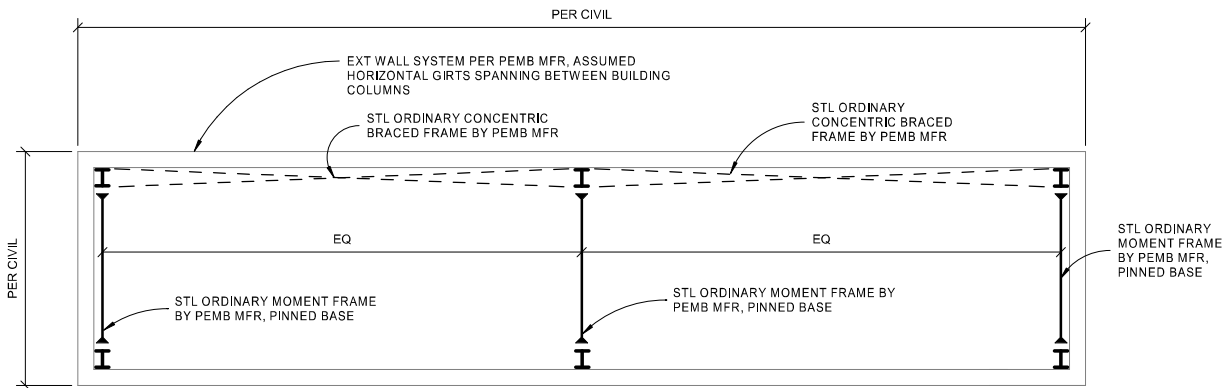
- NOTE**
1. DRAIN PIPES MUST DISCHARGE BY GRAVITY OR MECHANICAL MEANS TO AN APPROVED DRAINAGE SYSTEM THAT COMPLIES WITH THE INTERNATIONAL PLUMBING CODE.
 2. MINIMUM SLOPE FOR DRAIN PIPE RUNS IS 0.25%.
 3. DO NOT CONNECT FOUNDATION DRAIN SYSTEMS TO RAIN LEADERS OR DOWN SPOUTS.
 4. PROVIDE APPROVED CLEANOUTS FOR ALL DRAIN PIPE AT ENDS, CORNERS, AND NO MORE THAN 120'-0" ON CENTER.
 5. SEE CIVIL DRAWINGS FOR GRADES, DISCHARGE POINTS, AND OTHER REQMTS. ALSO SEE GEOTECH REPORT FOR ADDITIONAL REQMTS.
 6. SUBMIT A FOUNDATION DRAINAGE PLAN TO CIVIL FOR REVIEW.
 7. DRAIN ROCK/BASE MATERIAL SHALL CONSIST OF GRAVEL OR STONE WITH LESS THAN 10% OF MATERIAL PASSING THROUGH A NUMBER 4 SIEVE.

7 TYP FOUNDATION DRAINS AT BLOWER BUILDING SLAB EDGE
1" = 1'-0"



- NOTE**
1. PLACE CONTROL JOINTS AND CONST JOINTS AT NO MORE THAN 12'-0" O/C, EW. ALIGN JOINTS WITH COLS AND SLAB CORNERS WHERE POSSIBLE.
 2. CUT CJs AFTER CONC HAS SUFFICIENTLY HARDENED TO PREVENT SPALLING OR TEARING BUT NO MORE THAN 12 HOURS AFTER POUR.
 3. SUBMIT A SLAB PLAN SHOWING CONTROL & CONSTRUCTION JOINT LAYOUT AND POUR SEQUENCE FOR APPROVAL.
 4. THE SUBGRADE SHOWN ABOVE SHALL BE THE MINIMUM REQUIRED AT ANY LOCATION, AND SHALL BE INCREASED OR MODIFIED AS RECOMMENDED IN THE GEOTECH REPORT.
 5. SEE SPEC AND ARCH/MECH/CIVIL DRAWINGS FOR SLAB EXTENTS, SLOPES, STEPS, DRAINS, ELEVATIONS, CHAMFERS, FINISHES AND OTHER RELATED INFORMATION. MAINTAIN MIN THICKNESS PER PLAN THROUGHOUT.
 6. WHERE SINGLE PENETRATION (OR COMBINED GROUPING OF MULTIPLE PENETRATIONS, SUCH AS AT GROUPS OF PIPES OR CONDUIT) LARGER THAN 0'-10" ANY DIRECTION, PROVIDE ADD'L #3 EA SIDE EXTENDING 18" MIN BEYOND THE PENETRATION EXTENTS.
 7. AT INT CORNERS, PROVIDE (2) ADD'L #3x38" @ 2" O/C, PLACED AT 45-DEG AND 2" CLR TO THE INT CORNER OF SOG.
 8. SIM AT EXT SOG'S/SIDEWALKS AROUND BLDG, EXCEPT OMIT VAPOR BARRIER, REDUCE C.J SPACING TO 10'-0" O/C MAX, PROVIDE AIR ENTRAINMENT PER SN.

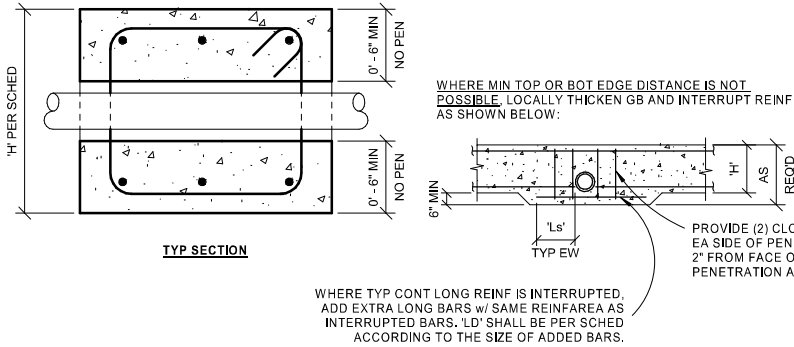
5 TYP SLAB-ON-GRADE DETAILS
1" = 1'-0"



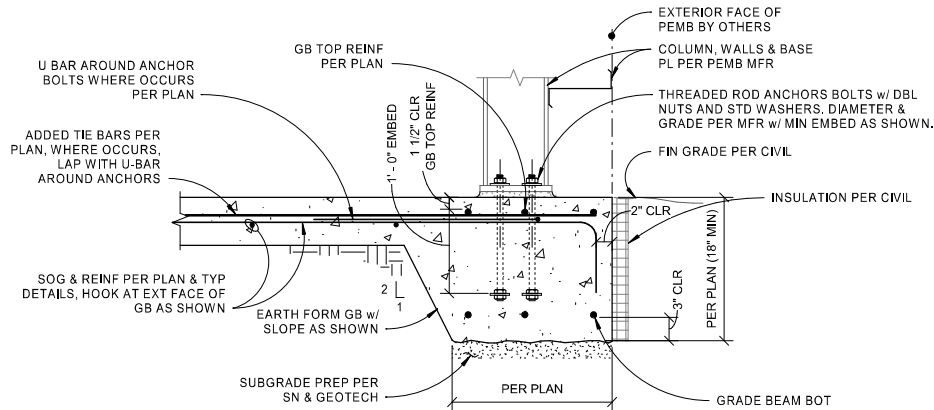
NOTE: THE LAYOUT SHOWN HERE WAS USED AS THE BASIS OF DESIGN FOR THE PEMB FOUNDATIONS, ORIENT LATERAL SYSTEM AS SHOWN.

2 BLOWER BLDG - PEMB LATERAL SYSTEM LAYOUT
1/4" = 1'-0"

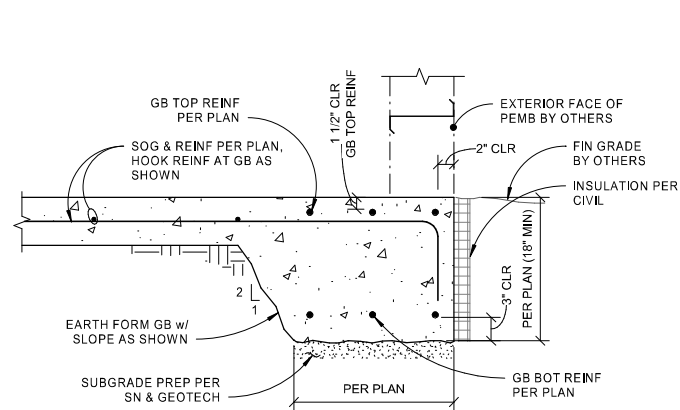
- NOTE**
1. PROVIDE SLEEVES w/ ID 2.5" LARGER THAN PIPE OD. CENTER PIPE AND WRAP w/ COMPRESSIBLE MATERIAL.
 2. ALL PROPOSED GB PENETRATIONS SHALL BE COORDINATED BY THE GC. PROVIDE 6" MIN CONC BETWEEN ADJACENT PENETRATIONS. CLEARLY INDICATE ALL PENETRATIONS ON THE SUBMITTED REINF SHOP DRAWINGS; THE ENGINEER SHALL REVIEW THE SHOP DRAWING FOR ADEQUACY AND ADD REINF AS REQUIRED. FOR BIDDING PURPOSES ONLY, ASSUME THAT ALL CONT LONG REINF IS LAP SPLICED WITH EQUIVALENT BAR AREA.
 3. SECTIONS SHOWS ACCEPTABLE PENETRATION LOCATIONS THROUGH AND BENEATH GB'S.
 4. DO-NOT INTERRUPT REINF IN ANY CASE. MAINTAIN COVER PER STRUCTURAL NOTES AT ALL REINFORCEMENT.



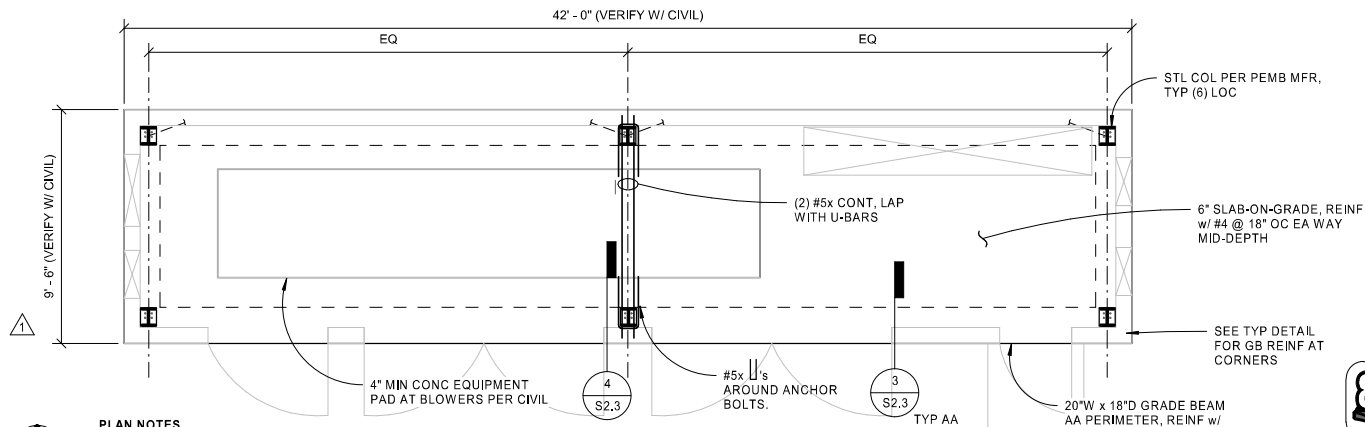
6 TYP GB PENETRATIONS
1 1/2" = 1'-0"



4 TYP GRADE BEAM SECTION AT COLUMNS
1" = 1'-0"



3 TYP GRADE BEAM SECTION
1" = 1'-0"



- PLAN NOTES**
1. SEE CIVIL FOR INFORMATION NOT SHOWN.
 2. PRE ENGINEERED METAL BUILDING (PEMB) IS DESIGNED BY OTHERS. SEE DEFERRED SUBMITTAL SECTION OF STRUCTURAL NOTES AND SPECIFICATION. FOUNDATION DESIGN IS SUBJECT TO CHANGE AFTER PEMB DESIGN IS FINALIZED.

1 BLOWER BLDG FOUNDATION PLAN
1/4" = 1'-0"

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22



BOT OF GRADE BEAM
PIPE DIA -OR- 3" MIN
FILL VOID w/ CONC UNLESS 'W' IS LESS THAN 1'-6" (IN WHICH CASE, OK TO FILL w/ GRAVEL OR SAND) DEPTH EQUAL GRADE BEAM WIDTH.



Verify Scale:
Enclosing box measures 1/2 inch tall x 2 inch wide when drawings are printed at full scale.

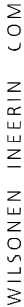
DESIGNED BY	Blomson	DRAWN BY	CKinney	CHECKED BY	Blomson
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EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1
BLOWER BUILDING FOUNDATION PLAN & DETAILS

DATE	10-15-2021	SCALE	As indicated (on 22"x34")	JOB NUMBER	Wilson: 2020-070
SHEET	S2.3	PAGE	59	OF	96

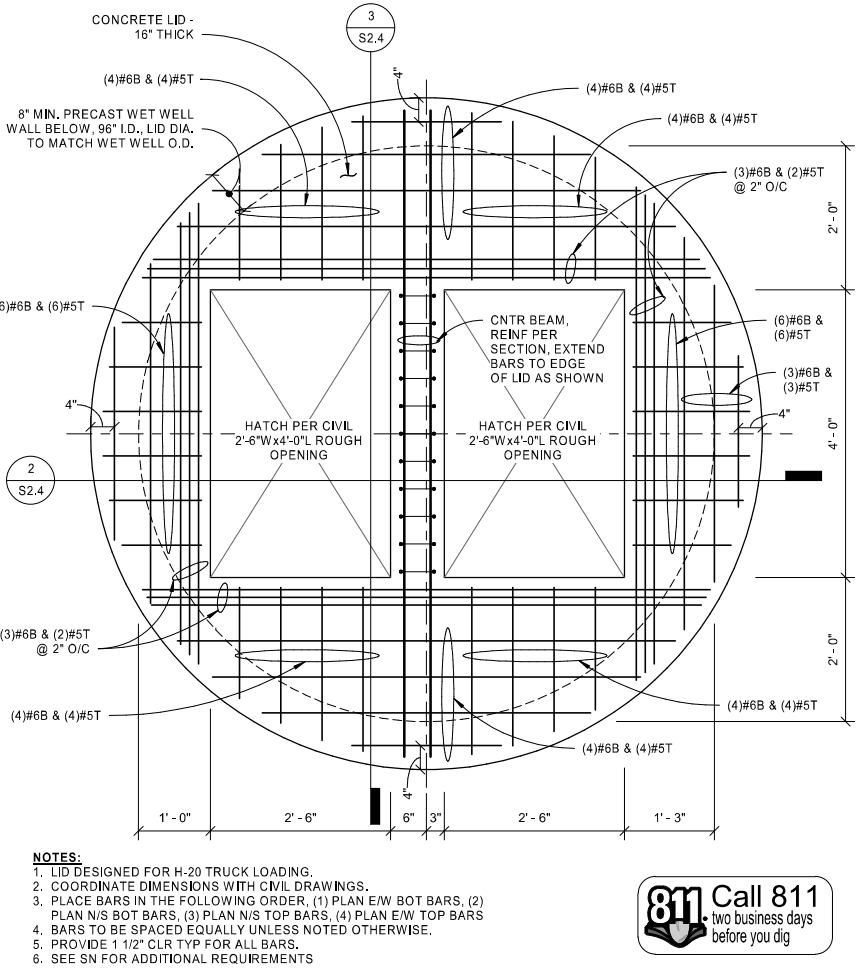
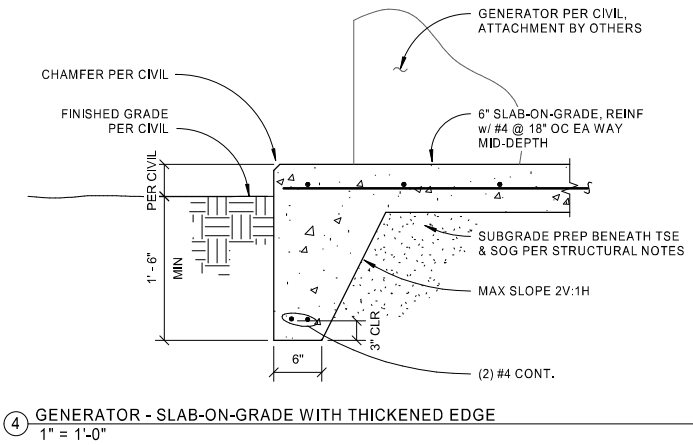


BID SET



WASTEWATER TREATMENT PLANT UPGRADE – PHASE 1
PUMP STATION LID, GENERATOR PAD, & FLOW SPLITTER PAD

SHEET S2.4	DATE	10-15-2021
	SCALE	As indicated (on 22"x34")
	JOB NUMBER Wilson: 2020-070 KW: 21131	
PAGE	60	OF 96



BID SET

NO.	REVISIONS	BY	DATE
1	UV Disinfection	TH	5/4/22
2	Bid Set	TH	9/6/22



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360.714.8246
www.king-works.com

Verify Scale:
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when drawings are printed at full scale.

DESIGNED BY	THEILDERSDORF	DRAWN BY	CKINNEY	CHECKED BY	THEILDERSDORF
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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY WASHINGTON

WASTEWATER TREATMENT PLANT UPGRADE - PHASE 1

TYP STRUCTURAL DETAILS

DATE
10-15-2021

SCALE
As indicated
(on 22"x34")

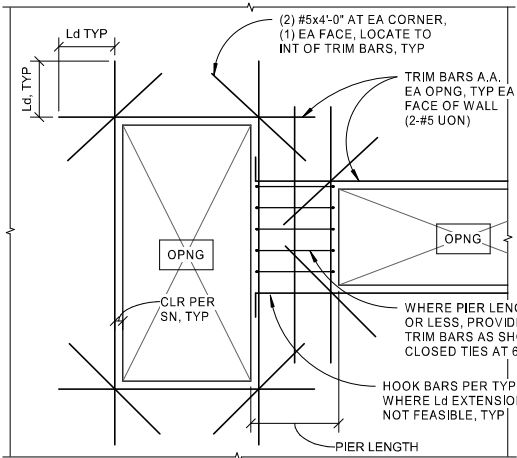
JOB NUMBER
Wilson: 2020-070
KW: 211.31

SHEET
S4.1

PAGE
61 OF 96

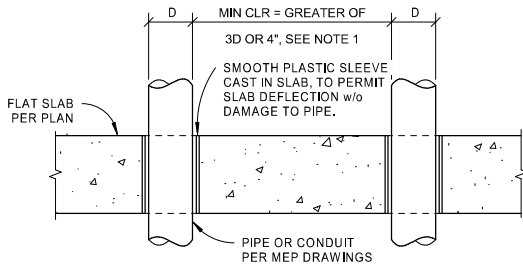
BID SET

811 Call 811
two business days
before you dig



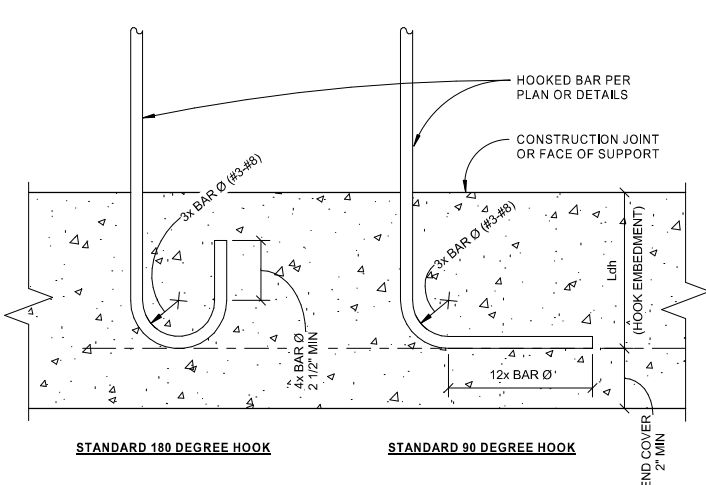
- NOTES**
1. DETAIL APPLIES TO ALL OPENINGS WITH EITHER HEIGHT OR WIDTH GREATER THAN 2'-0".
 2. Ld IS DEVELOPMENT LENGTH PER TYPICAL DETAILS.
 3. REINF SHOWN ABOVE IS TYP MINIMUM. OMIT ADDED REINFORCEMENT ABOVE WHEN SPECIAL REINF INDICATED ON PLANS OR DETAILS EXCEEDS THIS REINF.
 4. CONTRACTOR SHALL VERIFY ALL OPENINGS NOT SHOWN IN THE STRUCTURAL DRAWINGS WITH THE STRUCTURAL ENGINEER PRIOR TO SUBMITTAL DEVELOPMENT AND INSTALLATION.

7 TYP LARGE PENETRATIONS IN CONCRETE
1/2" = 1'-0"



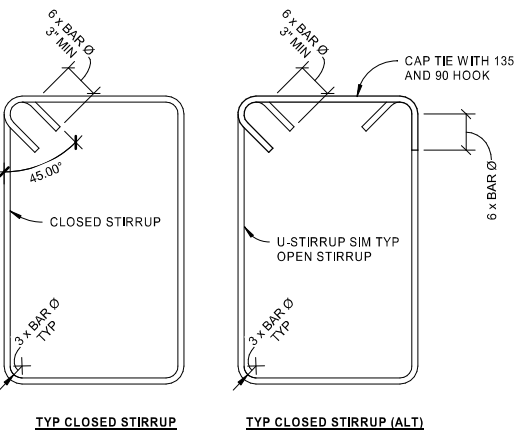
- NOTES**
1. IF MINIMUM CLR SPACING CANNOT BE ACHIEVED, REINF AROUND COMBINED EXTENTS PER TYP LARGE OPENING DETAIL.
 2. DETAIL APPLIES TO D= 9" OR LESS.

6 TYP SMALL PENETRATIONS IN CONCRETE
3/4" = 1'-0"



- NOTES**
1. FOR CONC STRENGTHS NOT INDICATED, USE NEXT LOWEST TABULATED f_c VALUE.
 2. PROVIDE 2 1/2" SIDE COVER AND 2" END COVER FOR HOOKS.
 3. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.

STANDARD HOOK EMBEDMENT (Ldh)	
BAR SIZE	CONC DESIGN STRENGTH (f _c)
#3	4,500 PSI
#4	6"
#5	7"
#6	9"
#7	10"
#8	12"
#8	14"

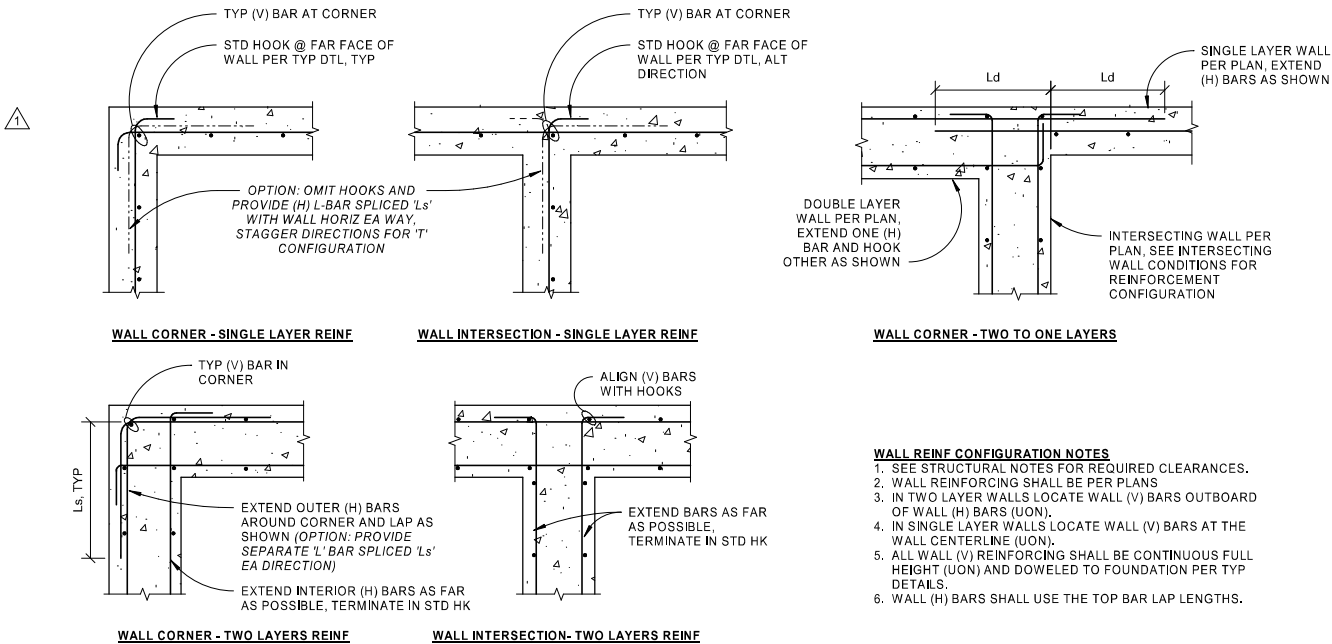


4 TYP CONCRETE BEAM STIRRUPS
1 1/2" = 1'-0"

4,500 PSI CONCRETE				
BAR SIZE	DEVELOPMENT LENGTHS (Ld)		LAP SPICE LENGTHS (Ls)	
	TOP	OTHER	TOP	OTHER
#3	16 in	12 in	18 in	16 in
#4	20 in	15 in	24 in	20 in
#5	28 in	22 in	36 in	28 in
#6	38 in	30 in	50 in	38 in
#7	60 in	46 in	78 in	60 in
#8	74 in	57 in	96 in	74 in
#9	90 in	69 in	116 in	90 in
#10	108 in	83 in	140 in	108 in
#11	126 in	97 in	164 in	126 in

- NOTES**
1. TABULATED VALUES ARE APPLICABLE FOR GRADE 60 UNCOATED REINF BARS.
 2. BAR C/C SPACING MUST BE GREATER THAN TWICE THE BAR DIAMETER AND COVER GREATER THAN ONE BAR DIAMETER.
 3. "TOP" BARS ARE DEFINED AS HORIZONTAL BARS WITH MORE THAN 12 INCHES OF FRESH CONCRETE CAST BELOW THE BARS, INCLUDING WALL HORIZONTALS.
 4. AT LOCATIONS WHERE Ld IS CALLED OUT BUT CANNOT BE ACHIEVED, EXTEND BAR AS FAR AS POSSIBLE AND PROVIDE STANDARD HOOK PER TYP DETAIL.

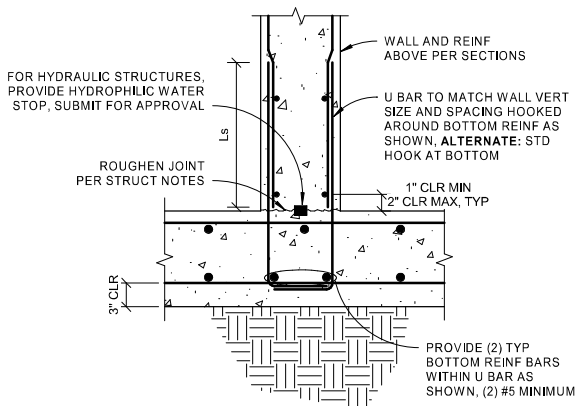
3 TYP CONCRETE REINFORCING DEVELOPMENT / SPICE LENGTHS
3/4" = 1'-0"



- WALL REINF CONFIGURATION NOTES**
1. SEE STRUCTURAL NOTES FOR REQUIRED CLEARANCES.
 2. WALL REINFORCING SHALL BE PER PLANS
 3. IN TWO LAYER WALLS LOCATE WALL (V) BARS OUTBOARD OF WALL (H) BARS (UON).
 4. IN SINGLE LAYER WALLS LOCATE WALL (V) BARS AT THE WALL CENTERLINE (UON).
 5. ALL WALL (V) REINFORCING SHALL BE CONTINUOUS FULL HEIGHT (UON) AND DOWELED TO FOUNDATION PER TYP DETAILS.
 6. WALL (H) BARS SHALL USE THE TOP BAR LAP LENGTHS.

2 TYP CONCRETE WALL REINFORCING AT CORNERS & INTERSECTIONS
3/4" = 1'-0"

NOTE SEE PLANS AND STRUCTURAL NOTES FOR REQUIRED CLEARANCES OF REINF IN SLAB AND WALL.

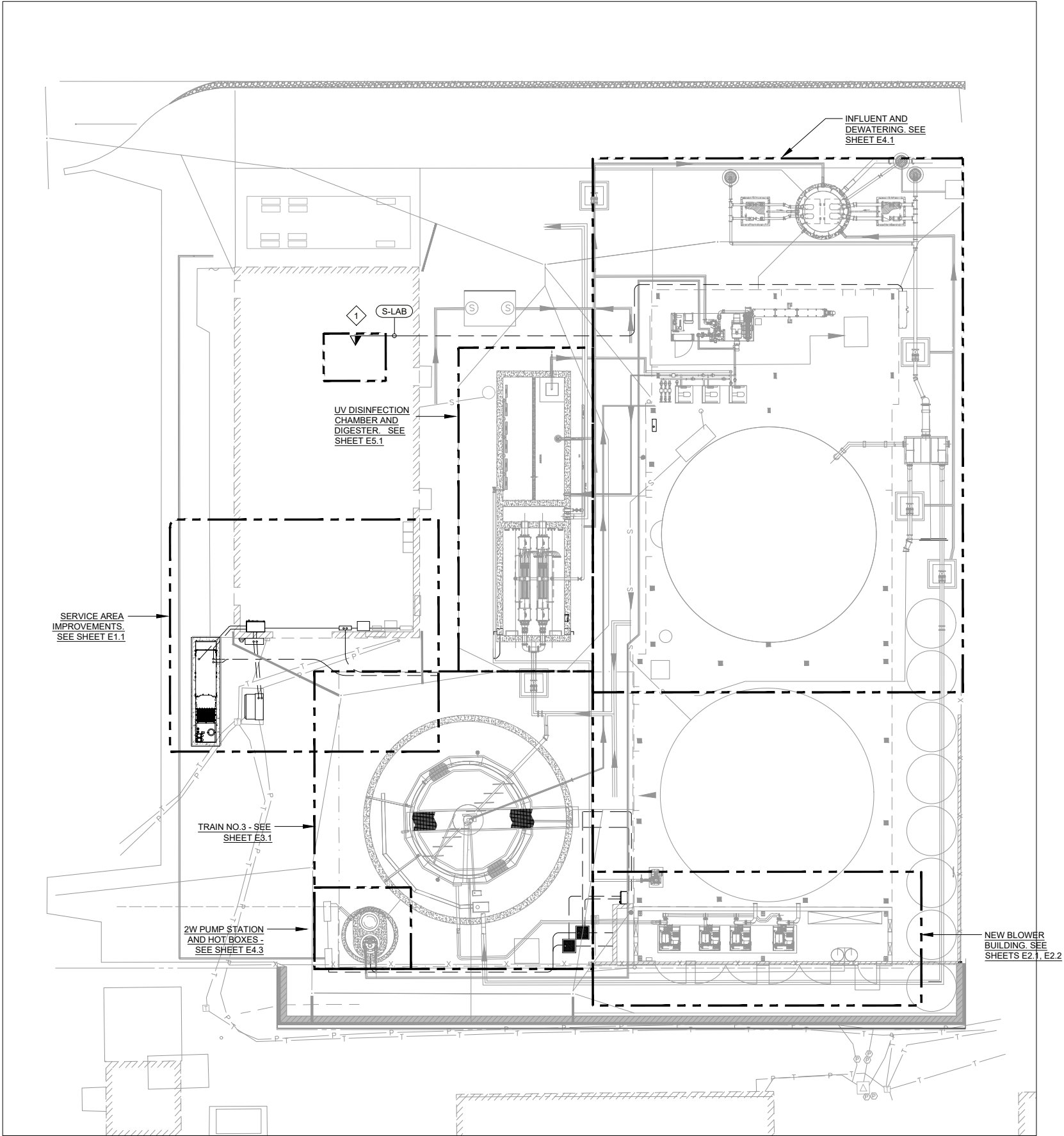


1 TYP CONCRETE WALL REINFORCING TO MAT SLAB
1" = 1'-0"

RACEWAYS AND CONDUCTORS		CALLOUTS AND DESIGNATIONS		CONTROLS AND INSTRUMENTATION		STANDARD ABBREVIATIONS		PROJECT GENERAL NOTES:	

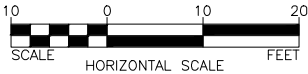


BID SET



OVERALL ELECTRICAL SITE PLAN

SCALE: 1" = 10' AT FULL SCALE



GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED IN DIVISION 26. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.

KEY NOTES:

1. PROVIDE NEW SCADA COMPUTER SYSTEM PER SPECIFICATIONS. COORDINATE WITH OWNER FOR FINAL LOCATION IN LAB. INSTALL NEW CAT 6 NETWORK JUNCTION BOX WITH COMBO NETWORK/TELEPHONE OUTLET. EXTEND CAT6 CABLING TO NETWORK SWITCH LOCATED IN INFLUENT PUMP STATION CONTROL PANEL. WIRE EXISTING TELEPHONE CIRCUIT TO JBOX FOR SCADA ALARM DIALER.

NO.	REVISIONS	BY	DATE



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

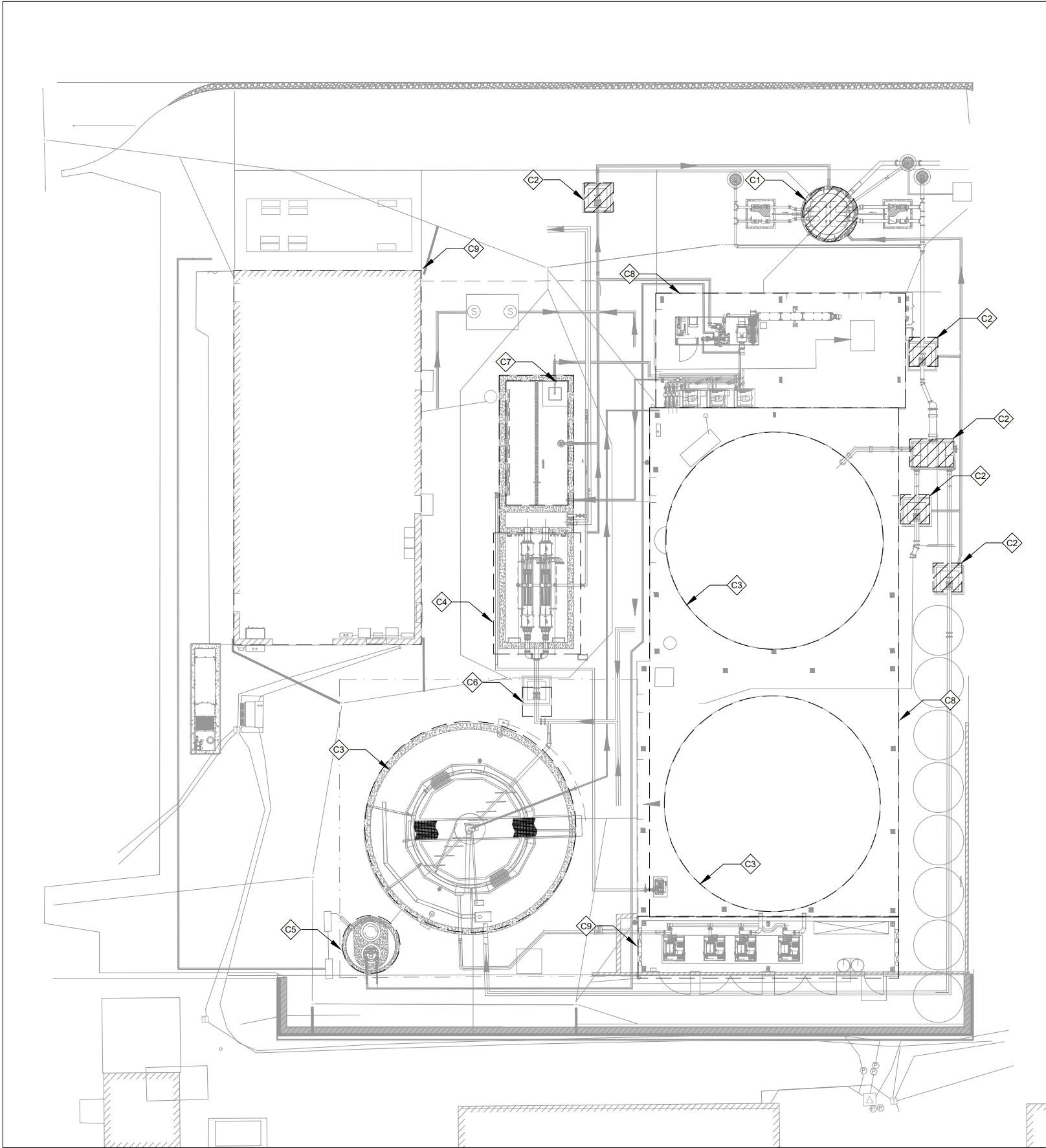
OVERALL ELECTRICAL SITE PLAN

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET	E0.3
PAGE	64 OF 96

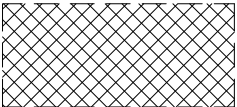


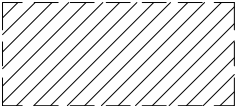
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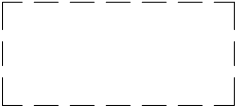


AREA CLASSIFICATION SITE PLAN
SCALE: 1" = 10' AT FULL SCALE
10 0 10 20
SCALE HORIZONTAL SCALE FEET

AREA CLASSIFICATION LEGEND:

CLASS I, DIVISION 1

CLASS I, DIVISION 2

UNCLASSIFIED

GENERAL NOTES:

1. AREA CLASSIFICATIONS BASED ON NFPA 820, STANDARD FOR FIRE PROTECTION IN WASTEWATER TREATMENT AND COLLECTION FACILITIES, 2020 EDITION.
2. PROCESS BASED ON TREATMENT OF WASTEWATER FROM RESIDENTIAL STEP STATIONS PER NFPA 820-2020, TABLE 4.2.2, ROW 9.
3. ELECTRICAL INSTALLATION IN CLASSIFIED AREAS SHALL COMPLY WITH THE REQUIREMENTS OF NFPA 70, NATIONAL ELECTRICAL CODE, AND OTHER STATE AND AGENCY REQUIREMENTS AS REQUIRED BY THE AUTHORITY HAVING JURISDICTION.

CLASSIFIED AREA NOTES:

- C1 INFLUENT PUMP STATION: PRIMARILY RESIDENTIAL SEWER - CLASS I, DIV 2 PER NFPA 820-2020 TABLE 4.2.2, ROW 10a. EXTENT OF CLASSIFIED LOCATION: WITHIN ENCLOSED SPACE.
- C2 BELOW GRADE METERING VAULT WITH CLOSED PIPING SYSTEM: CLASS I, DIV 2 PER NFPA 820-2020 TABLE 4.2.2, ROW 10a AND 34a. EXTENT OF CLASSIFIED LOCATION: WITHIN ENCLOSED SPACE.
- C3 AEROBIC TREATMENT OF WASTEWATER PRECEDED BY PRIMARY TREATMENT: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.11.
- C4 UV DISINFECTION UNITS: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.26.
- C5 2W PUMP STATION: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.21.
- C6 EFFLUENT FLOW METER VAULT: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.27.
- C7 AERATED DIGESTER OPEN TO ATMOSPHERE: AEROBIC TREATMENT OF WASTEWATER PRECEDED BY PRIMARY TREATMENT WITH SKIMMING: UNCLASSIFIED PER NFPA 820-2020 TABLE 5.2.2.11.
- C8 DEWATERING BUILDING WITH DEWATERING PRESS: UNCLASSIFIED PER NFPA 820-2020 TABLE 6.2.2, ROW 12a WITH TYPE 'C' VENTILATION, CONTINUOUSLY VENTILATED AT 6 AIR CHANGES PER HOUR AND IN ACCORDANCE WITH NFPA 820-2020 CHAPTER 9.
- C9 OFFICE BUILDING AND BLOWER BUILDING NOT CLASSIFIED.

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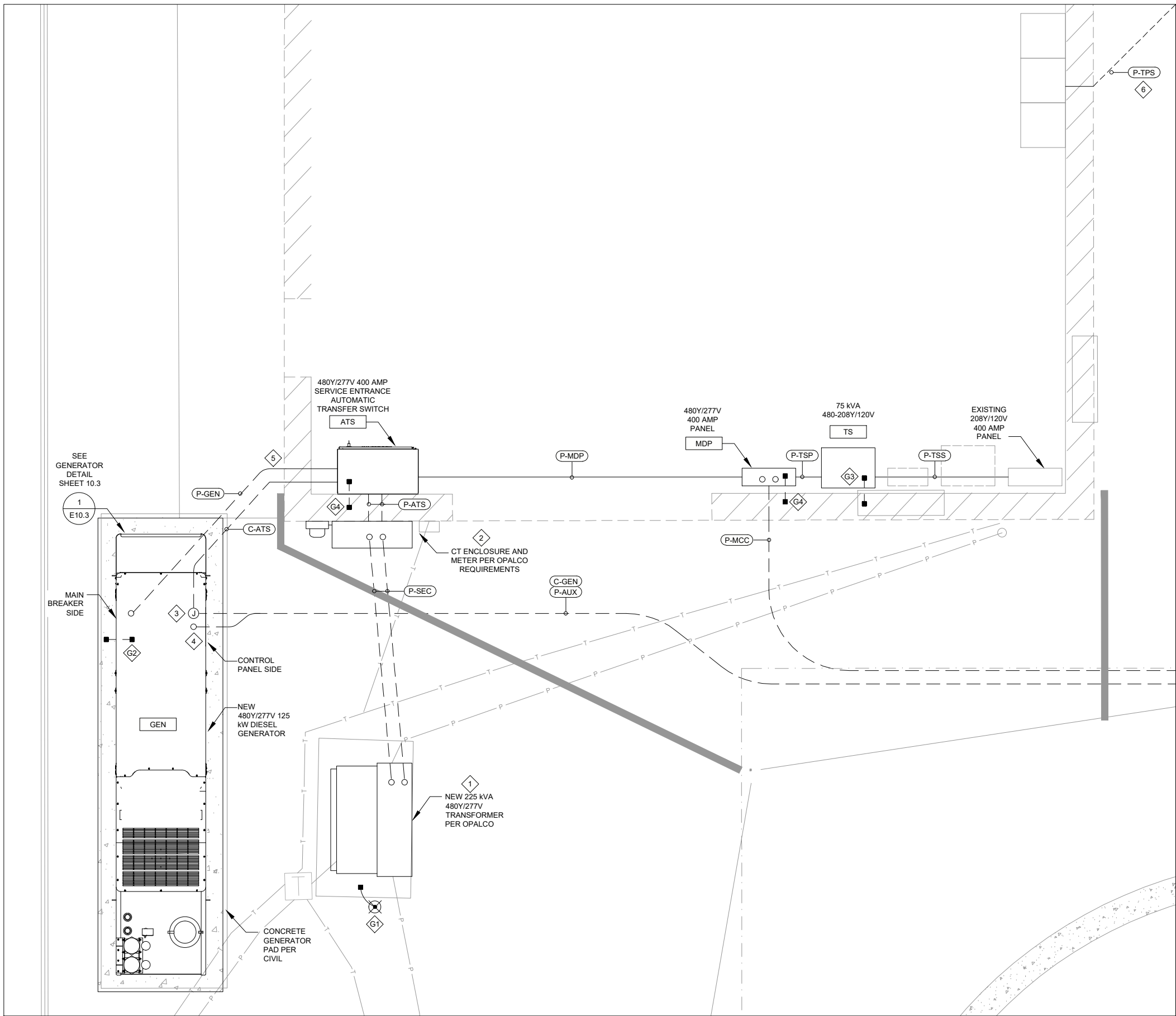
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SHEET	E0.4	DATE	09-12-2022		
		SCALE	AS SHOWN		
		JOB NUMBER	2020-070		
		PAGE		65 OF 96	
		EASTSOUND SEWER AND WATER DISTRICT		SAN JUAN COUNTY WASHINGTON	
		WASTEWATER TREATMENT PLANT UPGRADE - PHASE I		AREA CLASSIFICATION SITE PLAN	



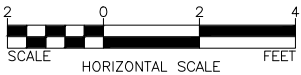
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ELECTRICAL SERVICE AREA PLAN

SCALE: 1" = 2' AT FULL SCALE



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GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. CONDUCTOR AND CONDUIT SIZING SHALL BE AS REQUIRED BY NEC.
3. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.

KEY NOTES:

1. COORDINATE WITH OPALCO FOR REMOVAL OF EXISTING 208Y/120V UTILITY TRANSFORMER AND INSTALLATION OF NEW 480Y/277V TRANSFORMER. CONTRACTOR SHALL MODIFY EXISTING PAD AND PREP FOR INSTALLATION PER OPALCO REQUIREMENTS. SECONDARY CONDUIT AND WIRE BY CONTRACTOR. OPALCO SHALL FURNISH TRANSFORMER AND INSTALL ON MODIFIED PAD. PRIMARY WIRE AND CONNECTIONS BY OPALCO.
2. CONTRACTOR TO PROVIDE NEW CT ENCLOSURE, LANDING PADS AND METER BASE PER OPALCO REQUIREMENTS. METER AND CTS BY UTILITY.
3. PROVIDE J-BOX TO COMBINE ATS AND GENERATOR STATUS SIGNALS AND WIRE TO PLC CONTROL PANEL IN BLOWER BUILDING.
4. PROVIDE 120V 20 AMP CIRCUIT FROM LOAD CENTER IN MCC TO GENERATOR AUX POWER CONNECTIONS TO POWER BATTERY CHARGER AND COOLANT HEATER.
5. REPAIR CONCRETE AND BUILDING WALLS AND STRUCTURE AS NEEDED TO ROUTE NEW CONDUITS INTO EXISTING BUILDING.
6. REPLACE EXISTING 200 AMP 208Y/120V POWER FEEDER FROM EXISTING SWITCHGEAR TO EXISTING TPS PANEL IN TREATMENT BUILDING. SEE SITE PLAN AND SCHEDULES.

GROUNDING NOTES:

- G1. PROVIDE GROUNDING FOR UTILITY TRANSFORMER PER NEC AND OPALCO REQUIREMENTS.
- G2. GROUND CONCRETE ENCASED GENERATOR PAD REBAR TO GENERATOR AND CONNECT TO GROUNDING ELECTRODE SYSTEM.
- G3. PROVIDE GROUNDING FOR TRANSFORMER AS SEPARATELY DERIVED SYSTEM.
- G4. INSTALL NEW GROUNDING ELECTRODE SYSTEMS AND TIE INTO EXISTING BUILDING GROUNDING ELECTRODE SYSTEM.



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

ELECTRIC SERVICE AREA PLAN

DATE
09-12-2022

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PAGE
66 OF 96

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SHEET	E2.2				DATE	09-12-2022				SCALE	AS SHOWN				JOB NUMBER	2020-070																										
	PAGE					68					OF					96																										
	PROJECT					EASTSOUND SEWER AND WATER DISTRICT					SAN JUAN COUNTY					WASHINGTON																										
																				WASTEWATER TREATMENT PLANT UPGRADE - PHASE I																						
																				BLOWER BUILDING LIGHTING AND HVAC PLAN																						



BID SET

GENERAL NOTES:

- ROUTE ALL BRANCH CIRCUITS TO LIGHTING PANEL(S) CIRCUIT SHOWN ON THE PLANS. ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODE.
- BRANCH CIRCUITS NOT SHOWN ON CONDUIT SCHEDULE SHALL BE SIZED AS REQUIRED BY NEC.
- ALL ELECTRICAL CONDUIT AND BOXES SHALL BE UNDERGROUND OR CONCEALED IN WALLS. COORDINATE INSTALLATION WITH CMU INSTALLER WHERE NECESSARY.

KEY NOTES:

- PROVIDE PHOTOCELL ON EXTERIOR FIXTURE. FIXTURE SHALL BE ON DEDICATED SWITCH AND DARY SKY COMPLAINT, SEE SCHEDULE.
- WIRE EMERGENCY LIGHTS TO INTERIOR LIGHTING CIRCUIT.
- PROVIDE OCCUPANCY SENSOR FOR INTERIOR LIGHTING. COMPLY WITH WASHINGTON STATE ENERGY CODE REQUIREMENTS.
- CONTRACTOR SHALL PROVIDE COMPLETE VENTILATION CONTROL PANEL, INCLUDING RELAY CONTROLS AND THERMOSTATS FOR EXHAUST FAN AND LOUVER CONTROL. ONE THERMOSTAT FOR EACH EXHAUST FAN. MOTORIZED LOUVER SHALL OPERATE WHEN EITHER FAN IS RUNNING.

PANEL: LP1				PANEL SCHEDULE							PROJECT: Eastsound WWTP				
120/240V, 1Ph, 3W.				100A Bus				100A M.C.B.				SURFACE MOUNTED			
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	C.B. TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO			
1	BLOWER ROOM RECEPTACLES	900	R	20	1	A	1	20	C	1,000	MAIN CONTROL PANEL	2			
3	LOUVER AND HVAC CONTROL	500	M	20	1	B	1	20	L	239	BLOWER BUILDING LIGHTING	4			
5	EXHAUST FAN EF-1 (BLOWER BUILDING)	1,127	M	20	1	A	1				SPARE	6			
7	EXHAUST FAN EF-2 (BLOWER BUILDING)	1,127	M	20	1	B	1	20	H	1,300	GENERATOR - HEATER AND BATTERY CHGR	8			
9	UV1 POWER	2,520	C	30	1	A	1	20	H	1,000	HOT BOX 1 - GFEP	10			
11	UV2 POWER	2,520	C	30	1	B	1	20	H	1,000	HOT BOX 2 - GFEP	12			
13	UV1 CONTROLLER	600	C	20	1	A	1	20	M	500	EFFLUENT SAMPLER	14			
15	UV2 CONTROLLER	600	C	20	1	B	1	20	M	500	HYPOCHLORITE FEED PUMP	16			
17	SPARE			20	1	A	1	20			SPARE	18			
19	SPACE					B					SPACE	20			
21	SPACE					A					SPACE	22			
23	SPACE					B					SPACE	24			
TOTAL CONNECTED LOAD:		PH A	7,647 VA	63.7 AMPS		DATE: September 15, 2022									
TOTAL CONNECTED LOAD:		PH B	7,786 VA	64.9 AMPS											
MAX PHASE CONNECTED LOAD:		PH B	7,786 VA			PANEL RATING:		10,000 AIC							
TOTAL CONNECTED LOAD (2 x MAX):			15.6 kVA	64.9 AMPS		TOTAL DEMAND LOAD:		17.3 kVA		72.1 AMPS					

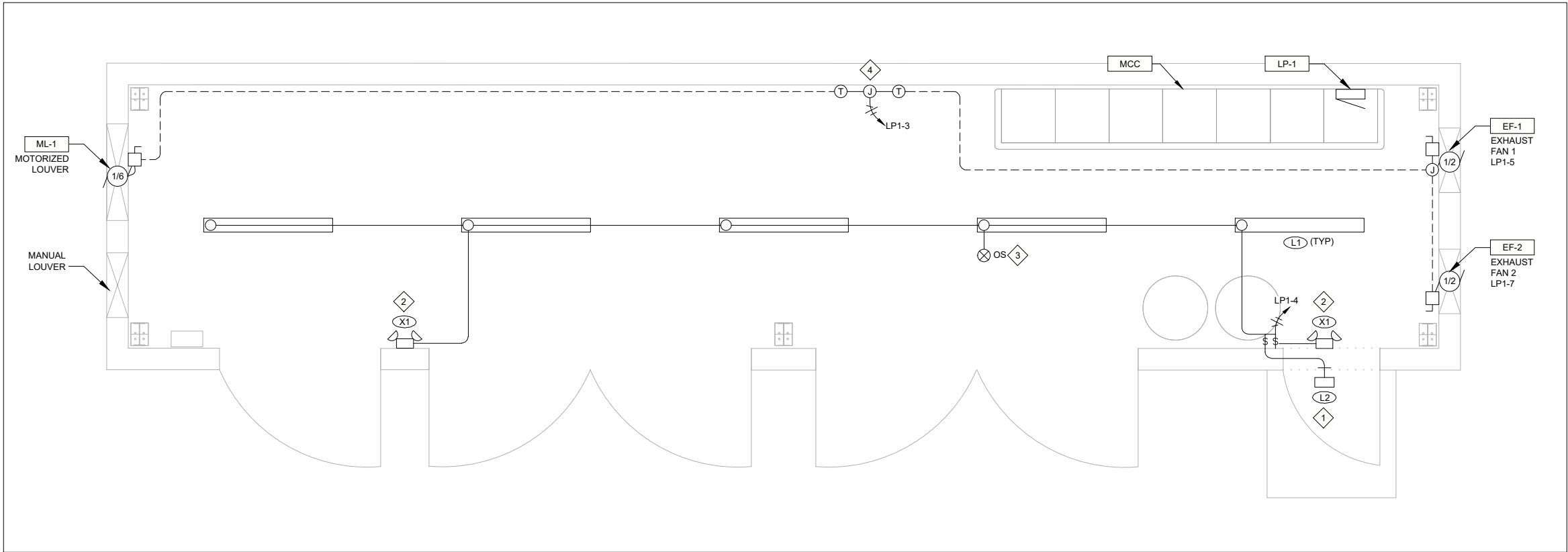
PANEL LP1 (MCC) SCHEDULE

SCALE: NONE

LIGHTING FIXTURE SCHEDULE						
ID	DESCRIPTION	LAMPS	VOLTAGE	VA	MANUFACTURER	CATALOG NUMBER
L1	NARROW LED 4' FIXTURE, VAPOR TIGHT	LED	120V	42	ACUITY	CSVT-L48-AL03-MVOLT-40K-80CRI-STSL
L2	LED WALL PACK, WET LOCATION WITH PHOTOCELL	LED	120V	17	ACUITY	ARC1LED-P2-30K-MVOLT-FAO-PE-DEBBD
X1	EMERGENCY LIGHTING W/BATTERY PACK	LED	120V	6	LITHONIA	ECC EMERGENCY COMBO UNIT

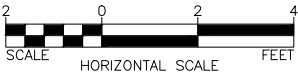
BLOWER BUILDING LIGHTING FIXTURE SCHEDULE

SCALE: NONE



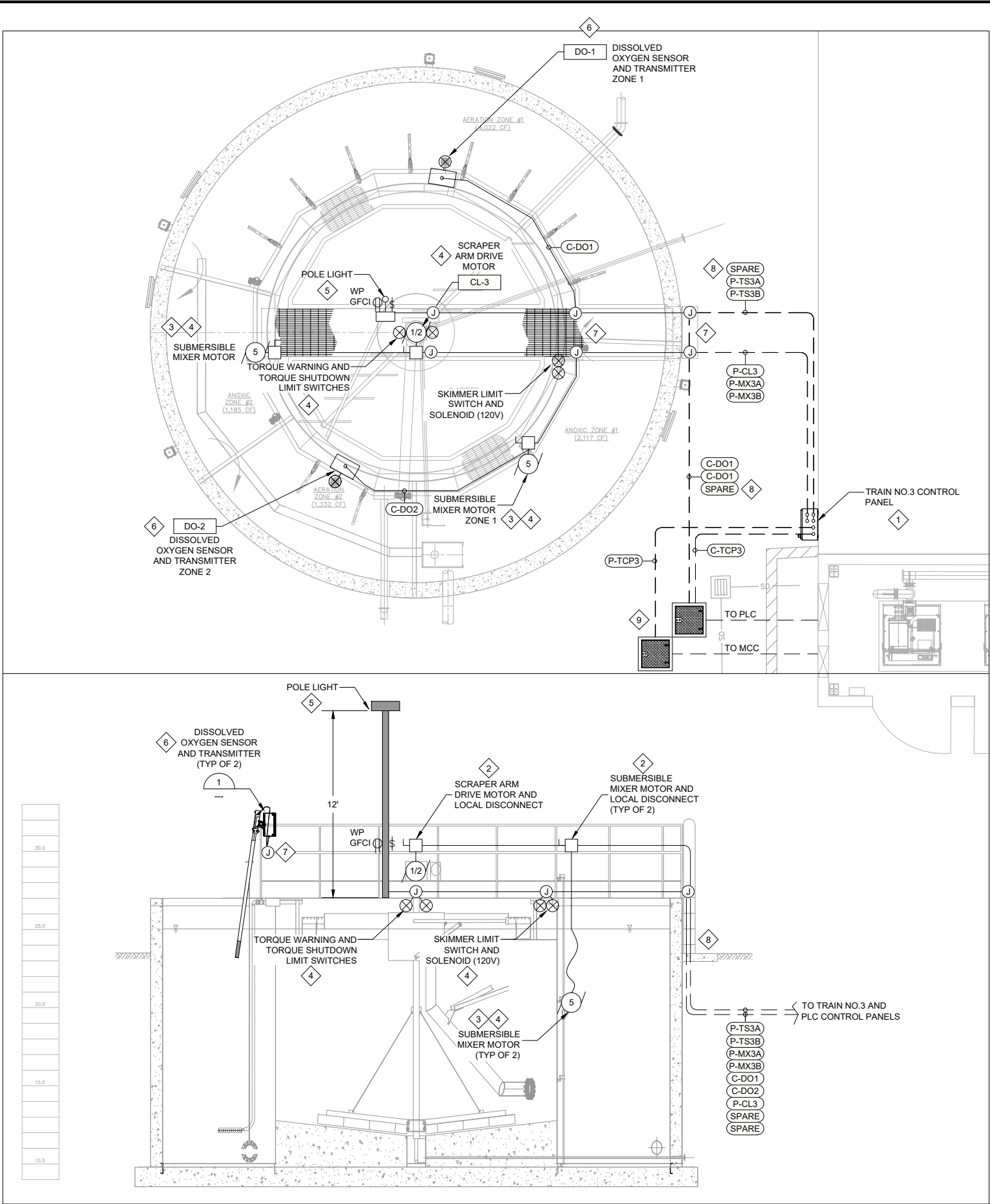
BLOWER BUILDING - LIGHTING AND HVAC PLAN

SCALE: 1" = 2' AT FULL SCALE

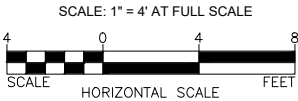


E2.2

68 OF 96



TRAIN NO.3 ELECTRICAL PLAN AND ELEVATION

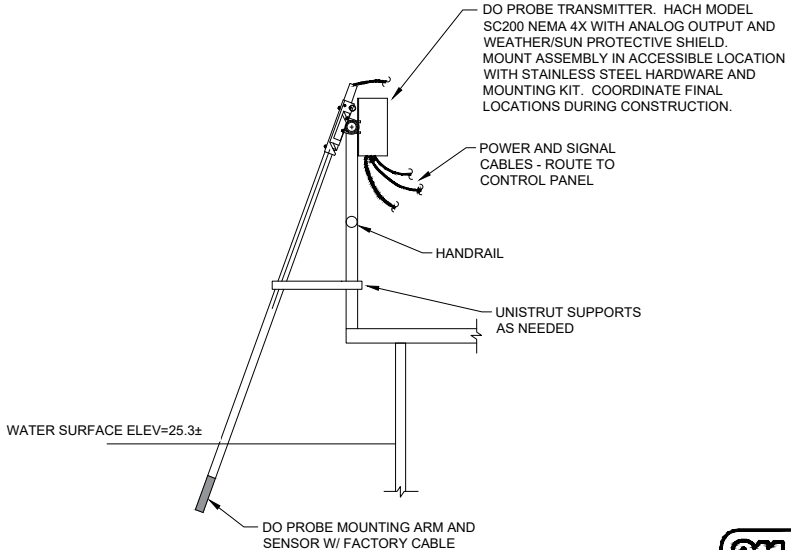


GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. CONDUCTOR AND CONDUIT SIZING SHALL BE AS REQUIRED BY NEC.
3. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
4. ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

KEY NOTES:

1. TREATMENT SYSTEM PROVIDER SHALL PROVIDE COMPLETE TREATMENT SYSTEM PER SPECIFICATIONS. CONTROL PANEL SHALL ACCEPT 480V 3 PHASE INPUT AND INCLUDE ALL WIRING AND CONTROLS FOR THE COMPLETE TREATMENT SYSTEM. FEATURES SHALL INCLUDE BUT NOT BE LIMITED TO:
 1. NEMA 4X TYPE 316 STAINLESS STEEL ENCLOSURE SIZED FOR THE APPLICATION. INCLUDE DRIP SHIELD FOR OUTDOOR PANELS.
 2. NEMA FULL VOLTAGE MOTOR STARTERS FOR BOTH CLARIFIER DRIVE MOTOR AND TWO SUBMERSIBLE MIXERS.
 3. SEAL FAIL AND OVERTEMP SENSOR FOR SUBMERSIBLE MOTOR PROTECTION.
 4. 480-120V STEP DOWN TRANSFORMER FOR CONTROL CIRCUITS.
 5. HARD WIRE RELAY LOGIC CONTROL CIRCUITS FOR SYSTEM CONTROL AND PROTECTION.
 6. ALARM LIGHT AND HORN WITH SILENCE PUSH BUTTON.
 7. INDICATOR LIGHTS, PUSH BUTTONS, HAND-OFF-AUTO SWITCHES FOR MOTOR CONTROL.
 8. WIRING TERMINALS FOR FIELD MOUNTED DEVICES INCLUDING TORQUE SWITCHES, LIMIT SWITCHES, SOLENOIDS AND ANY OTHER REQUIRED DEVICES.
 9. DRY RELAY CONTACTS FOR CUSTOMER CONNECTION INCLUDING: RUN AND FAIL FOR EACH MOTOR, SEAL FAIL AND OVERTEMP FOR MIXER, ALARM AND SHUTDOWN FOR CLARIFIER.
 10. PANEL HEATER WITH FAN TO PREVENT BUILDUP OF CONDENSATION IN PANEL.
 11. 120V CIRCUIT FOR POLE LIGHT AND RECEPTACLE.
2. ELECTRICAL CONTRACTOR SHALL PROVIDE LOCAL MOTOR DISCONNECTS FOR DRIVE MOTOR AND MIXER. NEMA 4X TYPE 316 STAINLESS STEEL.
3. ROUTE SEAL FAIL AND OVERTEMP MOTOR WIRING BACK TO CONTROL PANEL SENSOR. COMBINE WIRING IN CONTROL RACEWAY AS REQUIRED.
4. CONTRACTOR SHALL INSTALL ALL TREATMENT SYSTEM COMPONENTS PROVIDED BY MANUFACTURER AND PROVIDE WIRING AND RACEWAY BACK TO TREATMENT CONTROL PANEL. RATINGS, RACEWAY AND CONDUCTORS SHOWN ARE ESTIMATED; ADJUST DURING SUBMITTAL PERIOD BASED ON TREATMENT SYSTEM EQUIPMENT SUBMITTAL REQUIREMENTS.
5. CONTRACTOR SHALL PROVIDE POLE LIGHT ON PLATFORM. SECURE TO HANDRAIL WITH STAINLESS STEEL HARDWARE. PROVIDE 4" SQUARE ALUMINUM POLE 12' TALL WITH RECEPTACLE, PHOTOCELL AND LOCAL ON/OFF SWITCH. LITHONIA LIGHTING #DXS0-LED-P3-30K-T3M-MVOL-T-SPA-DNAXD WITH PHOTOCELL, OR APPROVED EQUAL.
6. PROVIDE (2) COMPLETE DISSOLVED OXYGEN MONITORS FOR EACH CELL. MOUNT TO TOP OF WALL WITH FACTORY HARDWARE TO ALLOW FOR TRANSMITTER ACCESS AND CLEANING. HACH MODEL LDO PROBE WITH SC200 TRANSMITTER AND ACCESSORIES. 24VDC POWERED. SEE SPECIFICATIONS.
7. PROVIDE JUNCTION BOXES AS REQUIRED TO ROUTE ALL POWER AND SIGNAL CIRCUITS TO TRAIN NO.3 CONTROL PANEL AND PLC CONTROL PANEL.
8. STUB SPARE CONDUITS ABOVE GROUND NEAR TREATMENT CELL WALLS. ONE FROM TREATMENT PANEL AND ONE TO PLC PANEL. CAP FOR FUTURE USE.
9. NOTE: POWER AND SIGNAL CONDUITS TO UV AND DIGESTER AREA AND 2W PUMP STATION ROUTE THRU THIS AREA AND HANDHOLES BUT ARE NOT SPECIFICALLY INDICATED. FIELD COORDINATE ROUTING, SEE SHEETS E5.1 AND E4.3.



DO PROBE ELEVATION DETAIL
SCALE: NONE

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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

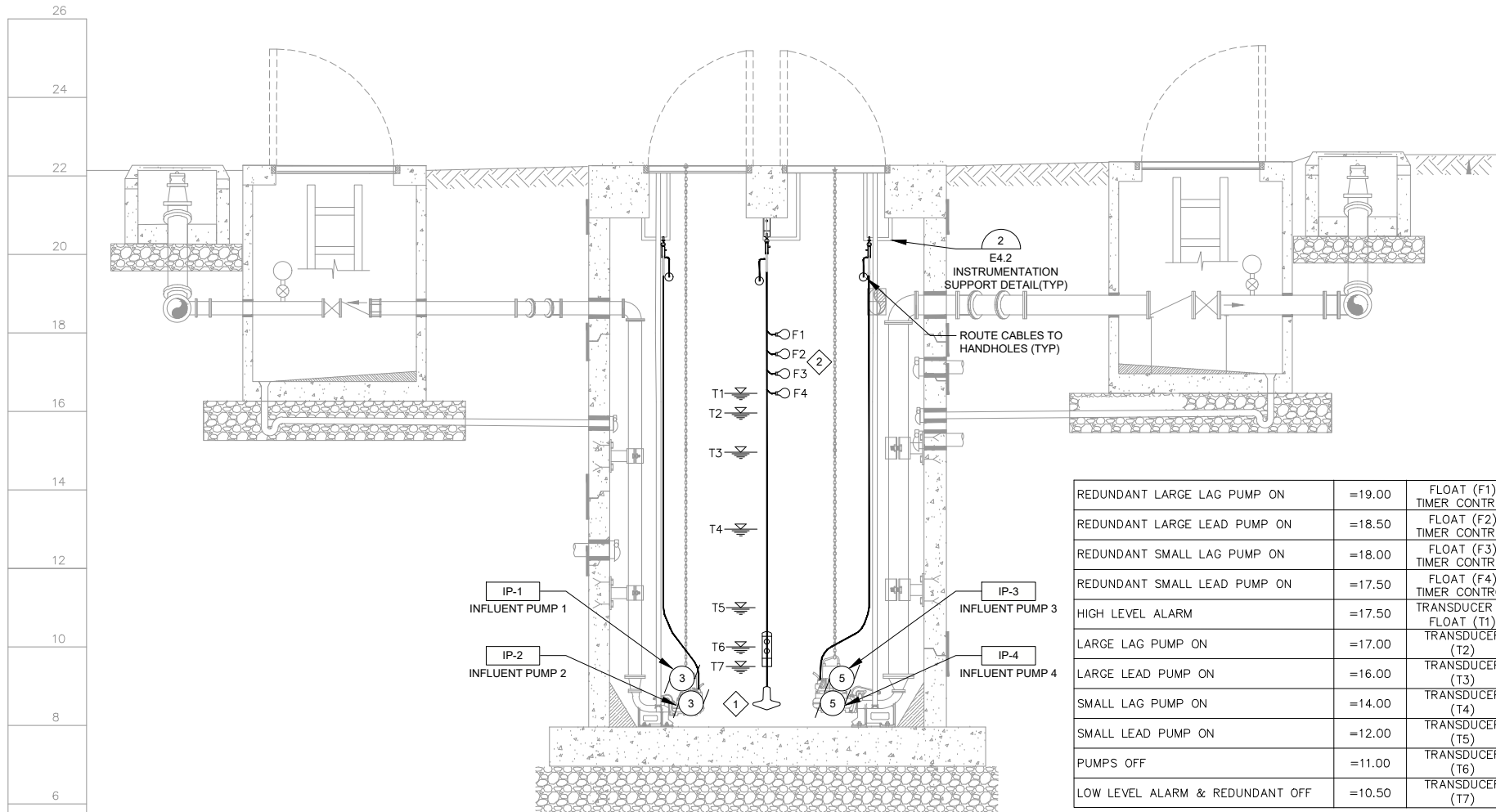
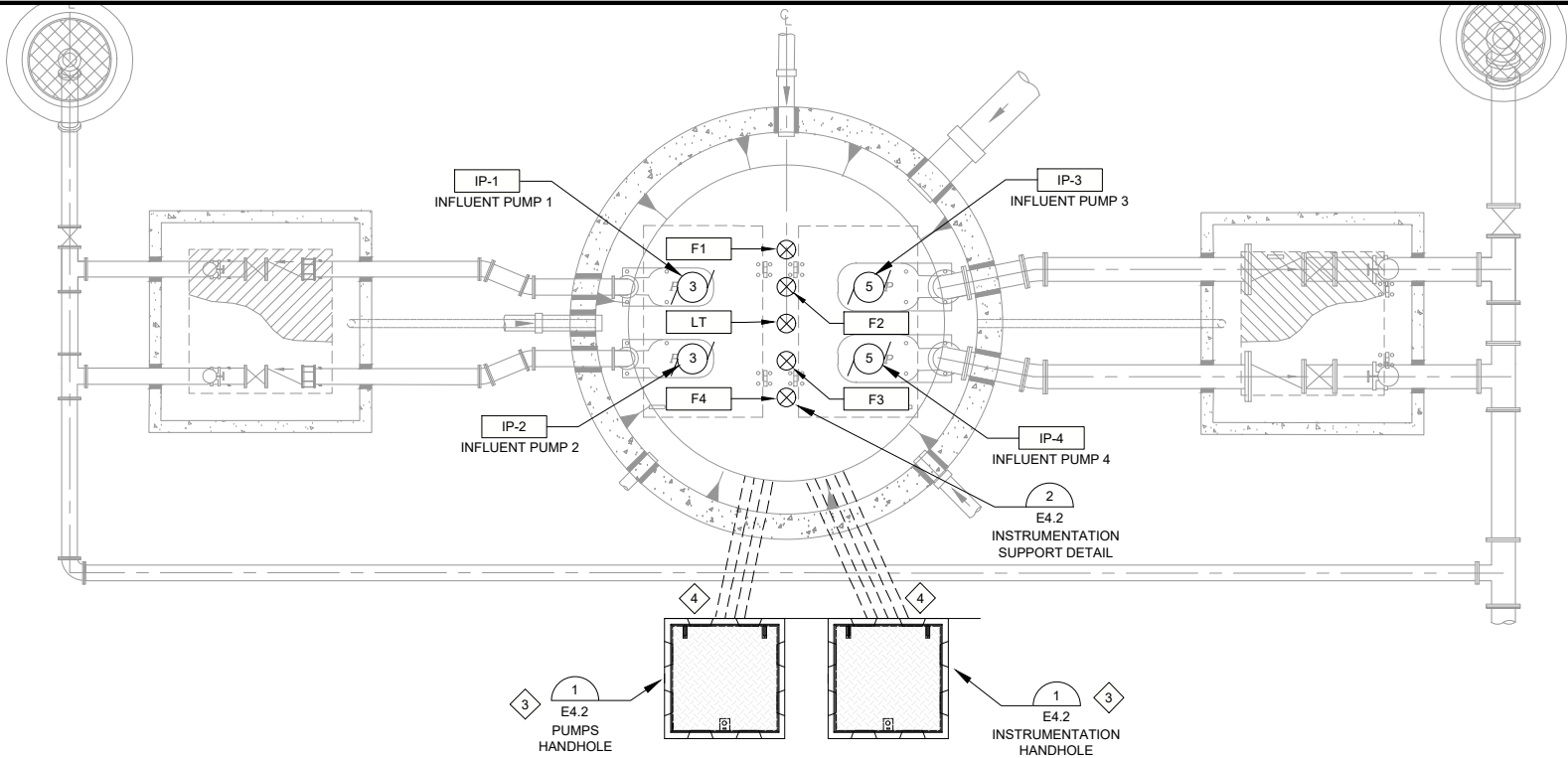
TRAIN NO.3 ELECTRICAL PLAN AND ELEVATION

SHEET	DATE	SCALE	JOB NUMBER
E3.1	09-12-2022	AS SHOWN	2020-070
PAGE	OF		
69	96		



BID SET





REDUNDANT LARGE LAG PUMP ON	=19.00	FLOAT (F1) TIMER CONTROL
REDUNDANT LARGE LEAD PUMP ON	=18.50	FLOAT (F2) TIMER CONTROL
REDUNDANT SMALL LAG PUMP ON	=18.00	FLOAT (F3) TIMER CONTROL
REDUNDANT SMALL LEAD PUMP ON	=17.50	FLOAT (F4) TIMER CONTROL
HIGH LEVEL ALARM	=17.50	TRANSDUCER & FLOAT (T1)
LARGE LAG PUMP ON	=17.00	TRANSDUCER (T2)
LARGE LEAD PUMP ON	=16.00	TRANSDUCER (T3)
SMALL LAG PUMP ON	=14.00	TRANSDUCER (T4)
SMALL LEAD PUMP ON	=12.00	TRANSDUCER (T5)
PUMPS OFF	=11.00	TRANSDUCER (T6)
LOW LEVEL ALARM & REDUNDANT OFF	=10.50	TRANSDUCER (T7)

INFLUENT PUMP STATION PLAN AND ELEVATION
SCALE: 1" = 2' AT FULL SCALE

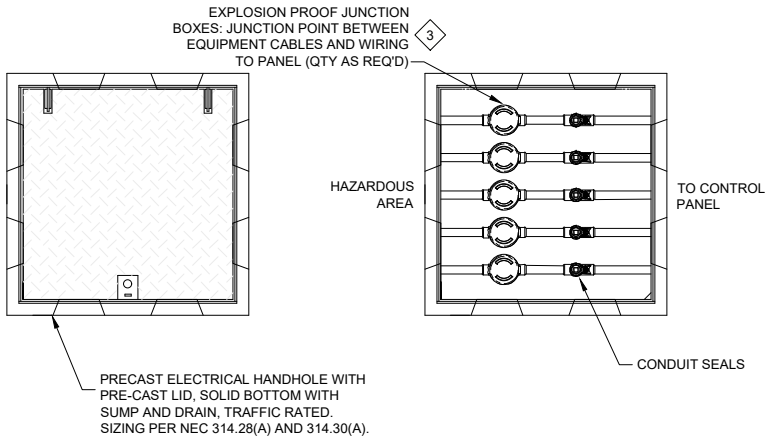
2 0 2 4
SCALE HORIZONTAL SCALE FEET

GENERAL NOTES:

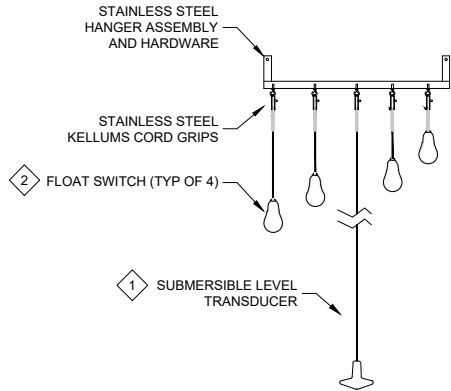
- ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
- EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
- ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
- UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
- ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

KEY NOTES:

- LEVEL TRANSDUCER: WIKA SUBMERSIBLE PRESSURE TRANSMITTER, 316SS BODY, 0-15 PSI RANGE WITH LEVELGUARD, POLYURETHANE CABLE, 4-20 mA LOOP POWERED, TERMINATION ENCLOSURE, FOR USE WITH VENTED PRESSURE TRANSMITTER, NEMA 4X, INTEGRALLY MOUNTED MOISTURE PROTECTION.
- HIGH LEVEL FLOATS: ECO-FLOAT MODEL GP60NONC, NON-MERCURY, NO/NC CONTACT, POLYPROPYLENE HOUSING AND CLAMP, PIPE MOUNT.
- PROVIDE SERVICE LOOPS FOR EXTRA PUMP AND INSTRUMENT CABLES INSIDE WET WELL. ROUTE TO HANDHOLES FOR CONNECTION POINT FOR CABLES TO PERMANENT WIRING. MAKE CONNECTIONS WITH APPROVED CONNECTORS. SEAL CONDUITS LEAVING HANDHOLES TO INFLUENT PUMP STATION PER NEC ART 500.
- PROVIDE INDIVIDUAL CONDUITS FOR EACH PUMP, FLOAT, AND TRANSDUCER CABLE. OVERSIZE CONDUITS AS REQUIRED BETWEEN WET WELL AND HANDHOLE FOR EASY REMOVAL AND REPLACEMENT OF FACTORY PUMP AND INSTRUMENT CABLES.



HAZARDOUS AREA CONNECTION DETAIL
SCALE: NONE



INSTRUMENTATION SUPPORT DETAIL
SCALE: NONE



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NO.	REVISIONS	BY	DATE



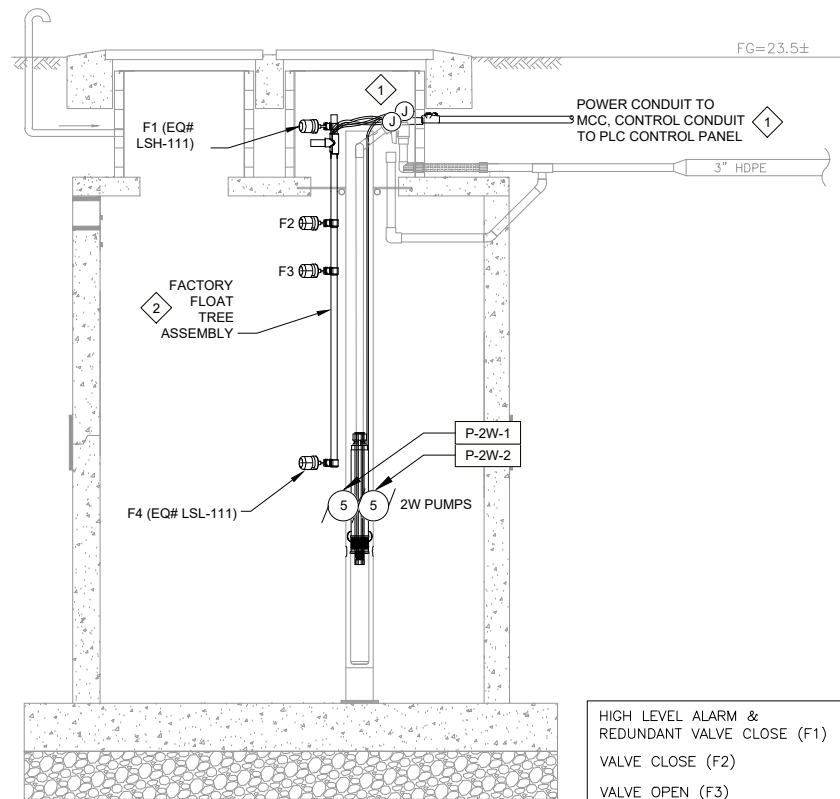
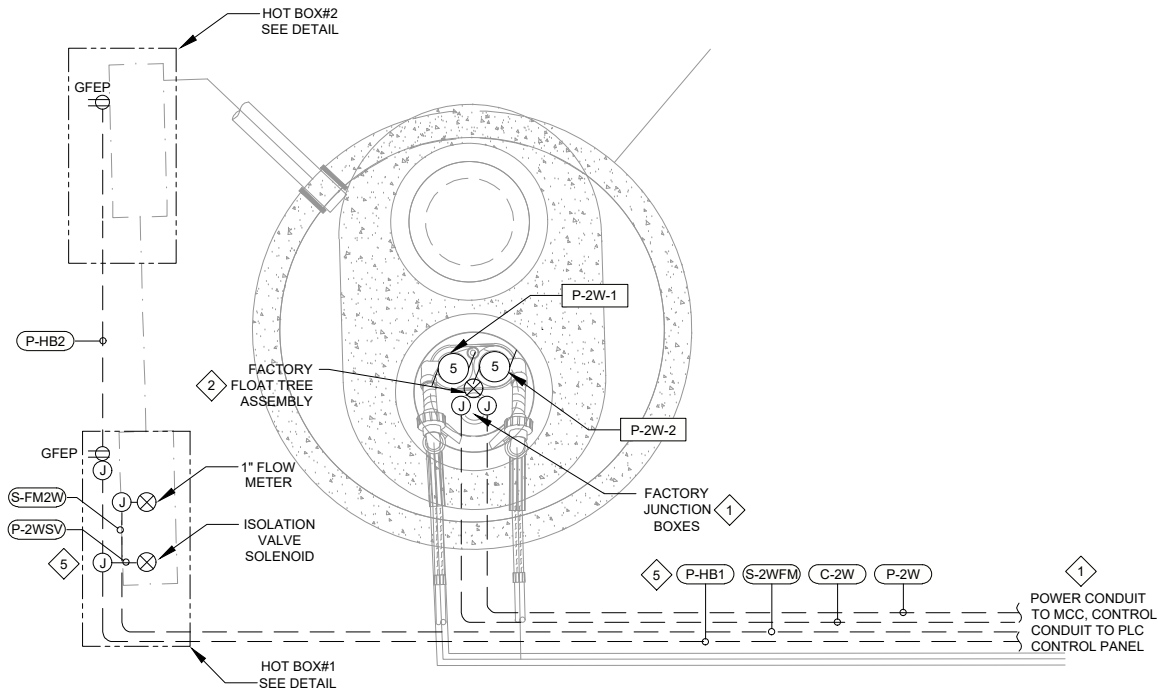
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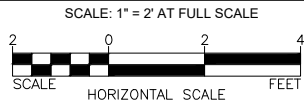
EASTSOUND SEWER AND WATER DISTRICT
WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
INFLUENT PUMP STATION ELECTRICAL PLAN AND ELEVATION

SHEET	DATE	SCALE	JOB NUMBER
E4.2	09-12-2022	AS SHOWN	2020-070
PAGE			71 OF 96



HIGH LEVEL ALARM & REDUNDANT VALVE CLOSE (F1)	22.00
VALVE CLOSE (F2)	20.00
VALVE OPEN (F3)	19.00
LOW LEVEL ALARM (F4) (& PUMP(S) "OFF" FLOAT SWITCH)	15.00

2W PUMP STATION PLAN AND ELEVATION

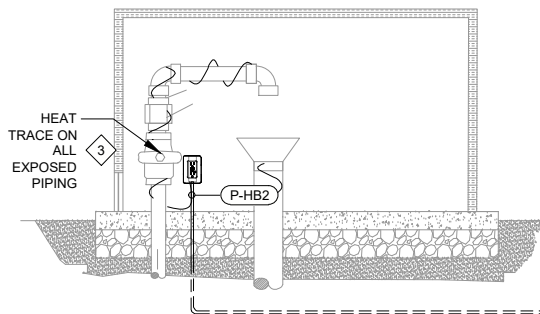


GENERAL NOTES:

1. ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
4. UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
5. ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

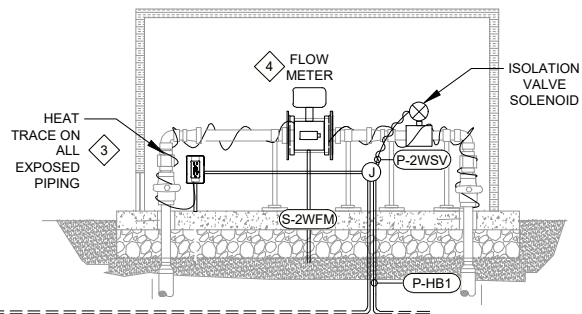
KEY NOTES:

1. 2W PUMP SYSTEM SHALL INCLUDE (2) FACTORY JUNCTION BOXES WITH CORD STRAIN RELIEF GRIPS FOR CABLES. PROVIDE ONE JBOX FOR THE MOTOR CABLES AND ONE FOR THE FLOAT SWITCHES. ROUTE POWER RACEWAY TO MCC BUCKETS, AND CONTROL WIRING TO PLC CONTROL PANEL.
2. PROVIDE FACTORY FLOAT TREE ASSEMBLY WITH 4 FLOATS.
3. PROVIDE HEAT TRACE ON ALL EXPOSED PIPING. RAY-CHEM BTV SELF REGULATING HEATING CABLE, OR EQUAL. SYSTEM SHALL INCLUDE GFEP BREAKER ON 120V 20AMP DEDICATED CIRCUIT. PROVIDE NEMA 4X THERMOSTAT AND END OF LINE TERMINATION KIT WITH INDICATOR LIGHT.
4. 2W FLOW METER: 1" MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, LOCAL DISPLAY, NEMA 4X, NPT 1/2". PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
5. ROUTE POWER WIRING FOR SOLENOID AND HEAT TRACE CIRCUITS TO JBOX IN HB#1 AND EXTEND TO HB#2.



HOT BOX #2 ELECTRICAL DETAIL

NOT TO SCALE



HOT BOX #1 ELECTRICAL DETAIL

NOT TO SCALE

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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

2W PUMP STATION ELECTRICAL PLAN AND ELEVATION

SHEET	DATE	SCALE	JOB NUMBER
E4.3	09-12-2022	AS SHOWN	2020-070
PAGE	OF		
72	96		



BID SET

NO.	REVISIONS	BY	DATE
1	ADD UV DISINFECTION	BZ	04/18/22



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EASTSOUND SEWER AND WATER DISTRICT

WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I
UV DISINFECTION CHAMBER & DIGESTER ELECTRICAL PLAN AND ELEVATION

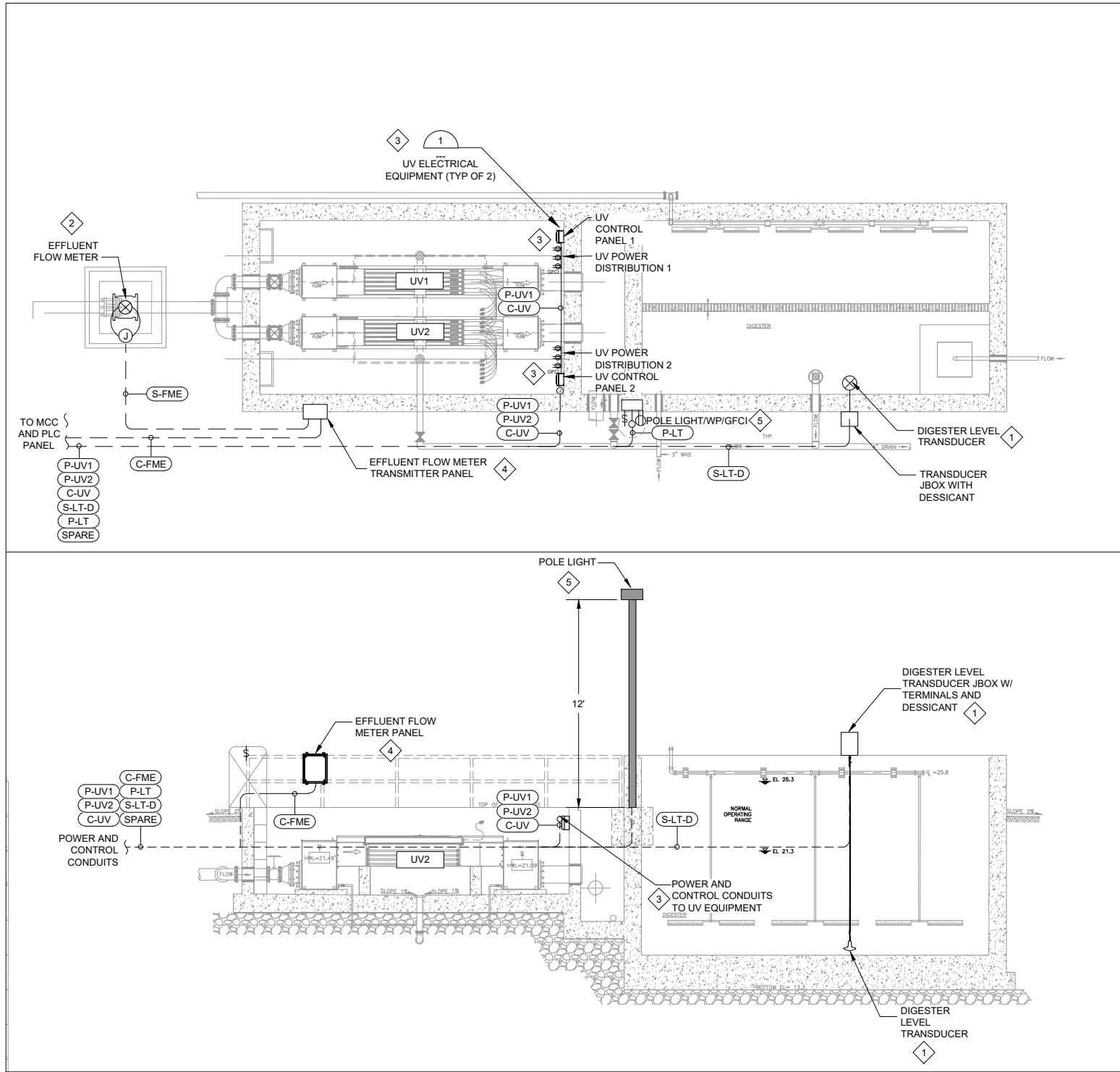
SHEET	DATE	SCALE	JOB NUMBER
E5.1	09-12-2022	AS SHOWN	2020-070
PAGE			
73			OF 96

GENERAL NOTES:

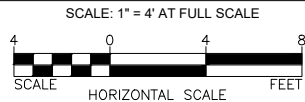
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2. EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
3. ELECTRIC HAND HOLES ARE NOT SHOWN ON THESE PLANS BUT SHALL BE REQUIRED TO MEET CODE AND AS SPECIFIED. COORDINATE ALL FINAL LOCATIONS WITH ENGINEER.
4. UPSIZE CONDUITS FROM WET WELLS TO JUNCTION BOXES AS REQUIRED FOR PUMP AND FLOAT CABLES TO MEET NEC FILL REQUIREMENTS.
4. ALL OUTDOOR CONTROL PANEL AND DISCONNECT ENCLOSURES SHALL BE TYPE 316 STAINLESS STEEL.

KEY NOTES:

1. LEVEL TRANSDUCER: WIKA SUBMERSIBLE PRESSURE TRANSMITTER, 316SS BODY, 0-15 PSI RANGE WITH LEVELGUARD, POLYURETHANE CABLE, 4-20 mA LOOP POWERED, TERMINATION ENCLOSURE, FOR USE WITH VENTED PRESSURE TRANSMITTER, DESSICANT, NEMA 4X, INTEGRALLY MOUNTED MOISTURE PROTECTION. MOUNT ENCLOSURE TO HANDRAIL WITH SS HARDWARE.
2. FLOW METERS: SIZE PER CIVIL DRAWINGS. MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 120-240VAC OR 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, REMOTE HOUSING, NPT 1/2, SENSOR IP68 TYPE 6, POTTED CORROSION PROTECTION AND SUBMERSIBLE. PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
3. INSTALL LOCAL UV CONTROL PANELS AND POWER DISTRIBUTION FOR UV SYSTEMS. CONTRACTOR SHALL PROVIDE WIRING GUTTER, JUNCTION BOXES AND (3) DUPLEX GFCI RECEPTACLES (PDR'S) WITH IN-USE WET LOCATION COVERS FOR EACH UV UNIT. MOUNT TO BASIN WALL WITH STAINLESS STEEL UNITSTRUT AND HARDWARE. CONDUIT, FITTINGS AND JUNCTION BOXES SHALL BE STAINLESS STEEL. PROVIDE ALL WIRING AND CONNECTIONS PER MANUFACTURER REQUIREMENTS.
4. PROVIDE LOCAL CONTROL PANEL TO MOUNT EFFLUENT FLOW METER TRANSMITTER, AND TERMINALS FOR LOCAL INSTRUMENTATION SIGNALS. ROUTE BACK TO PLC CONTROL PANEL. ENCLOSURE SHALL BE STAINLESS STEEL, MOUNT TO HANDRAIL OR RACK WITH SS HARDWARE.
5. CONTRACTOR SHALL PROVIDE POLE LIGHT TO ILLUMINATE UV AREA. PROVIDE CONCRETE POLE BASE AND SECURE TO SIDE OF BASIN WITH STAINLESS STEEL HARDWARE; FIELD COORDINATE WITH CIVIL. PROVIDE 4" SQUARE ALUMINUM POLE 12' TALL WITH PHOTOCELL AND LOCAL ON/OFF SWITCH. LITHONIA LIGHTING #DXS0-LED-P3-30K-T3M-MVOLT-SPA-DNAXD WITH PHOTOCELL, OR APPROVED EQUAL.

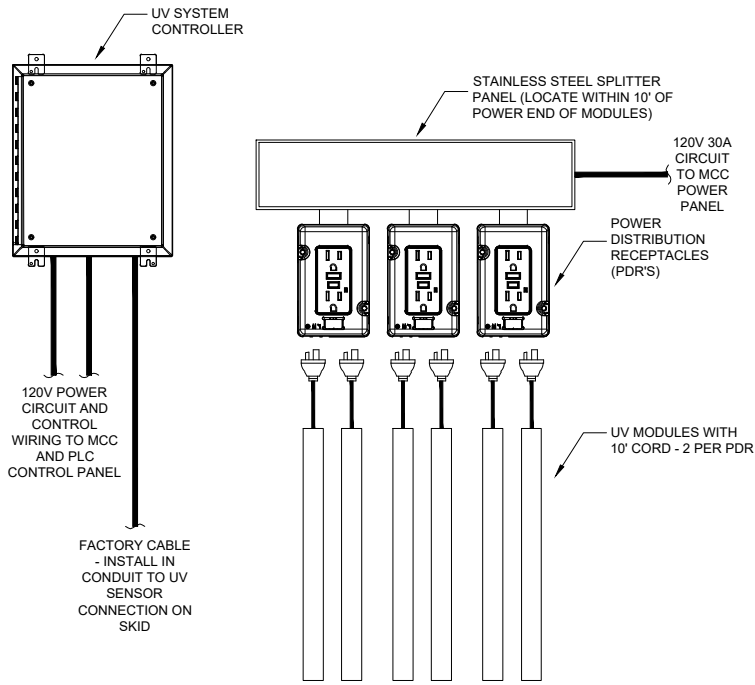


1 UV DISINFECTION CHAMBER AND DIGESTER ELECTRICAL PLAN AND ELEVATION



1 UV SYSTEM INTERCONNECTION DIAGRAM - TYP OF 2

SCALE: NONE



BID SET

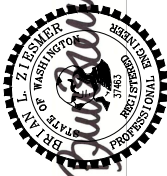
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EASTSOUND SEWER AND WATER DISTRICT

SAN JUAN COUNTY
WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

ELECTRICAL PANEL SCHEDULES

DATE
09-12-2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

SHEET
E6.2

PAGE
75 OF 96



BID SET

PANEL: MDP				PANEL SCHEDULE						PROJECT: Eastsound WWTP				
480Y/277V, 3Ph, 4W.				400A Bus			400A		M.L.O.		SURFACE MOUNTED			
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO		
1	BLOWER BUILDING MCC	60,000	M	400	3	A	3	125	G	15,000	TRANSFORMER FOR EXISTING BUILDING	2		
3	---	60,000	M	---	---	B	---	---	G	15,000	480-208Y/120V	4		
5	---	60,000	M	---	---	C	---	---	G	15,000		6		
7	SPACE					A					SPACE	8		
9	---					B					---	10		
11	---					C					---	12		
13	SPACE					A					SPACE	14		
15	---					B					---	16		
17	---					C					---	18		
19	SPACE					A					SPACE	20		
21	---					B					---	22		
23	---					C					---	24		
25	SPACE					A					SPACE	26		
27	---					B					---	28		
29	---					C					---	30		
DATE: September 15, 2022														
TOTAL CONNECTED LOAD:		PH A	75,000 VA	270.6 AMPS										
TOTAL CONNECTED LOAD:		PH B	75,000 VA	270.6 AMPS										
TOTAL CONNECTED LOAD:		PH C	75,000 VA	270.6 AMPS										
MAX PHASE CONNECTED LOAD:		PH A	75,000 VA			PANEL RATING:		42,000 AIC						
TOTAL CONNECTED LOAD (3 x MAX):			225.0 kVA	270.6 AMPS		TOTAL DEMAND LOAD:		225.0 kVA	270.6 AMPS					

PANEL MDP
SCALE: NONE

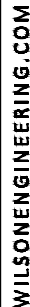
PANEL: LP1				PANEL SCHEDULE							PROJECT: Eastsound WWTP			
120/240V, 1Ph, 3W.				100A Bus			100A M.C.B.				SURFACE MOUNTED			
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO		
1	BLOWER ROOM RECEPTACLES	900	R	20	1	A	1	20	C	1,000	MAIN CONTROL PANEL	2		
3	LOUVER AND HVAC CONTROL	500	M	20	1	B	1	20	L	239	BLOWER BUILDING LIGHTING	4		
5	EXHAUST FAN EF-1 (BLOWER BUILDING)	1,127	M	20	1	A	1				SPARE	6		
7	EXHAUST FAN EF-2 (BLOWER BUILDING)	1,127	M	20	1	B	1	20	H	1,300	GENERATOR - HEATER AND BATTERY CHGR	8		
9	UV1 POWER	2,520	C	30	1	A	1	20	H	1,000	HOT BOX 1 - GFEP	10		
11	UV2 POWER	2,520	C	30	1	B	1	20	H	1,000	HOT BOX 2 - GFEP	12		
13	UV1 CONTROLLER	600	C	20	1	A	1	20	M	500	EFFLUENT SAMPLER	14		
15	UV2 CONTROLLER	600	C	20	1	B	1	20	M	500	HYPOCHLORITE FEED PUMP	16		
17	SPARE			20	1	A	1	20			SPARE	18		
19	SPACE					B					SPACE	20		
21	SPACE					A					SPACE	22		
23	SPACE					B					SPACE	24		
TOTAL CONNECTED LOAD: PH A 7,647 VA 63.7 AMPS													DATE: September 15, 2022	
TOTAL CONNECTED LOAD: PH B 7,786 VA 64.9 AMPS														
MAX PHASE CONNECTED LOAD: PH B 7,786 VA													PANEL RATING: 10,000 AIC	
TOTAL CONNECTED LOAD (2 x MAX): 15.6 kVA 64.9 AMPS													TOTAL DEMAND LOAD: 17.3 kVA 72.1 AMPS	

PANEL LP1 (MCC)
SCALE: NONE

PANEL: LP2				PANEL SCHEDULE						PROJECT: Eastsound WWTP			
120/240V, 1Ph, 3W.			100A Bus			60A M.C.B.			SURFACE MOUNTED				
CKT NO	DESCRIPTION / LOCATION	LOAD (VA)	LOAD TYPE	C.B. AMP	C.B. POLE	PHASE	C.B. POLE	C.B. AMP	LOAD TYPE	LOAD (VA)	DESCRIPTION / LOCATION	CKT NO	
1	PUMP STATION CONTROL CIRCUITS	1,000	R	20	1	A	1	20	C	500	H2S SENSOR	2	
3	WALL PACK LIGHTING	500	L	20	1	B	1	20	C	500	METHANE SENSOR	4	
5	VENTILATION CONTROL PANEL	500	G	20	1	A	1	20	M	500	POLYMER FEED PUMP RECEPTACLE	6	
7	EX-HAUST FAN EF-3 (TREATMENT BUILDING)	1,127	M	20	1	B	1	20	M	500	INFLUENT SAMPLER RECEPTACLE	8	
9	EX-HAUST FAN EF-4 (TREATMENT BUILDING)	1,127	M	20	1	A	1	20			SPARE	10	
11	EX-HAUST FAN EF-5 (TREATMENT BUILDING)	1,127	M	20	1	B	1	20			SPARE	12	
13	SPARE			20	1	A	1	20			SPARE	14	
15	SPARE			20	1	B	1	20			SPARE	16	
DATE: September 15, 2022													
TOTAL CONNECTED LOAD:		PH A	3,627 VA	30.2 AMPS									
TOTAL CONNECTED LOAD:		PH B	3,754 VA	31.3 AMPS									
MAX PHASE CONNECTED LOAD:		PH B	3,754 VA			PANEL RATING:		10,000 AIC					
TOTAL CONNECTED LOAD (2 x MAX):			7.5 kVA	31.3 AMPS		TOTAL DEMAND LOAD:		7.8 kVA		32.3 AMPS			

PANEL LP2 (INFLUENT PUMP STATION PANEL)
SCALE: NONE

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CONDUIT AND CONDUCTOR SCHEDULES

SCALE: NONESCALE: NONE

96 OF 96

The diagram illustrates a network configuration for an Allen Bradley CompactLogix PLC system. The PLC rack consists of ten slots: Slot 0 contains a power supply, Slot 1 contains a CPU, and Slots 2 through 9 contain various modules. A network switch is connected to the PLC rack via a network cable. The switch is also connected to three flow meters: a 2W Flow Meter, an Effluent Flow Meter, and a VFDs in MCC (Typical of 10). The flow meters are connected to the switch via network cables.

The diagram illustrates the hardware configuration of an Allen Bradley CompactLogix PLC system. The rack consists of six slots, labeled SLOT 0 through SLOT 5. The components are as follows:

- SLOT 0:** Power supply module (1768-ENBT).
- SLOT 1:** CPU module (1768-L30).
- SLOT 2:** Empty slot.
- SLOT 3:** I/O module (1768-IR128).
- SLOT 4:** I/O module (1768-IR128).
- SLOT 5:** Empty slot.

Below the rack, a **NETWORK SWITCH** is connected to the system. The switch has two rows of ports. The top row is connected to the PLC rack via fiber optic cables (labeled 1-4 and 5-8). The bottom row is connected to the PLC rack via Ethernet cables (labeled 1-4 and 5-8). Additionally, the switch is connected to a network of flow meters and a drain meter via Ethernet cables (labeled 1-4 and 5-8).

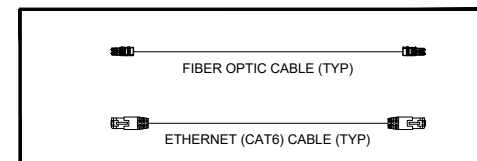
The network components are:

- TRAIN #1 FLOW METER (FUTURE)**
- TRAIN #2 FLOW METER**
- TRAIN #3 FLOW METER**
- PLANT DRAIN FLOW METER**
- INFLUENT FLOW METER**
- WAS FLOW METER**

The diagram illustrates an office environment with a dashed border. Inside, there is a tall server rack on the left and a large computer monitor on the right. The monitor displays the text "SCADA COMPUTER WITH ALARM DIALER". A line connects the server rack to the monitor. Below the monitor, there is a network switch labeled "NETWORK SWITCH" with four ports and a power button. A line connects the network switch to a telephone icon labeled "TELEPHONE (E)". Another line connects the network switch to an internet icon labeled "INTERNET (E)". The entire office environment is enclosed in a dashed box.

The diagram illustrates a compact PLC system architecture. At the top left, a 'NETWORK SWITCH' is shown with a rack of ports. A cable connects one of its ports to a 'COMPACT LOGIX PLC' at the bottom. The PLC is a multi-slot device with various modules, including a power supply, CPU, and I/O modules. To the right of the PLC is an 'OPERATOR INTERFACE', depicted as a large rectangular screen or panel. The entire system is enclosed within a dashed-line boundary.

CONTROL SYSTEM CABLE LEGEND



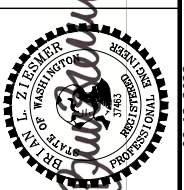
1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.
2. ALL ETHERNET ENABLED EQUIPMENT SHALL BE CONNECTED TO CONTROL SYSTEM COMMUNICATION NETWORK. PROVIDE QUANTITY OF PORTS AS REQUIRED.



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EASTSOUND SEWER AND WATER DISTRICT

WASHTENAW COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

CONTROL SYSTEM OVERVIEW

DATE	09-12-2022
------	------------

SCALE

JOB NUMBER
2020-070

E7.0
SHEET

PAGE 7 OF 96

811 Call 811
two business days
before you dig

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NO.	REVISIONS	BY	DATE

GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



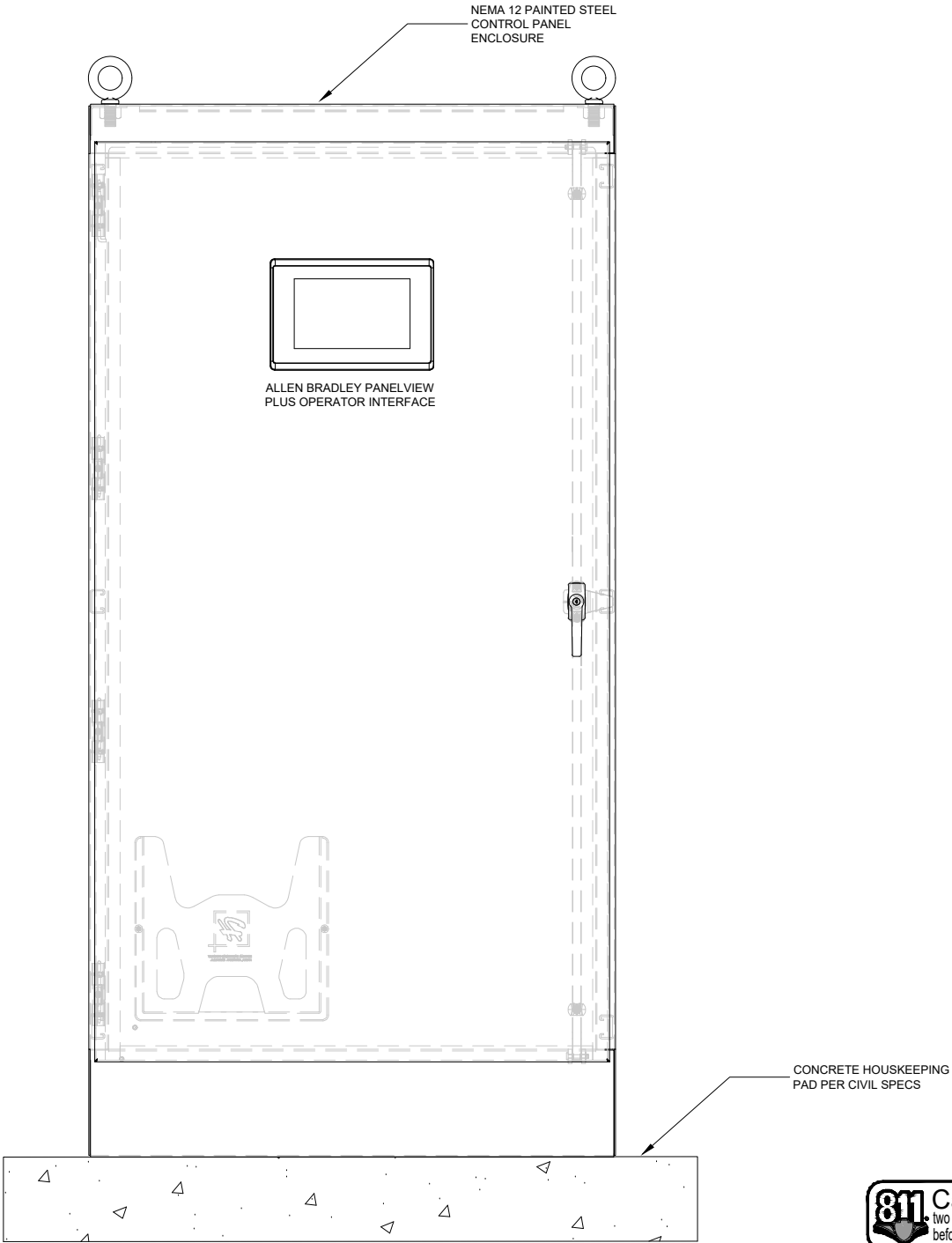
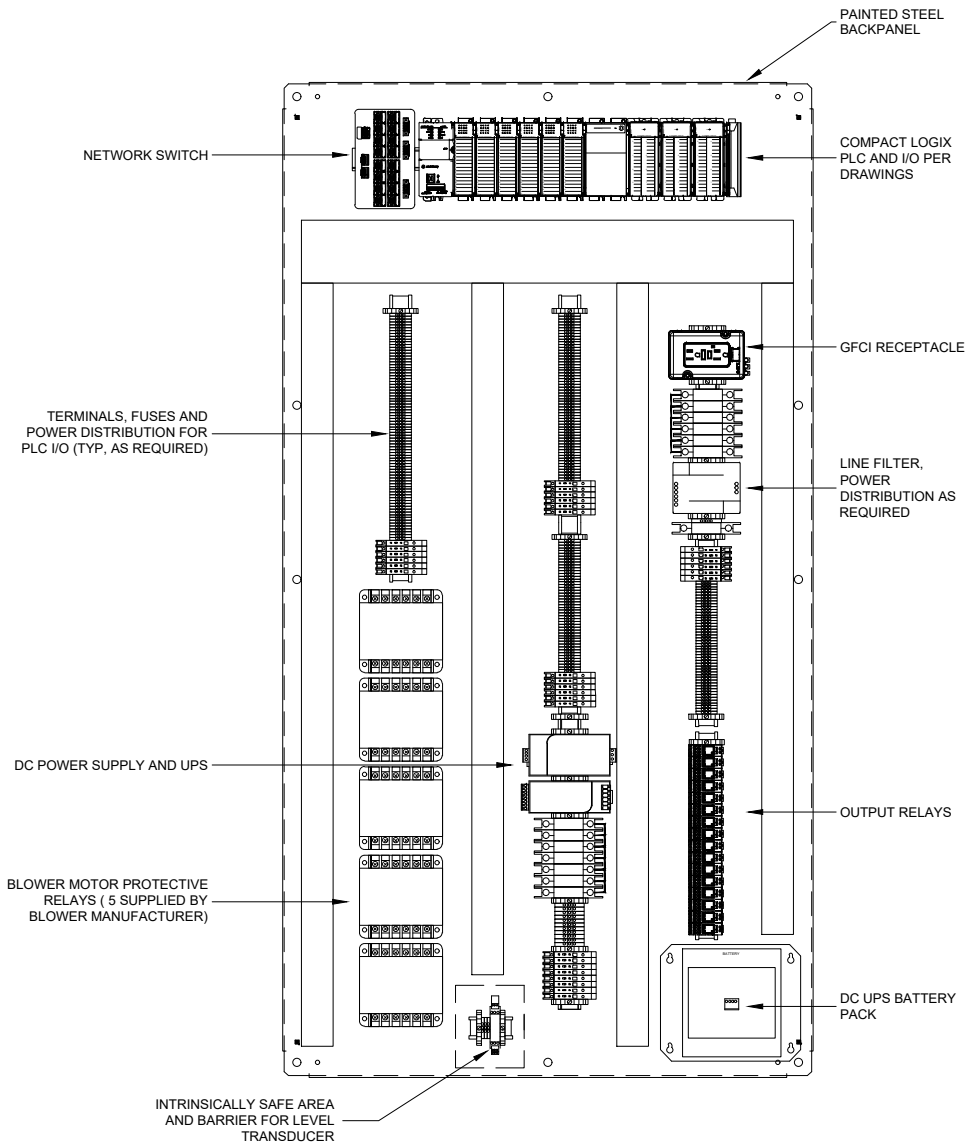
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MAIN PLC CONTROL PANEL ELEVATIONS
SCALE: NONE



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WASHINGTON
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

MAIN PLC CONTROL PANEL ELEVATION

DATE
09-12-2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

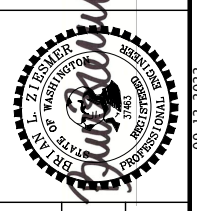
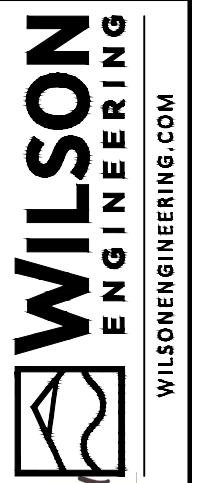
SHEET
E7.1

PAGE
78 OF 96

NO.	REVISIONS	BY	DATE



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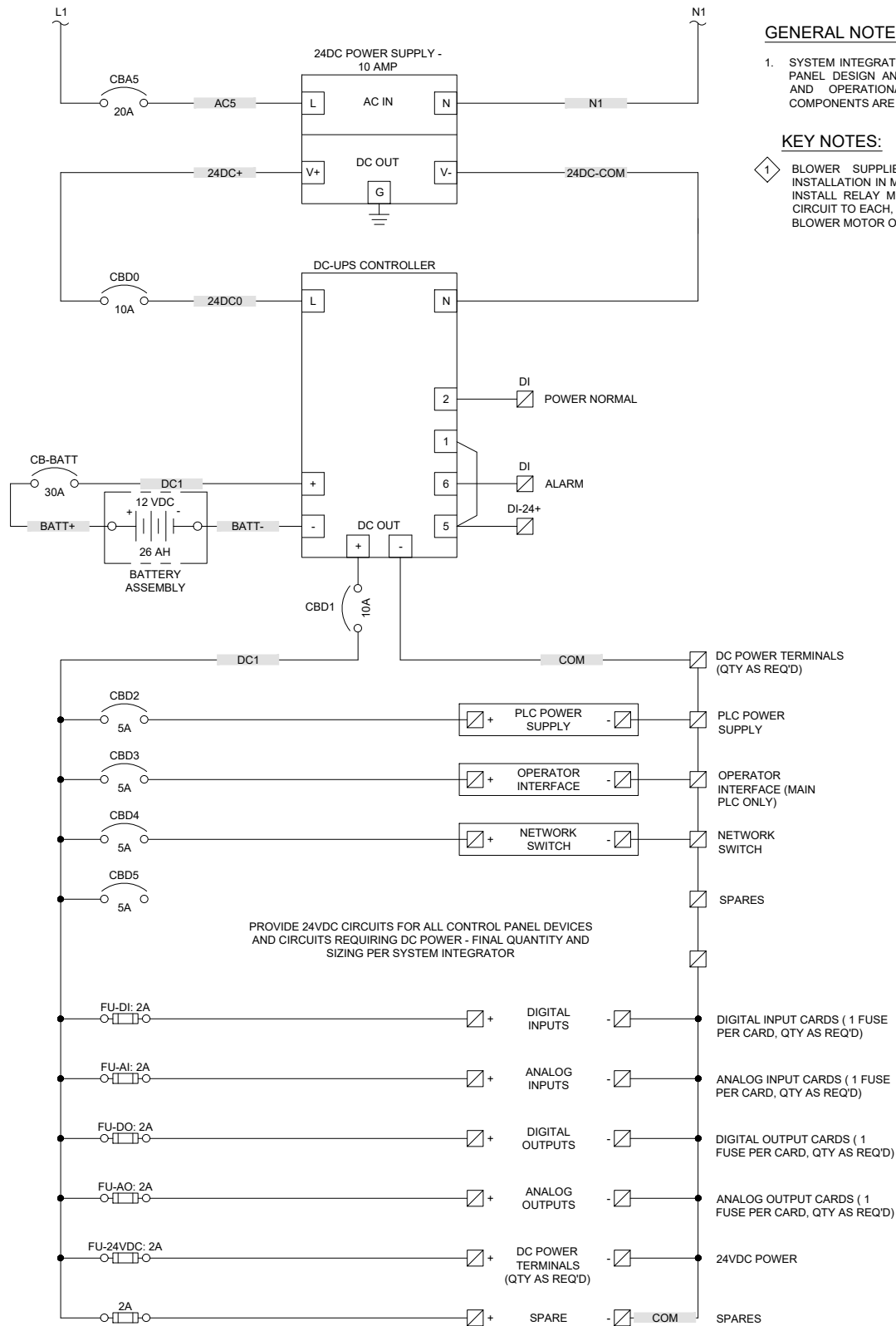
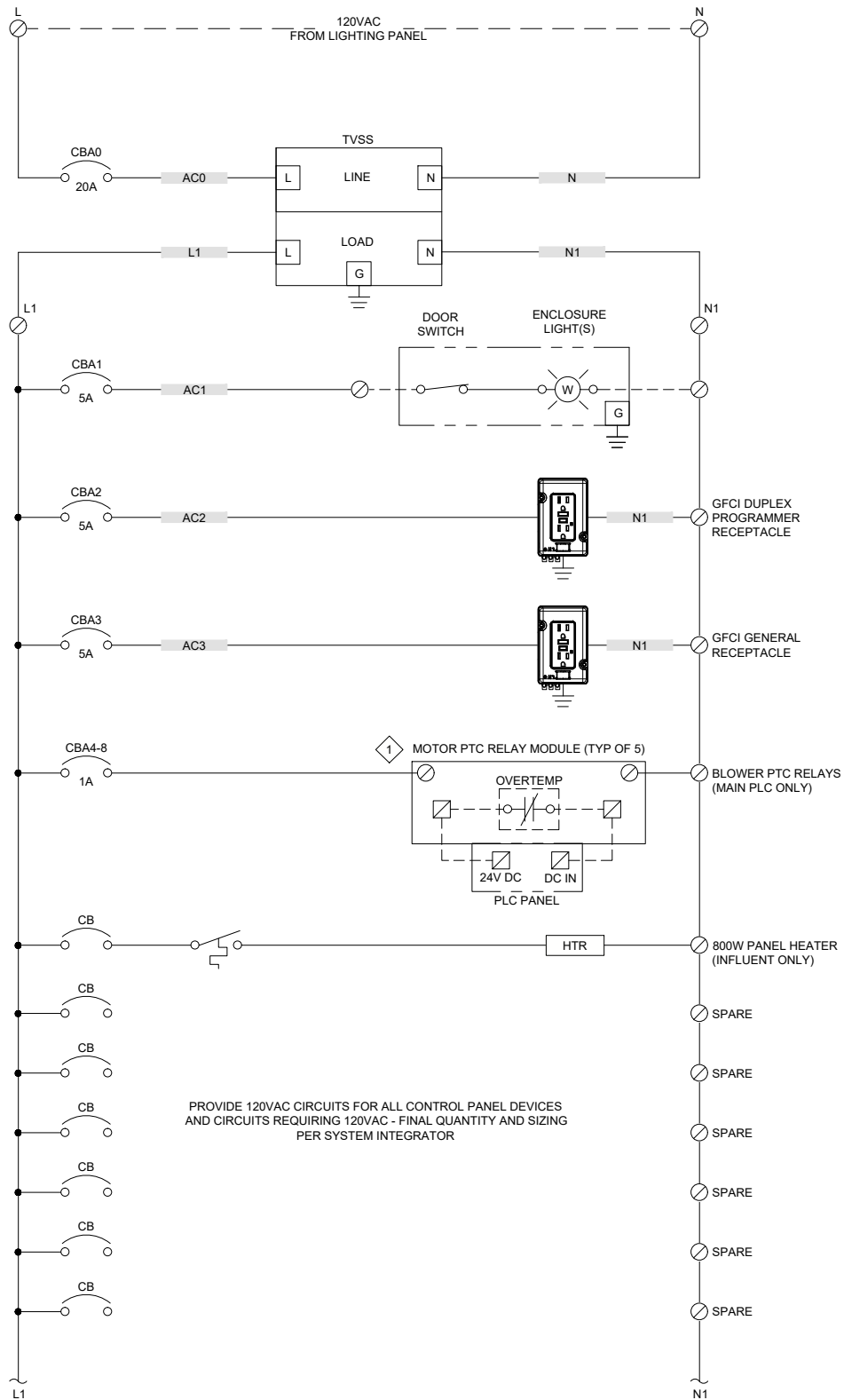
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CHECKED BY	BZ

SHEET	E7.2	
	PAGE	79 OF 96
	BID SET	

EASTSOUND SEWER AND WATER DISTRICT	
WASHINGTON	
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I	
MAIN PLC CONTROL PANEL - POWER WIRING DIAGRAM	

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070



GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.

KEY NOTES:

1. BLOWER SUPPLIER SHALL PROVIDE PTC RELAY MODULE FOR INSTALLATION IN MAIN PLC CONTROL PANEL. SYSTEM INTEGRATOR TO INSTALL RELAY MODULE IN CONTROL PANEL, PROVIDE 120V POWER CIRCUIT TO EACH, AND WIRE OVERTEMP SIGNAL TO PLC TO SHUT DOWN BLOWER MOTOR ON HIGH TEMPERATURE.

MAIN PLC AND INFLUENT CONTROL PANEL - POWER WIRING DIAGRAM (TYP)
SCALE: NONE



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – DIGITAL INPUTS

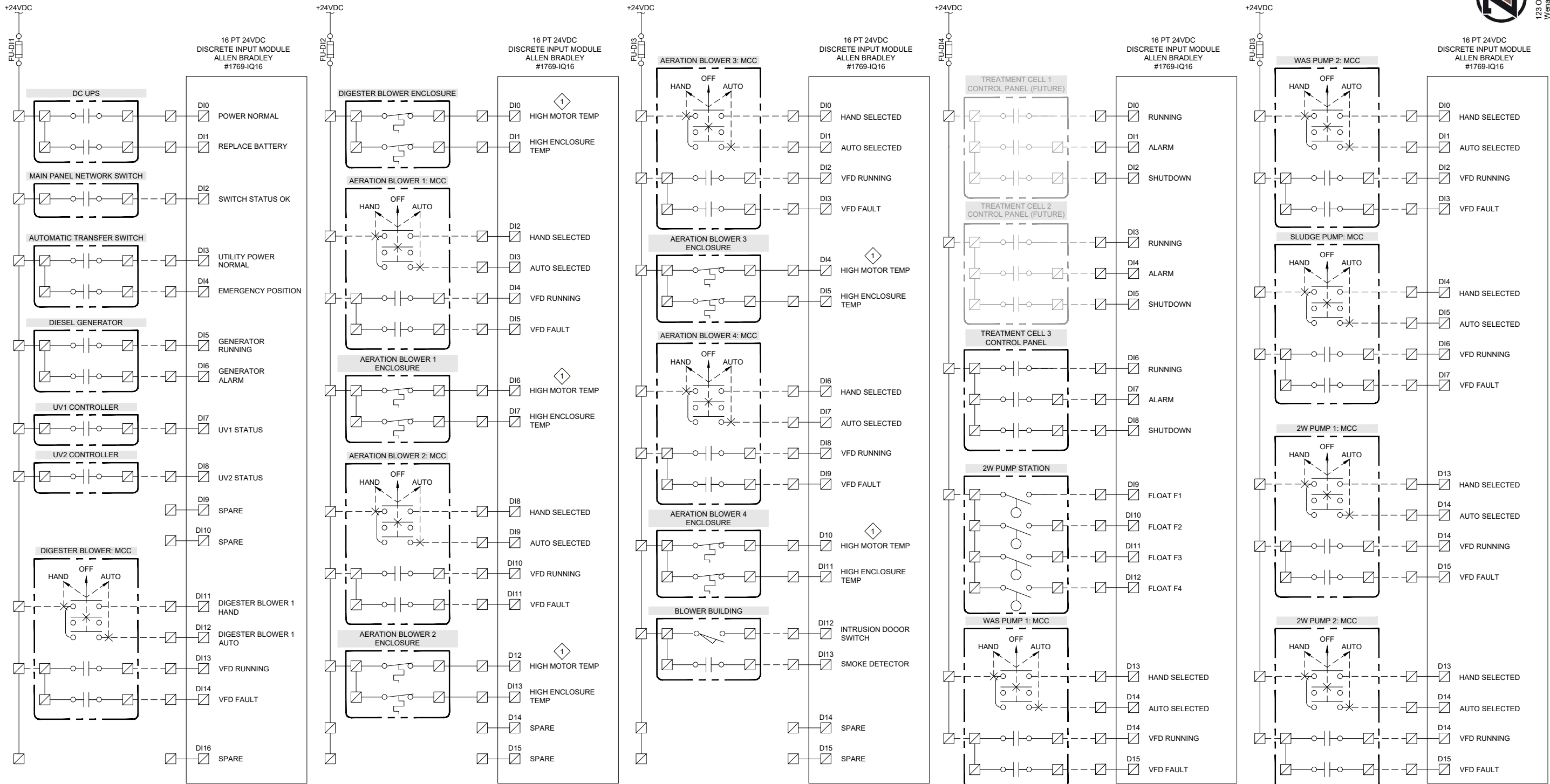
SHEET: E7.3
DATE: 09-12-2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070
PAGE: 80 OF 96

KEY NOTES:

1. BLOWER SUPPLIER SHALL PROVIDE PTC RELAY MODULE FOR INSTALLATION IN MAIN PLC CONTROL PANEL. SYSTEM INTEGRATOR TO INSTALL RELAY MODULE IN CONTROL PANEL, PROVIDE 120V POWER CIRCUIT TO EACH, AND WIRE OVERTEMP SIGNAL TO PLC TO SHUT DOWN BLOWER MOTOR ON HIGH TEMPERATURE.

GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



MAIN PLC CONTROL PANEL - DIGITAL INPUTS
SCALE: NONE



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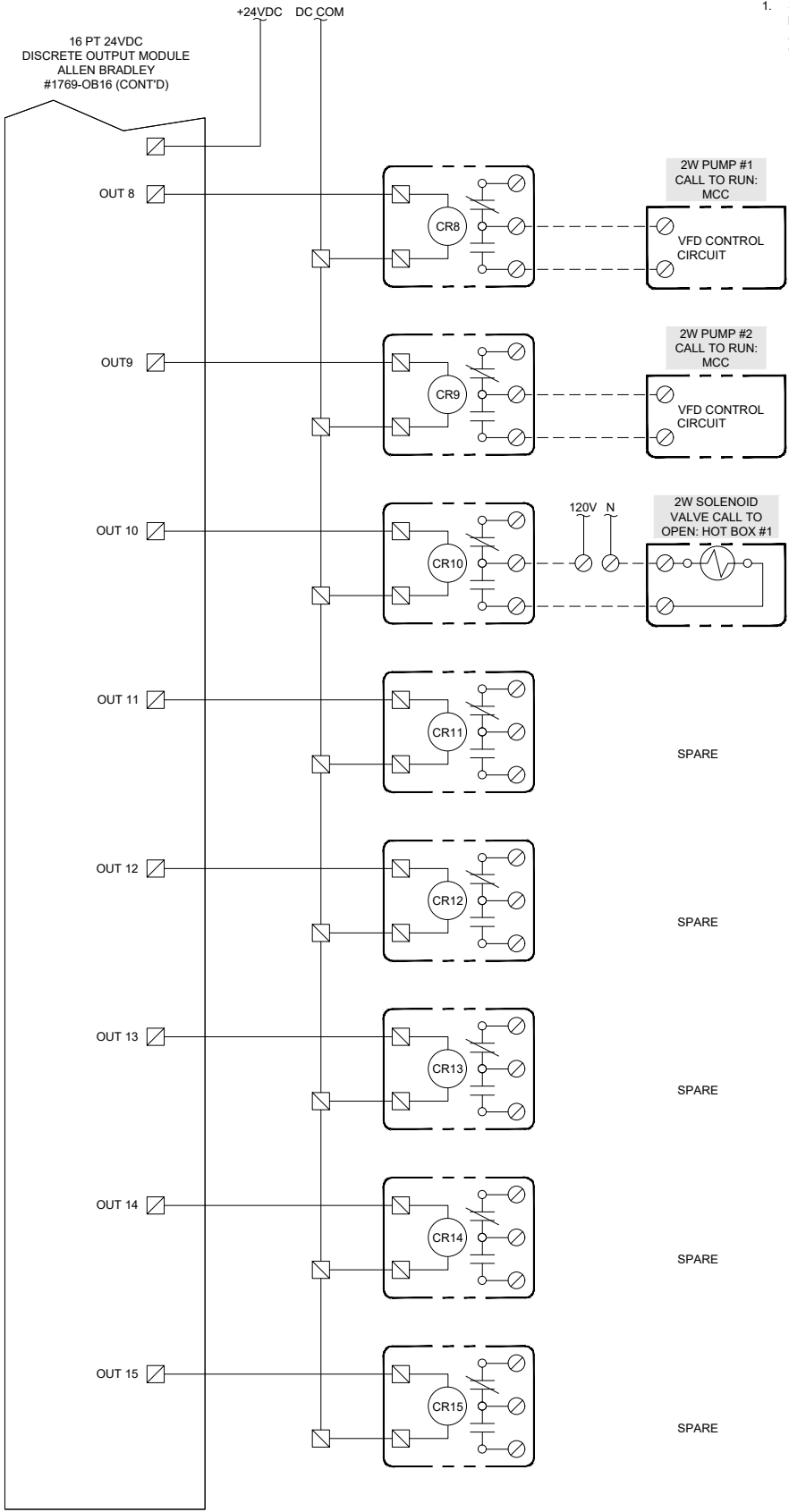
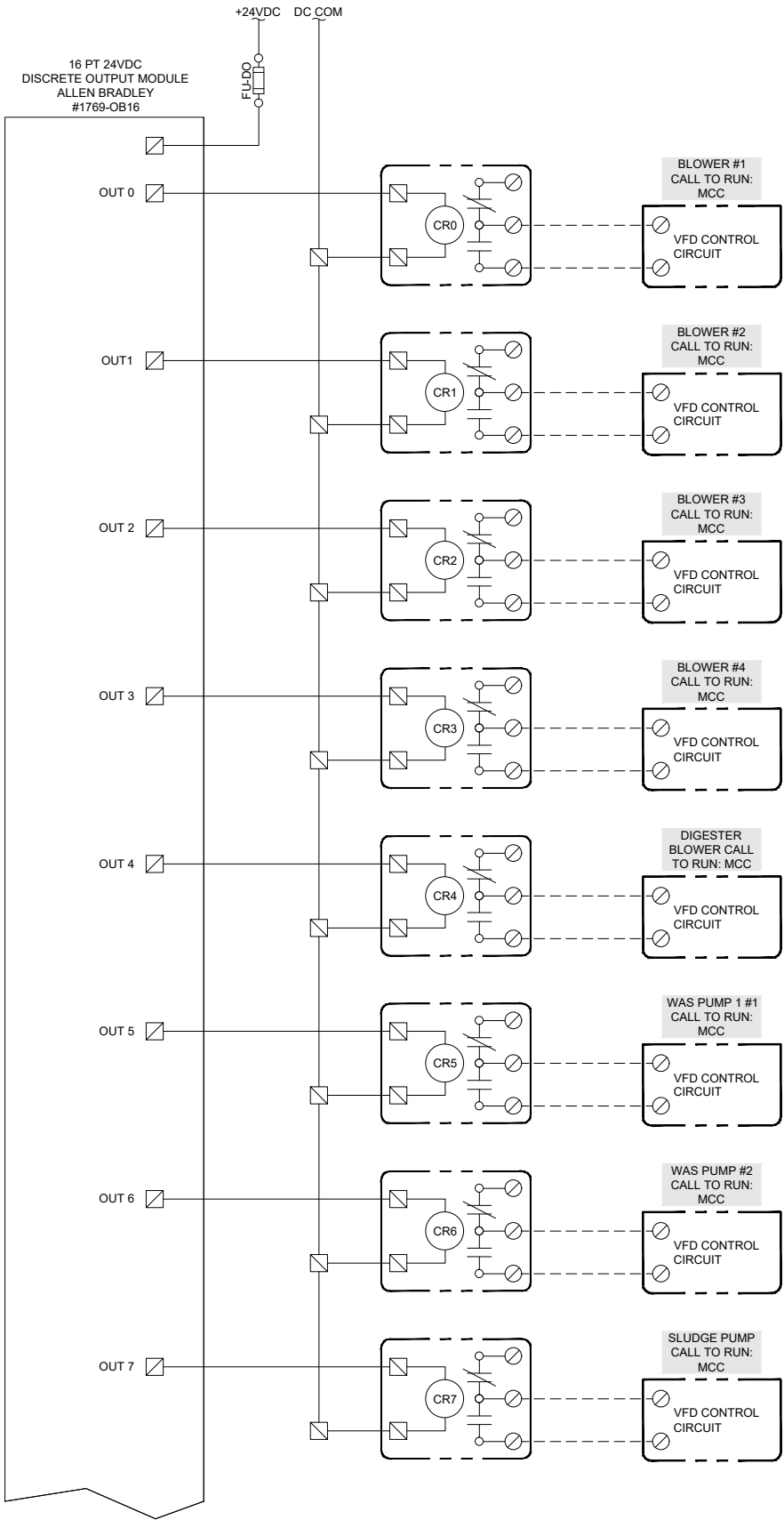
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GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



MAIN PLC CONTROL PANEL - DIGITAL OUTPUTS
SCALE: NONE



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SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

MAIN PLC CONTROL PANEL - DIGITAL OUTPUTS

DATE
09-12-2022

SCALE
AS SHOWN

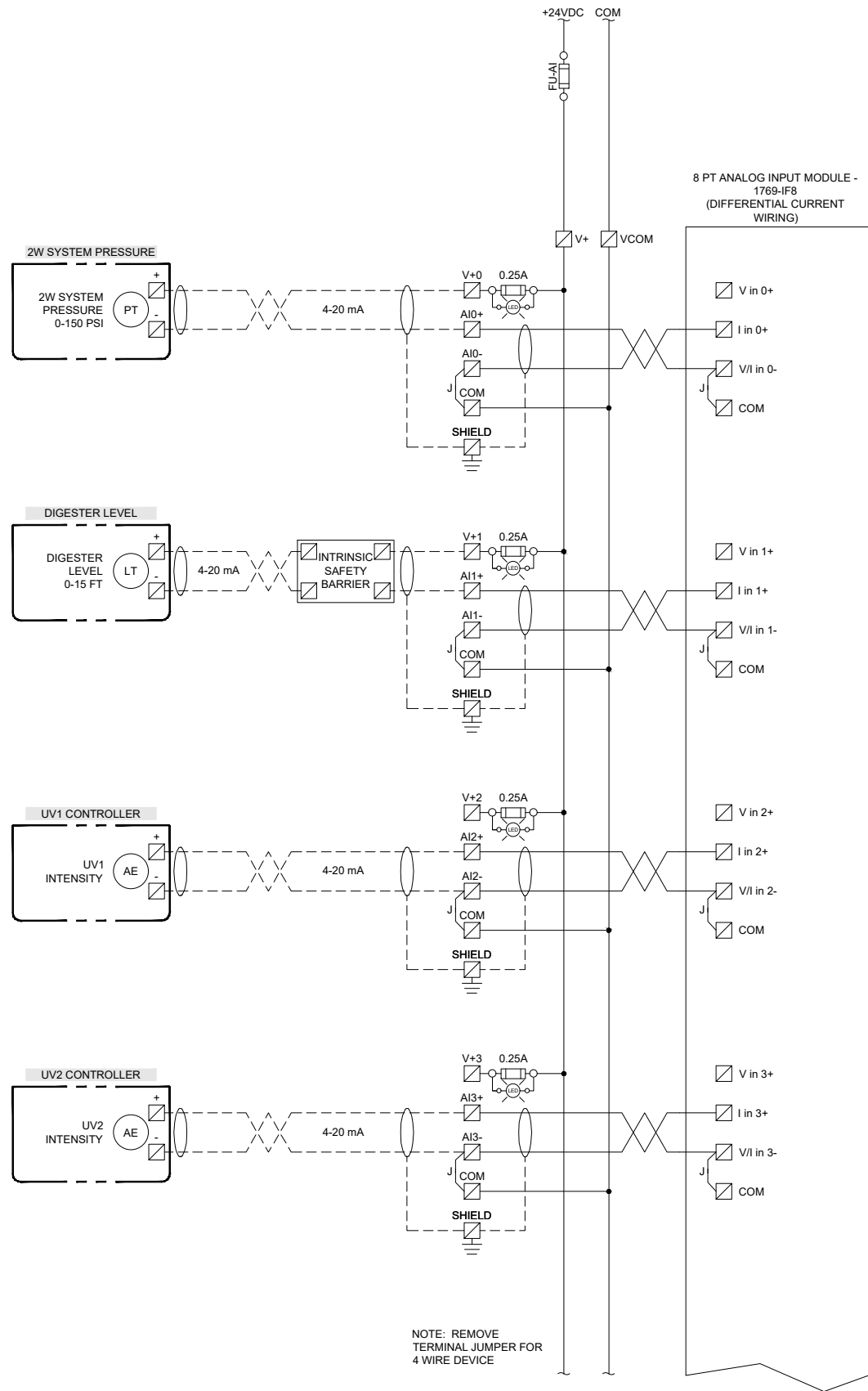
JOB NUMBER
2020-070

SHEET
E7.4

PAGE
81 OF 96

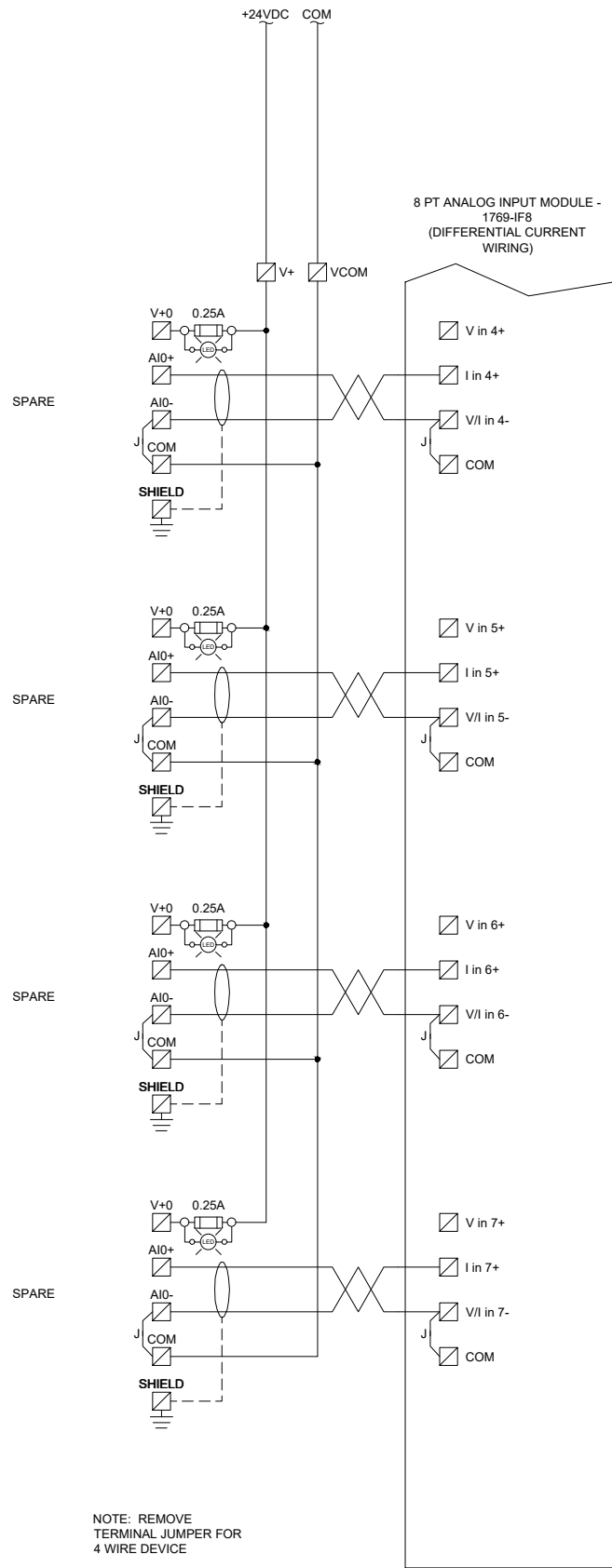
GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



NOTE: REMOVE
TERMINAL JUMPER FOR
4 WIRE DEVICE

MAIN PLC CONTROL PANEL - ANALOG INPUTS
SCALE: NONE



NOTE: REMOVE
TERMINAL JUMPER FOR
4 WIRE DEVICE

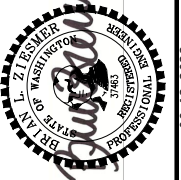
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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – ANALOG INPUTS – SHEET 1

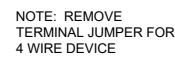
DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET	E7.5
PAGE	82 OF 96



BID SET

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



SCALE: NONE



10

SAN JUAN

SHEET

E7.6

PAGE

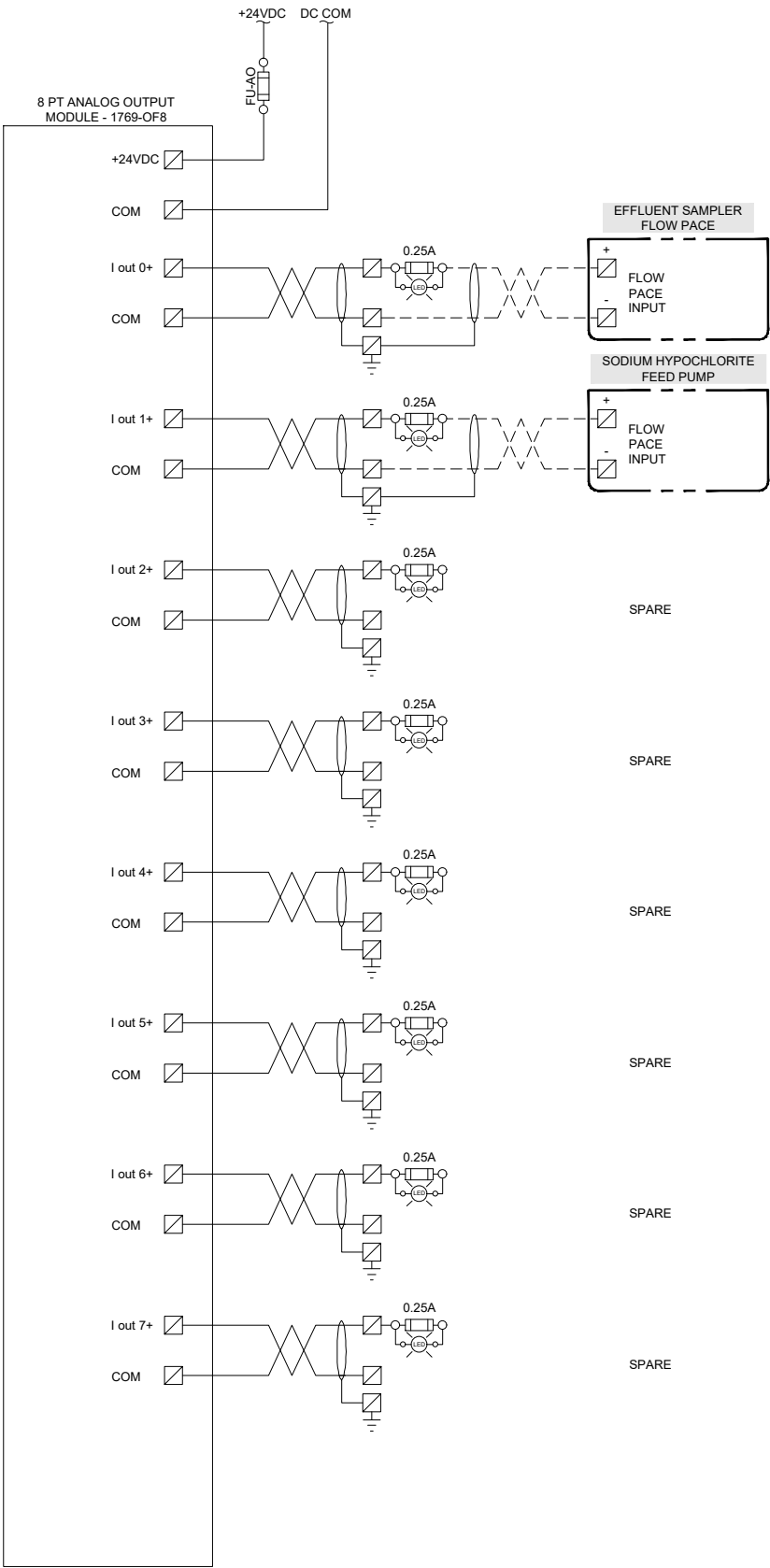
83 OF 96



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



MAIN PLC CONTROL PANEL - ANALOG OUTPUTS
SCALE: NONE

NO.	REVISIONS	BY	DATE



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

MAIN PLC CONTROL PANEL – ANALOG OUTPUTS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET	E7.7
PAGE	84 OF 96



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

INFLUENT PUMP STATION CONTROL PANEL ELEVATIONS

SHEET	DATE	SCALE	JOB NUMBER
E8.1	09-12-2022	AS SHOWN	2020-070
PAGE			
85		OF	96



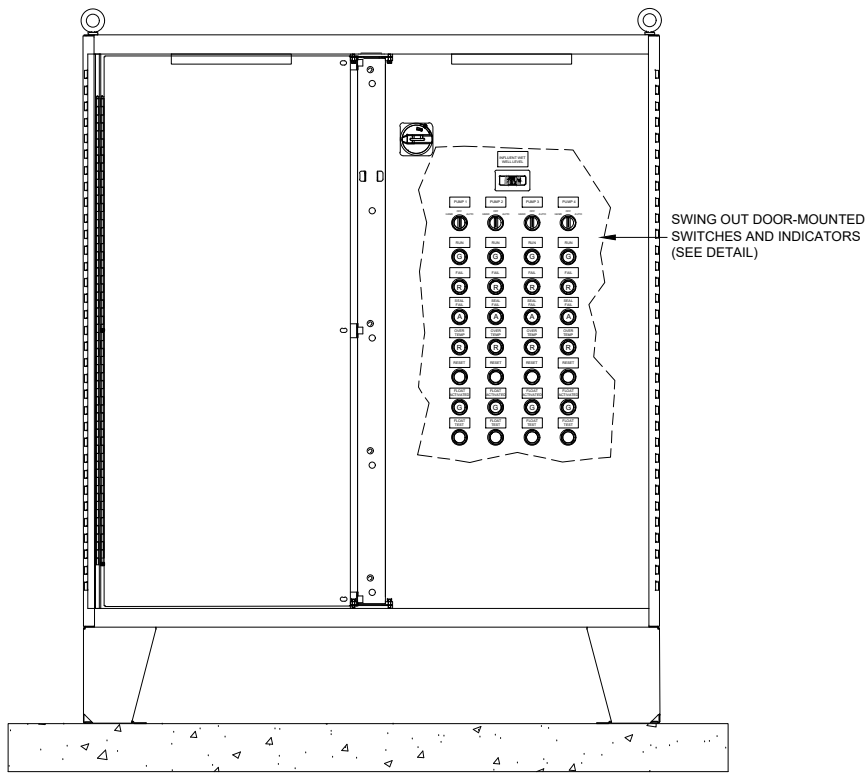
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GENERAL NOTES:

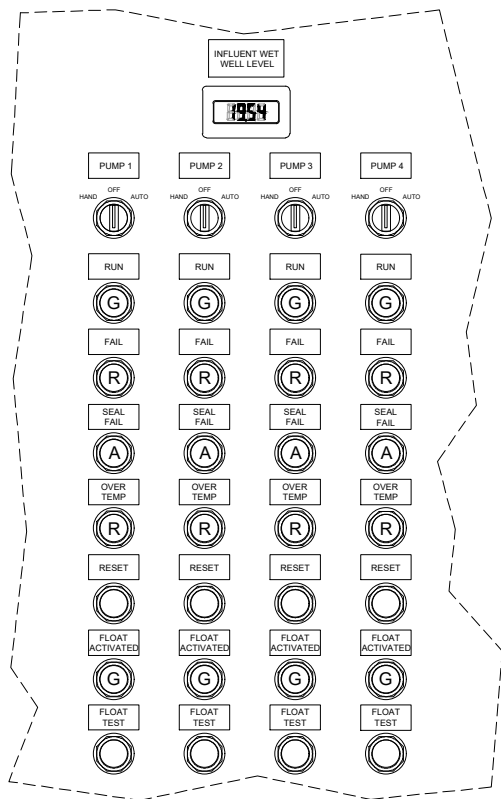
1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.

KEY NOTES:

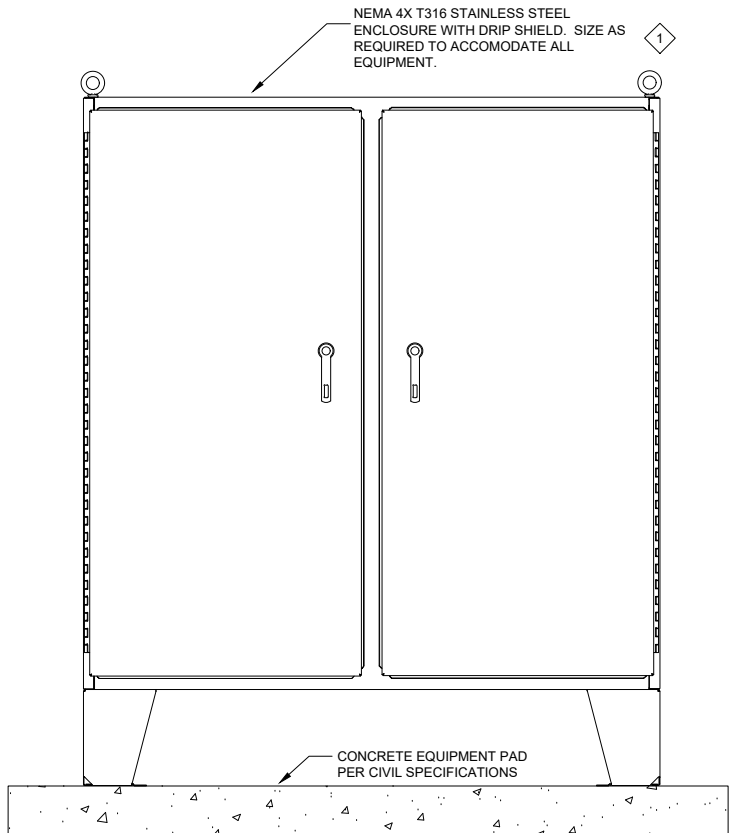
- 1 PROVIDE INFLUENT PUMP STATION CONTROL PANEL FOR LOCAL CONTROL OF INFLUENT PUMPS, SEAL FAIL AND OVERTEMP WIRING, AND INSTRUMENTATION. MOUNT CONTROLS AND INDICATORS ON INTERNAL SWING OUT DOOR, LOCKABLE OUTSIDE DOOR. SYSTEM INTEGRATOR SHALL SIZE CONTROL PANEL TO ACCOMMODATE ALL CONTROLS AND EQUIPMENT REQUIRED.
- 2 PROVIDE FLOW METER TRANSMITTERS FOR THE FOLLOWING LOCATIONS: INFLUENT, CELL 1 (FUTURE SPACE), CELL 2, CELL 3, PLANT DRAIN. SYSTEM INTEGRATOR SHALL SIZE CONTROL PANEL TO ACCOMMODATE ALL CONTROLS AND EQUIPMENT REQUIRED.
- 3 INSTALL SEAL FAIL AND OVERTEMP RELAYS AS PROVIDED BY PUMP MANUFACTURER; MINI-CAS OR EQUAL. SYSTEM INTEGRATOR SHALL COORDINATE WITH PUMP SUPPLIER FOR RELAYS.
- 4 ALL FIELD CONDUITS NOT SHOWN. CONTRACTOR SHALL ROUTE ALL RACEWAY SHOWN ON THE PLANS TO THE INFLUENT CONTROL PANEL.



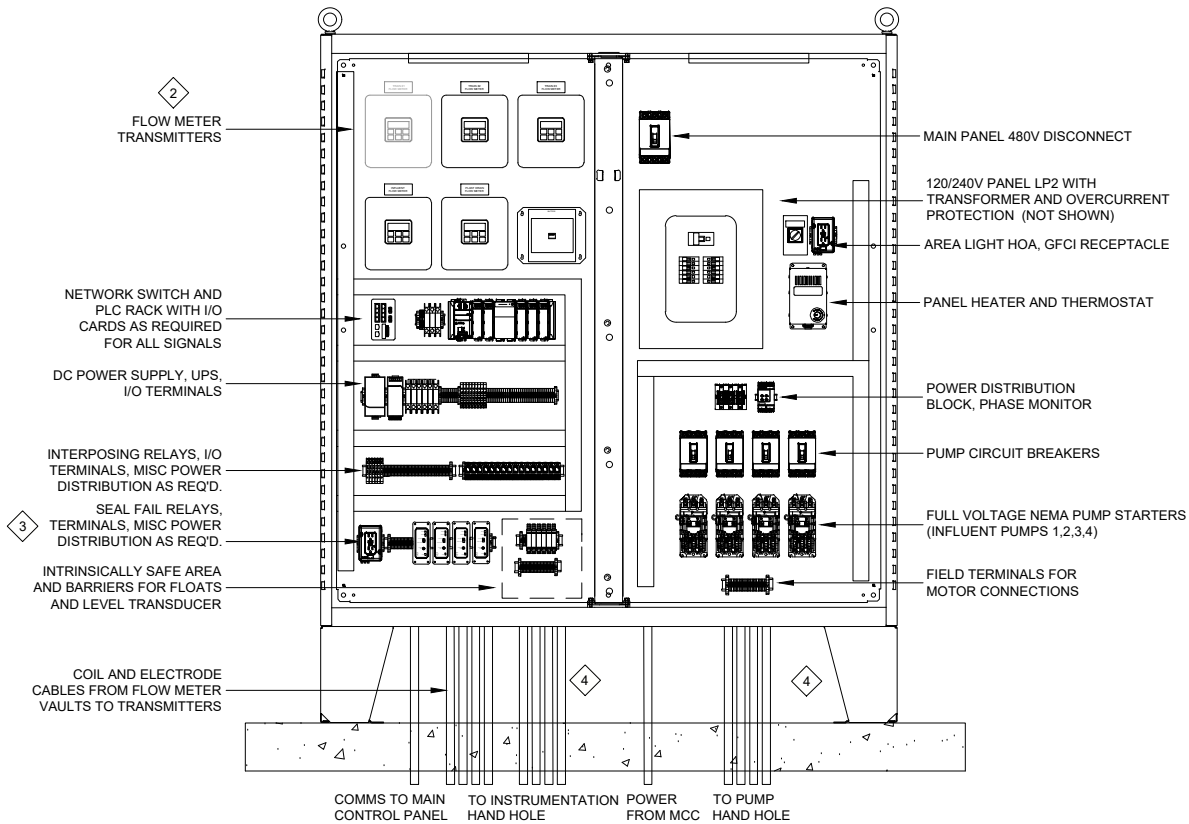
INFLUENT PUMP STATION PANEL - INTERIOR DOOR
SCALE: NONE



SWING OUT INTERIOR DOOR DETAIL
SCALE: NONE



INFLUENT PUMP STATION PANEL - EXTERIOR
SCALE: NONE



INFLUENT PUMP STATION PANEL - INTERIOR BACKPANEL
SCALE: NONE

NO.	REVISIONS	BY	DATE

GENERAL NOTES:

1. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM. ONLY MAJOR PANEL COMPONENTS ARE SPECIFIED.



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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
WAS

INFLUENT PUMP STATION CONTROL PANEL-POWER WIRING DIAGRAM

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

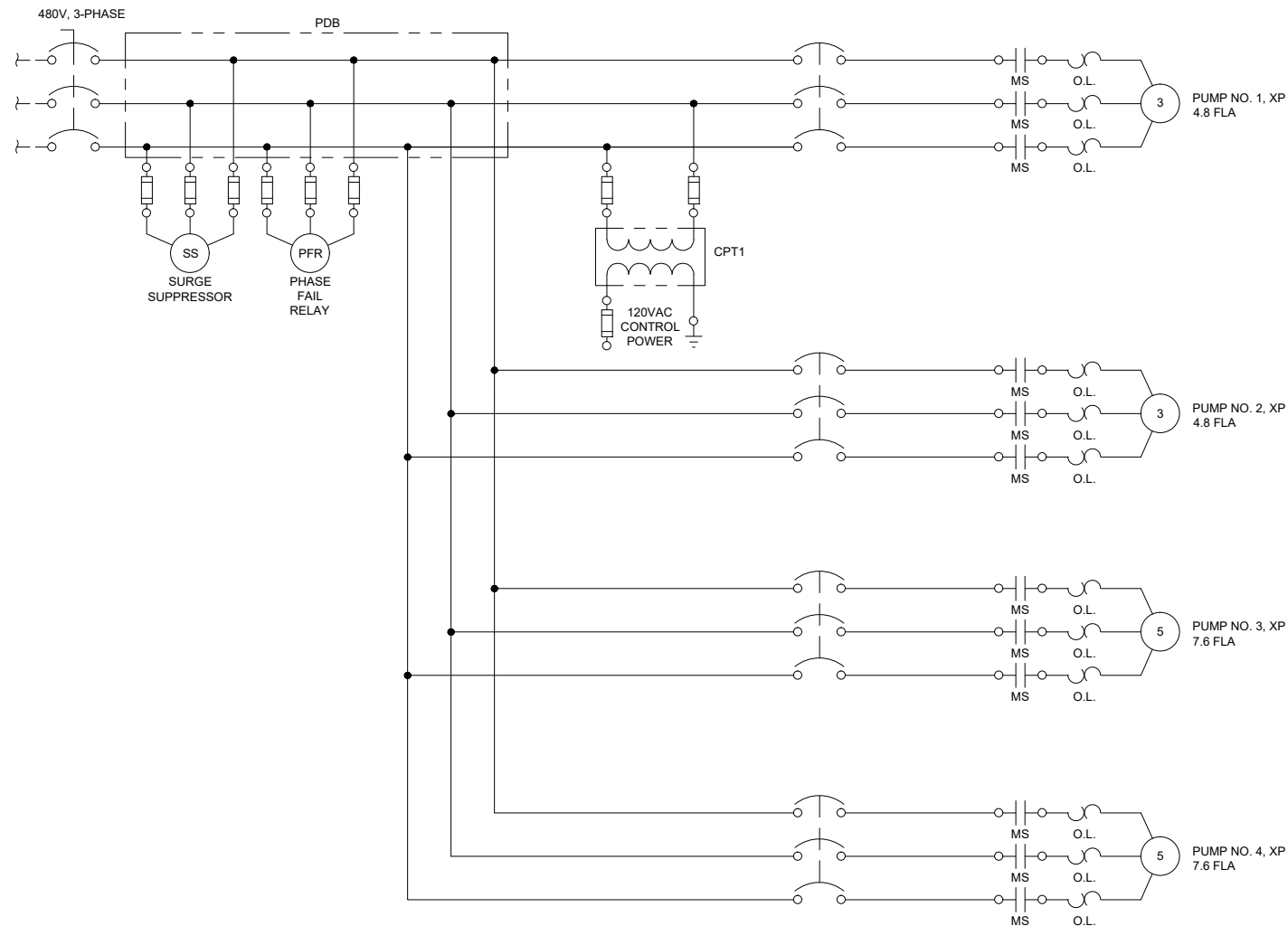
SHEET

E8.2

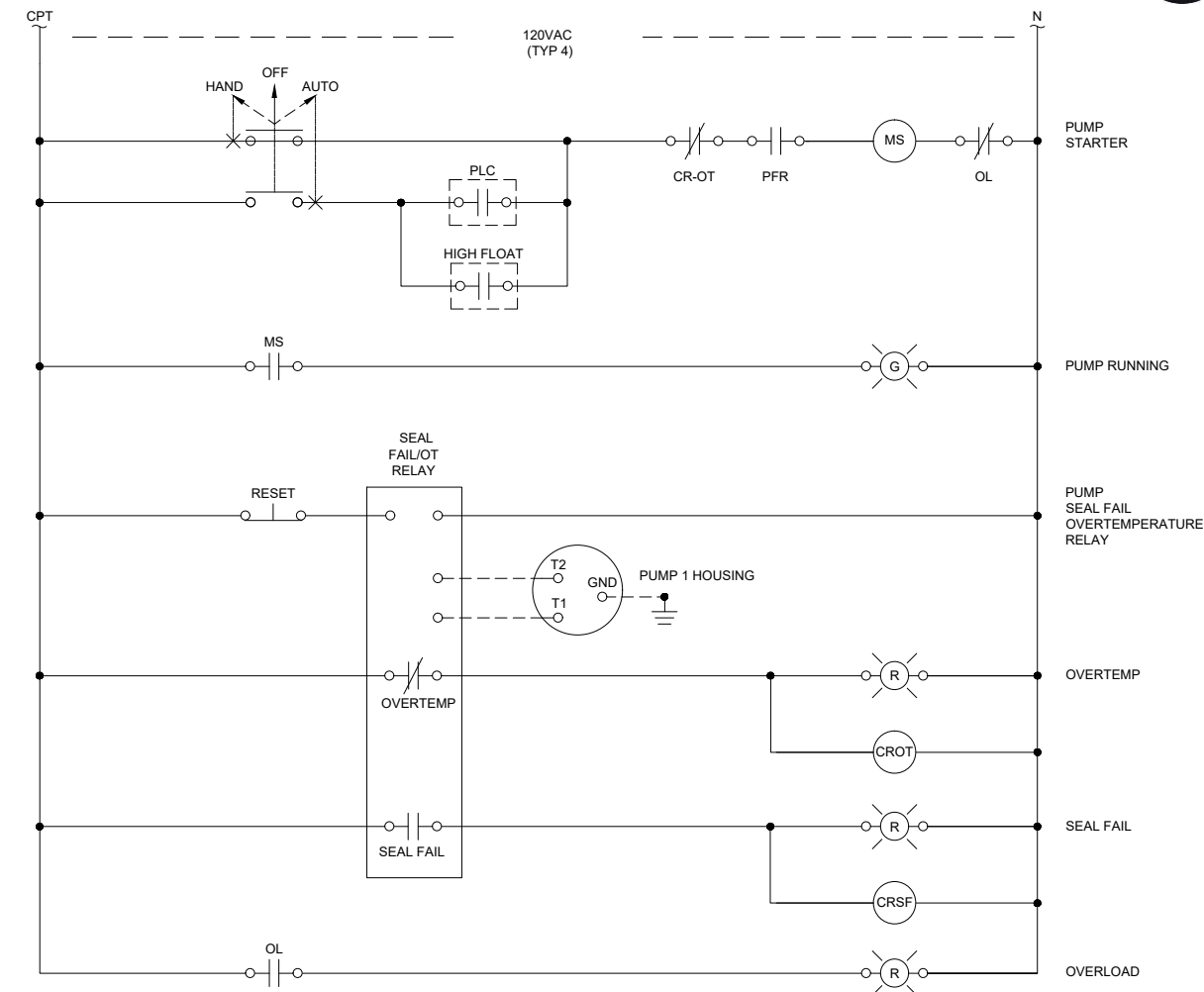
PAGE 86 OF 96



BID SET



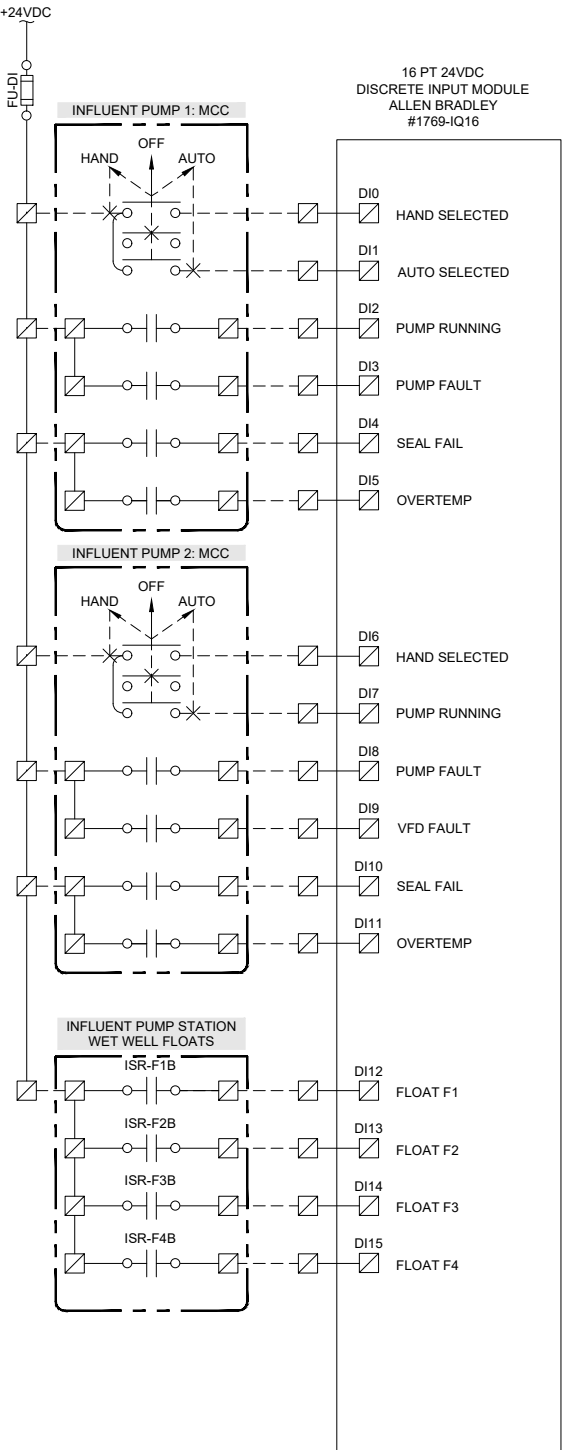
INFLUENT PUMP STATION CONTROL PANEL - SCHEMATIC POWER WIRING DIAGRAM
SCALE: NONE



PUMP CONTROL WIRING DIAGRAM (TYPICAL FOR EACH OF 4 PUMPS)
SCALE: NONE

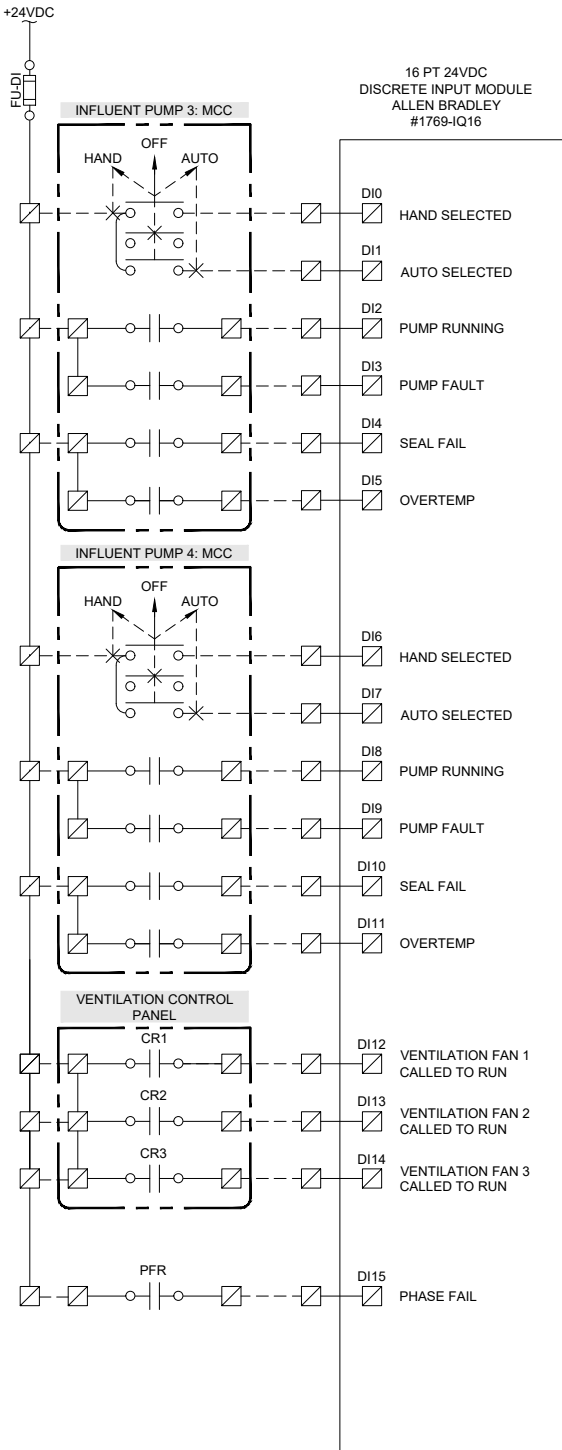
GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.
2. PROVIDE ADDITIONAL I/O CARDS AS NECESSARY FOR MINIMUM 20% SPARE INPUTS, NOT SHOWN.

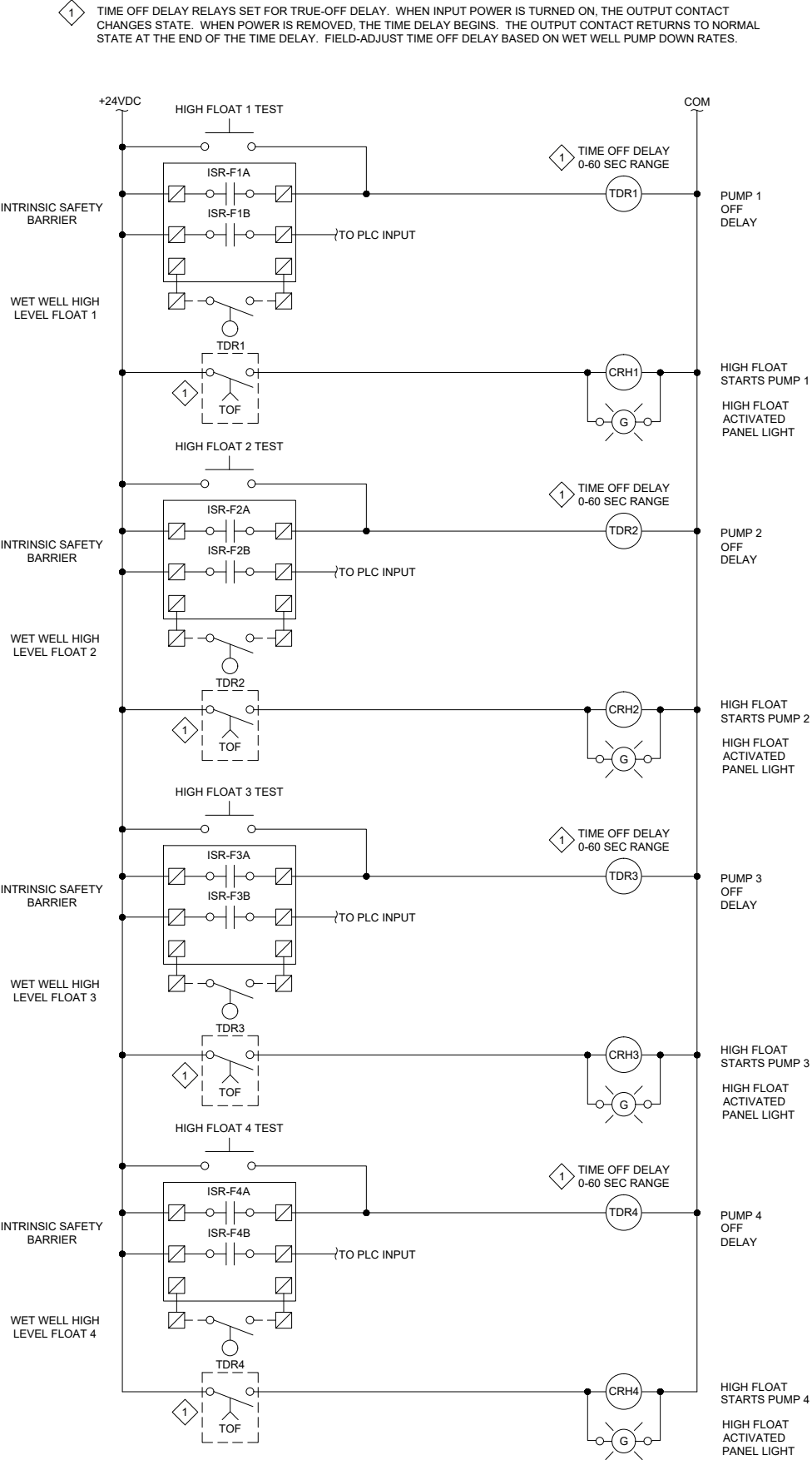


INFLUENT PUMP STATION CONTROL PANEL - DIGITAL INPUTS

SCALE: NONE



KEY NOTES:



INFLUENT PUMP STATION - HIGH LEVEL FLOAT CONTROL CIRCUIT

SCALE: NONE

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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - DIGITAL INPUTS

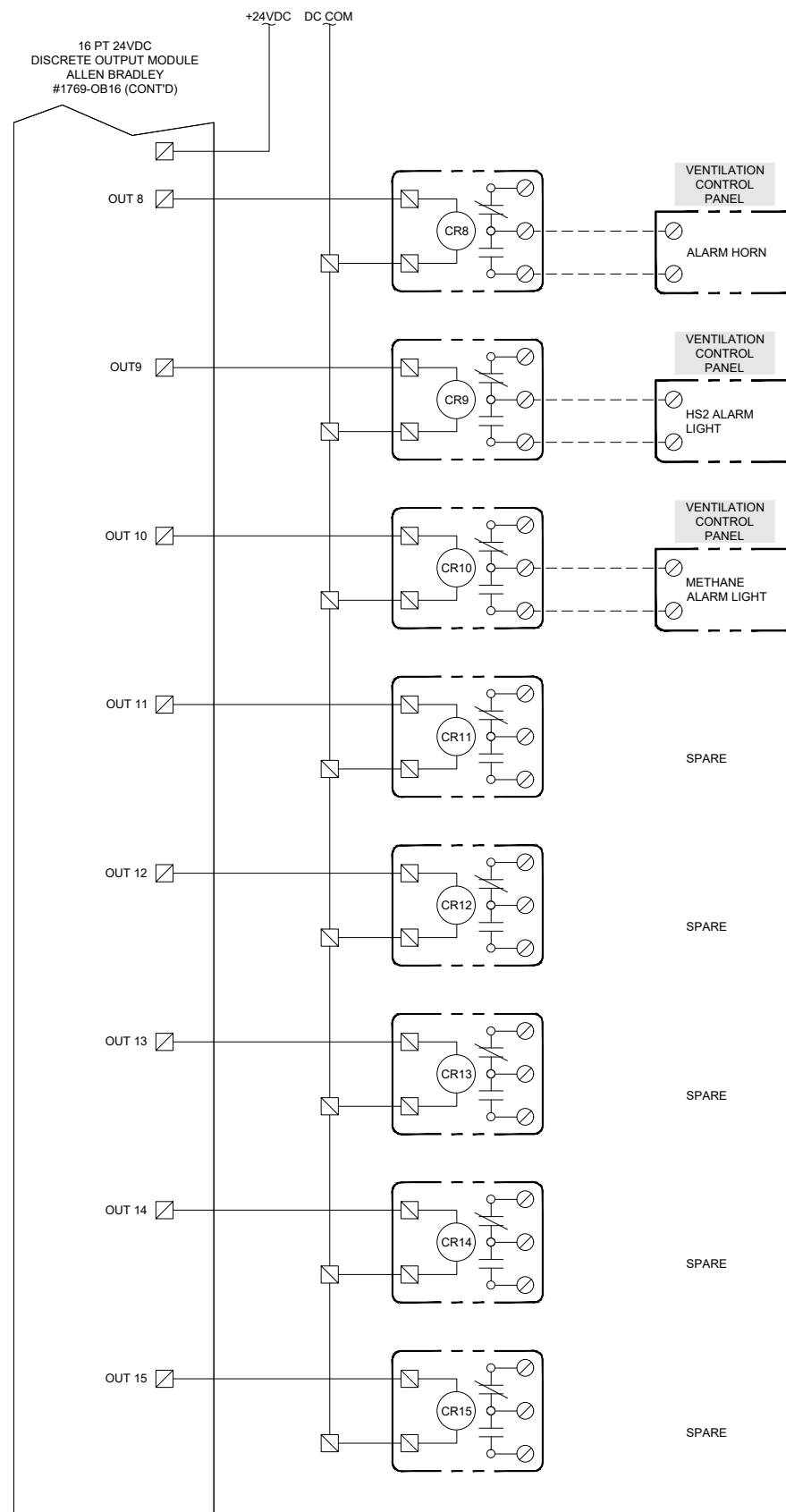
DATE: 09-12-2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070

SHEET: E8.3
PAGE: 87 OF 96



BID SET

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - DIGITAL OUTPUTS

DATE _____

09-12-2022

SCALE

AS SHOWN

JOB NUMBER

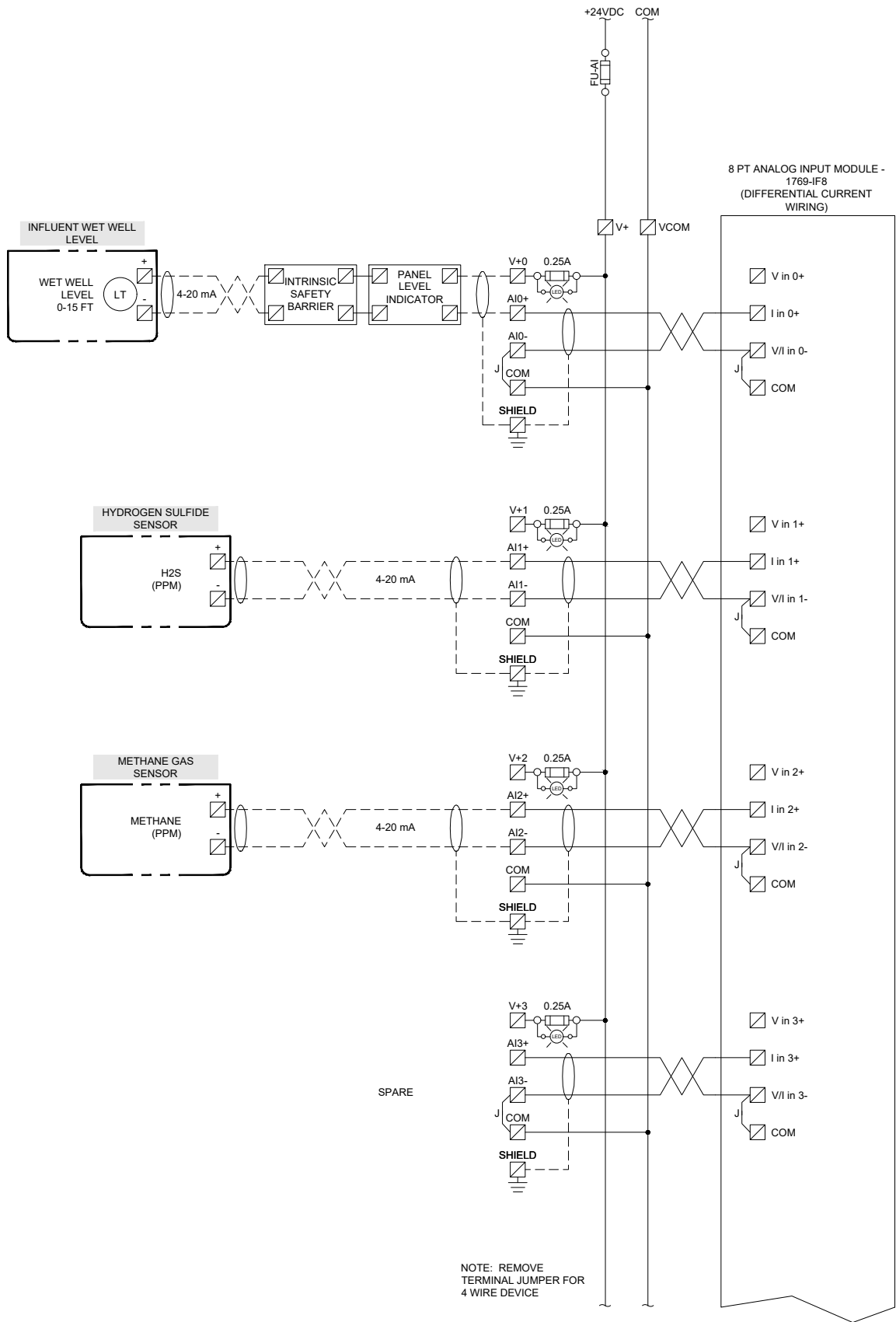
E8.4

PAGE

8 OF 96

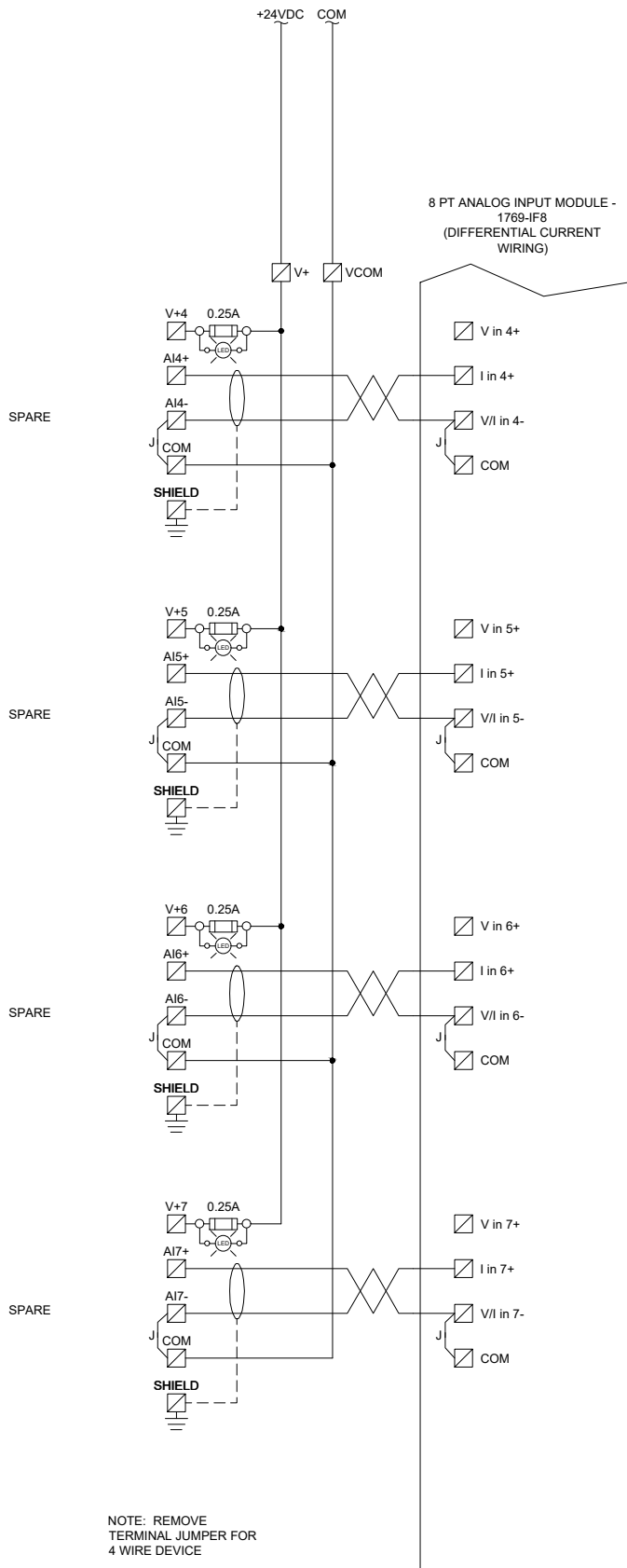
GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



INFLUENT PUMP STATION CONTROL PANEL - ANALOG INPUTS

SCALE: NONE



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

INFLUENT PUMP STATION CONTROL PANEL - ANALOG INPUTS

DATE
09-12-2022

SCALE
AS SHOWN

JOB NUMBER
2020-070

SHEET
E8.5

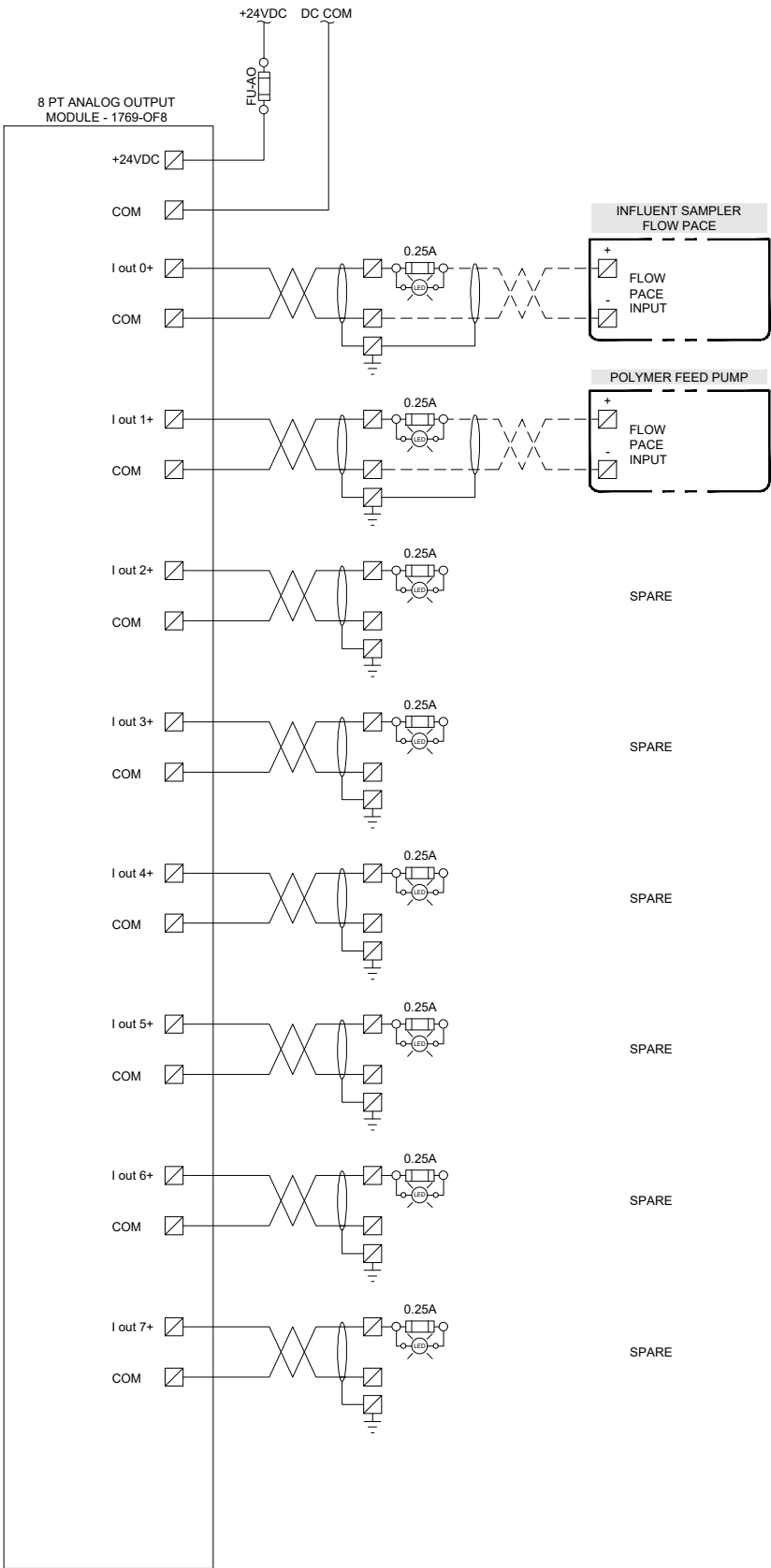
PAGE
89 OF 96



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



INFLUENT PUMP STATION CONTROL PANEL - ANALOG OUTPUTS
SCALE: NONE

NO.	REVISIONS	BY	DATE



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EASTSOUND SEWER AND WATER DISTRICT	DATE	09-12-2022
	SCALE	AS SHOWN
	JOB NUMBER	2020-070
SAN JUAN COUNTY		
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I		
INFLUENT PUMP STATION CONTROL PANEL – ANALOG OUTPUTS		

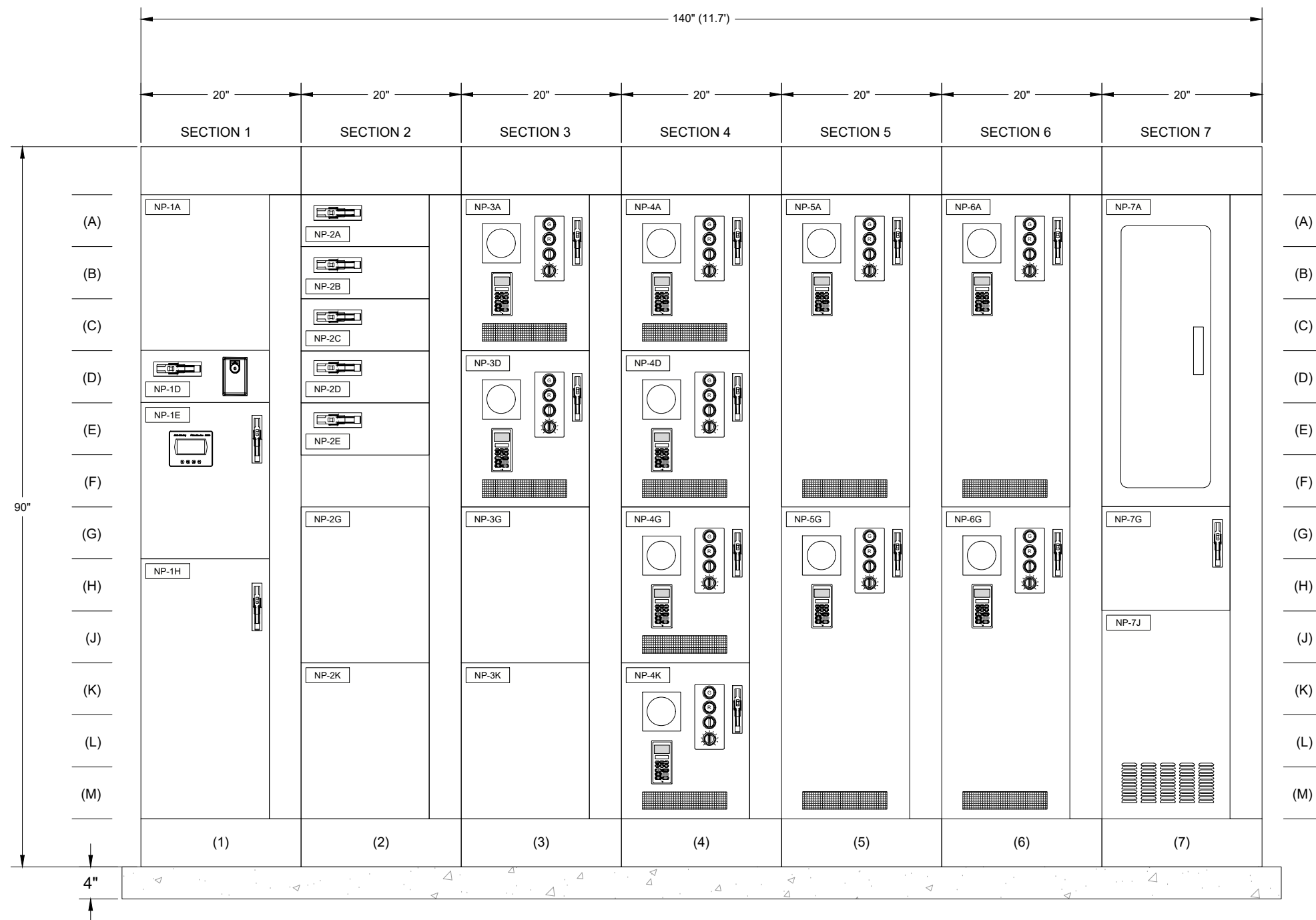


BID SET

SHEET	E8.6
PAGE	90 OF 96

1. CONTRACTOR SHALL VERIFY AND PROVIDE FINAL MCC EQUIPMENT LAYOUT TO MEET PROJECT REQUIREMENTS. MOTOR CONTROL CENTER SHALL BE ALLEN BRADLEY CENTERLINE 2100 INTELLICENTER MODEL.

		MCC NAMEPLATE SCHEDULE	
EQ ID	NAMEPLATE LABEL (LINE 1 / LINE 2 / LINE 3)	EQ ID	NAMEPLATE LABEL (LINE 1 / LINE 2 / LINE 3)
NP-1A	MCC NETWORK / ETHERNET SWITCH	NP-3G	SPACE
NP-1D	SURGE PROTECTOR	NP-3K	SPACE
NP-1E	POWER MONITOR	NP-4A	SLUDGE PUMP / 3 HP
NP-1H	MAIN BREAKER DISCONNECT / 400 AMP	NP-4D	WAS PUMP NO.1 / 3 HP
NP-2A	TRAIN NO.1 / PANEL FEEDER (FUTURE)	NP-4G	WAS PUMP NO.2 / 3 HP
NP-2B	TRAIN NO.2 / PANEL FEEDER (FUTURE)	NP-4K	DIGESTER BLOWER / 5 HP
NP-2C	TRAIN NO.3 / PANEL FEEDER	NP-5A	AERATION BLOWER NO.1 / 20 HP
NP-2D	DEWATERING SKID / FEEDER	NP-5G	AERATION BLOWER NO.2 / 20 HP
NP-2E	INFLUENT PUMP STATION/FEEDER	NP-6A	AERATION BLOWER NO.3 / 20 HP
NP-2G	SPACE	NP-6G	AERATION BLOWER NO.4 / 20 HP
NP-2K	SPACE	NP-7A	LIGHTING PANEL / 120-240V 1 PH
NP-3A	2W PUMP NO.1 / 5 HP	NP-7G	TRANSFORMER DISCONNECT
NP-3D	2W PUMP NO.2 / 5 HP	NP-7J	TRANSFORMER / 480-120/240V / 20 KVA

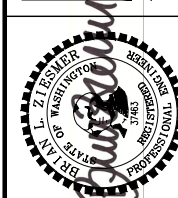


MOTOR CONTROL CENTER (MCC) ELEVATION
NOT TO SCALE

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RZ	

EASTSOUND SEWER AND WATER DISTRICT

COUNTY _____ WAS _____
WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

BLOWER BIII DING MOTOR CONTROL CENTER ELEVATION

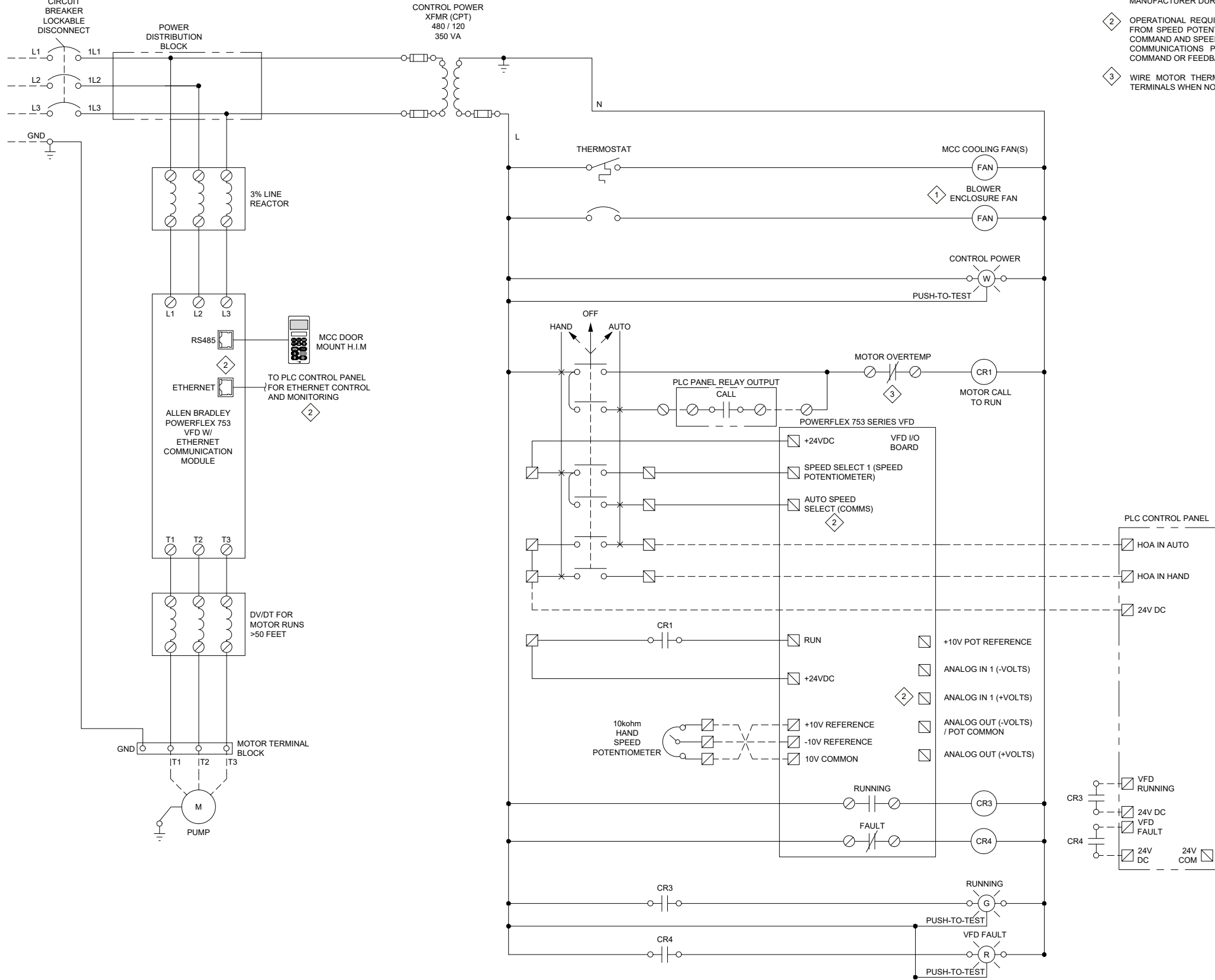
SHEET E9.1	DATE	09-12-2022
	SCALE	AS SHOWN
	JOB NUMBER	2020-070
PAGE 91 OF 96		



BID SET

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



KEY NOTES:

1. PROVIDE 120V BLOWER FAN CIRCUIT FROM EACH MCC FOR EACH BLOWER. BASIS OF DESIGN: FAN RATING 0.21HP AND 1.51 AMP PER FAN. CONFIRM WITH MANUFACTURER DURING SUBMITTALS.
2. OPERATIONAL REQUIREMENTS: IN HAND, SPEED CONTROL SHALL BE SELECTED FROM SPEED POTENTIOMETER ON MCC DOOR OR FIELD PANEL. IN AUTO, SPEED COMMAND AND SPEED FEEDBACK SHALL BE DIRECT TO THE PLC VIA THE ETHERNET COMMUNICATIONS PORT. HARDWIRING NOT REQUIRED FOR ANALOG SPEED COMMAND OR FEEDBACK.
3. WIRE MOTOR THERMAL CONTACT TO EACH VFD WHERE AVAILABLE. JUMPER TERMINALS WHEN NOT PROVIDED WITH MOTOR.



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WASHINGTON
SAN JUAN COUNTY
WASTEWATER TREATMENT PLANT UPGRADE – PHASE I
BLOWER BUILDING MOTOR CONTROL CENTER – VFD WIRING DIAGRAMS

DATE: 09-12-2022
SCALE: AS SHOWN
JOB NUMBER: 2020-070

SHEET: E9.2
PAGE: 92 OF 96

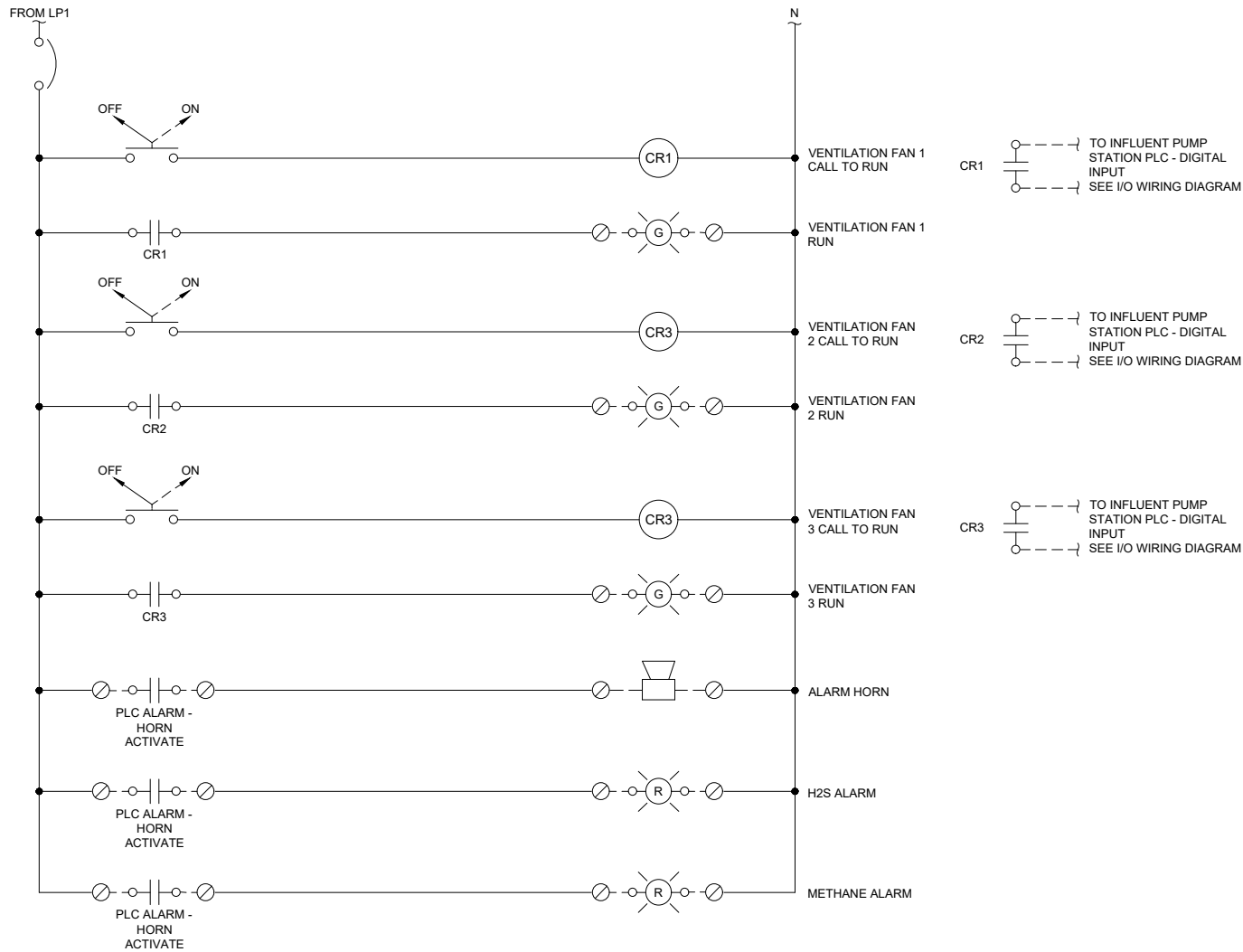
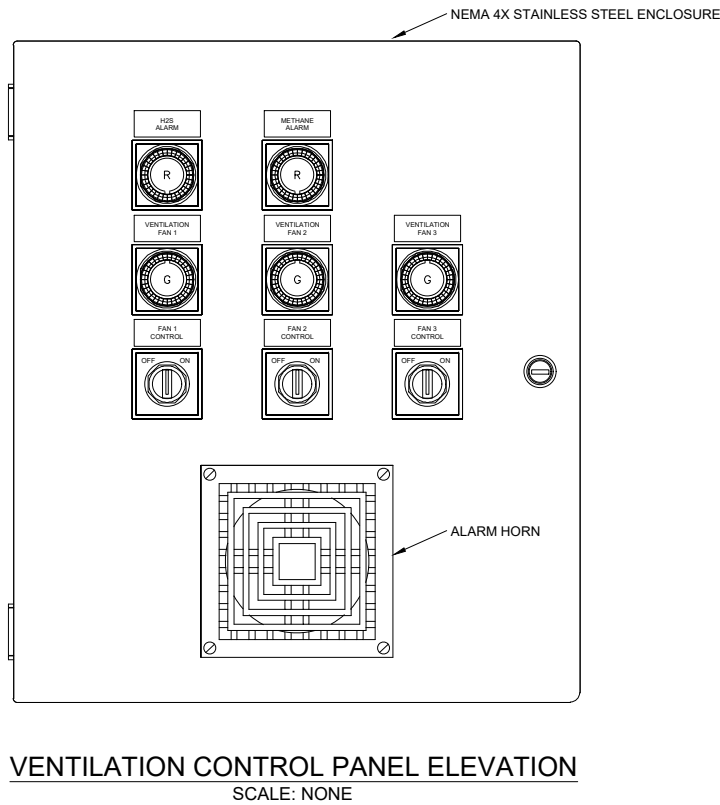


BID SET

VFD FUNCTIONAL WIRING DIAGRAM
SCALE: NONE

GENERAL NOTES:

1. THESE SCHEMATICS SHOW FUNCTIONAL REQUIREMENTS OF THE PLC CONTROL PANEL. SYSTEM INTEGRATOR SHALL PROVIDE DETAILED CONTROL PANEL DESIGN AND DOCUMENTATION FOR A COMPLETE AND OPERATIONAL SYSTEM.



VENTILATION CONTROL PANEL ELEVATION AND WIRING DIAGRAM
SCALE: NONE

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

VENTILATION CONTROL PANEL ELEVATION AND WIRING DIAGRAM

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET	E9.3
PAGE	93 OF 96



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EASTSOUND SEWER AND WATER DISTRICT

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WASTEWATER TREATMENT PLANT UPGRADE – PHASE I

ELECTRICAL DETAILS – SHEET 2

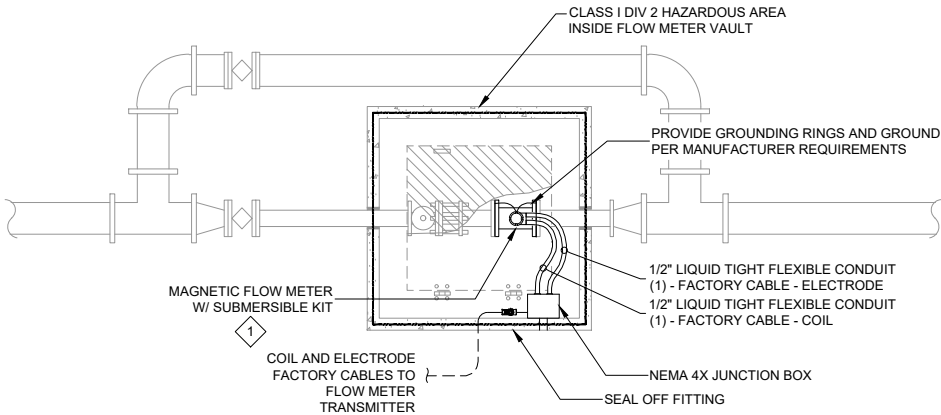
SHEET	DATE	SCALE	JOB NUMBER
E10.2	09-12-2022	AS SHOWN	2020-070
PAGE			
95			OF 96

GENERAL NOTES:

- ALL CONDUIT ROUTING IS NOT SHOWN, AND WHERE SHOWN, IS SCHEMATIC. ELECTRICAL CONTRACTOR SHALL DETERMINE THE BEST ROUTING PATH AND CIRCUIT COMBINATIONS BASED ON FIELD CONDITIONS AND ELECTRICAL CODES.
- EQUIPMENT LOCATIONS AND ARRANGEMENT ARE SCHEMATIC. CONTRACTOR SHALL COORDINATE WITH EQUIPMENT MANUFACTURER FOR DETAILED CONNECTION REQUIREMENTS AND PROVIDE MATERIALS AND INSTALLATION FOR A COMPLETE AND OPERATIONAL SYSTEM.

KEY NOTES:

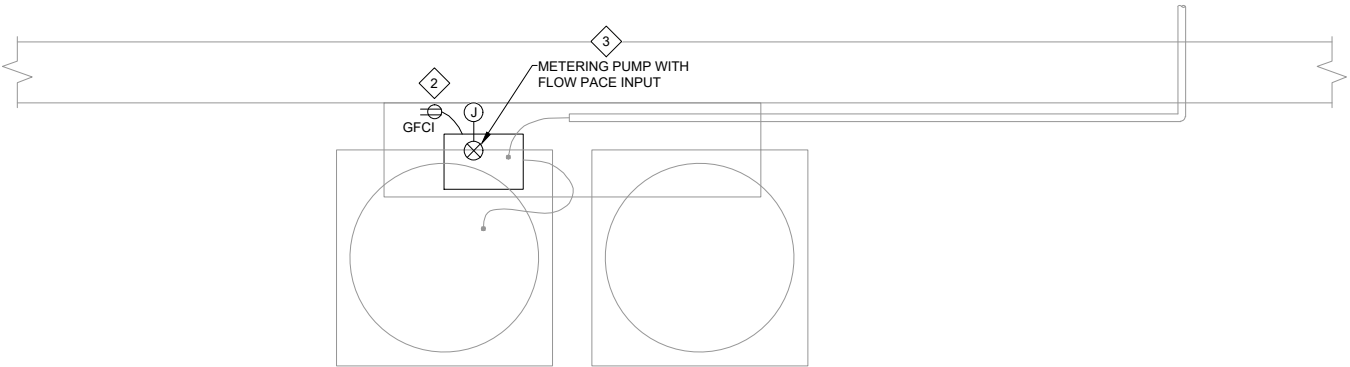
- FLOW METERS: SIZE PER CIVIL DRAWINGS. MAGNETIC FLOW METER, ENDRESS-HAUSER PROMAG W400, CLASS I DIV 2, CARBON STEEL FLANGE, 120-240VAC OR 24VAC/DC, ETHERNET/IP, POLYURETHANE LINER, 316L ELECTRODES, REMOTE HOUSING, NPT 1/2, SENSOR IP68 TYPE 6, POTTED CORROSION PROTECTION AND SUBMERSIBLE. PROVIDE (2) 316L GROUNDING RINGS FOR EACH FLOW TUBE.
- PROVIDE DEDICATED GFCI RECEPTACLE FOR METERING PUMP POWER CONNECTION. WIRE TO LIGHTING PANEL, SEE SCHEDULES.
- PROVIDE 4-20 mA OUTPUT FROM PLC TO METERING PUMP. SCALE OUTPUT PROPORTIONAL TO FLOW METER SIGNAL.



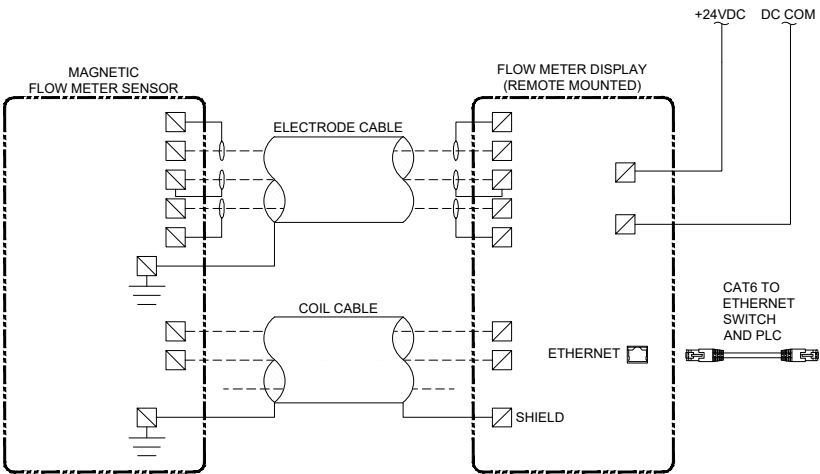
VAULT FLOW METER STANDARD DETAIL
SCALE: NONE

1 FLOW METER SUMMARY:

- INFLUENT FLOW METER. SIZE: 6". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- TRAIN #1 FLOW METER. SIZE: 8". (FUTURE PROJECT - NOT IN THIS PROJECT) PROVIDE SPACE FOR FUTURE TRANSMITTER IN INFLUENT CONTROL PANEL.
- TRAIN #2 FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- TRAIN #3 FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- PLANT DRAIN FLOW METER. SIZE: 4". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN INFLUENT CONTROL PANEL.
- EFFLUENT FLOW METER. SIZE: 8". INSTALL IN VAULT WITH SUBMERSIBLE KIT. TRANSMITTER LOCATED IN PANEL NEAR UV DISINFECTION CHAMBER.
- WAS FLOW METER. SIZE: 2". INSTALL ON PIPING NEAR WAS PUMPS, MOUNT TRANSMITTER TO FLOW TUBE.
- 2W FLOW METER. SIZE: 1". INSTALL ON PIPING IN HOT BOX #1, MOUNT TRANSMITTER TO FLOW TUBE.



CHEMICAL FEED EQUIPMENT ELECTRICAL DETAIL
NOT TO SCALE



GENERAL NOTES:

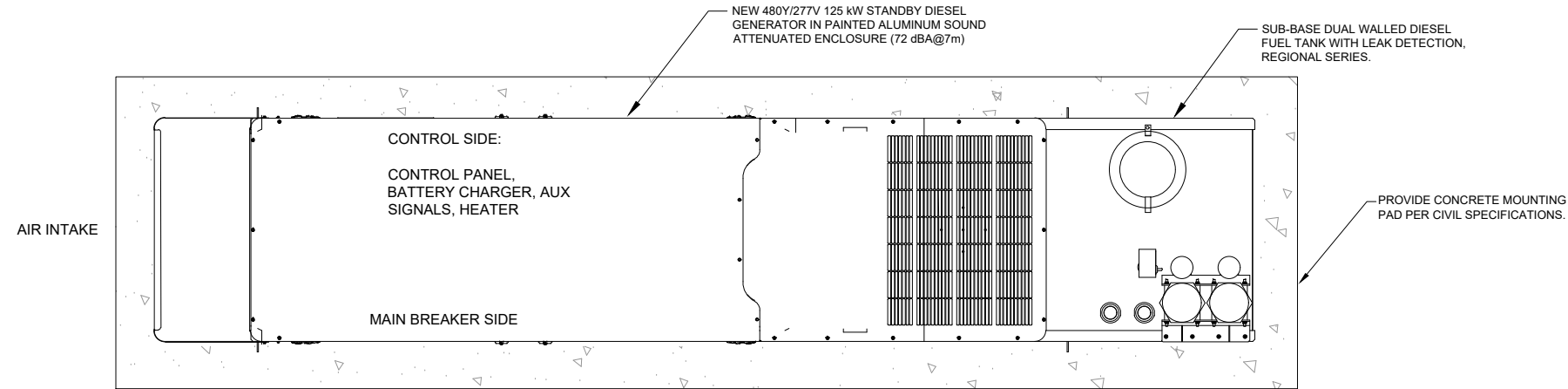
- KEEP UNSCREENED CABLE ENDS AS SHORT AS POSSIBLE.
- SEPARATE ELECTRODE AND COIL CABLES TO PREVENT INTERFERENCE.

REMOTE DISPLAY FLOW METER WIRING DETAIL
SCALE: NONE

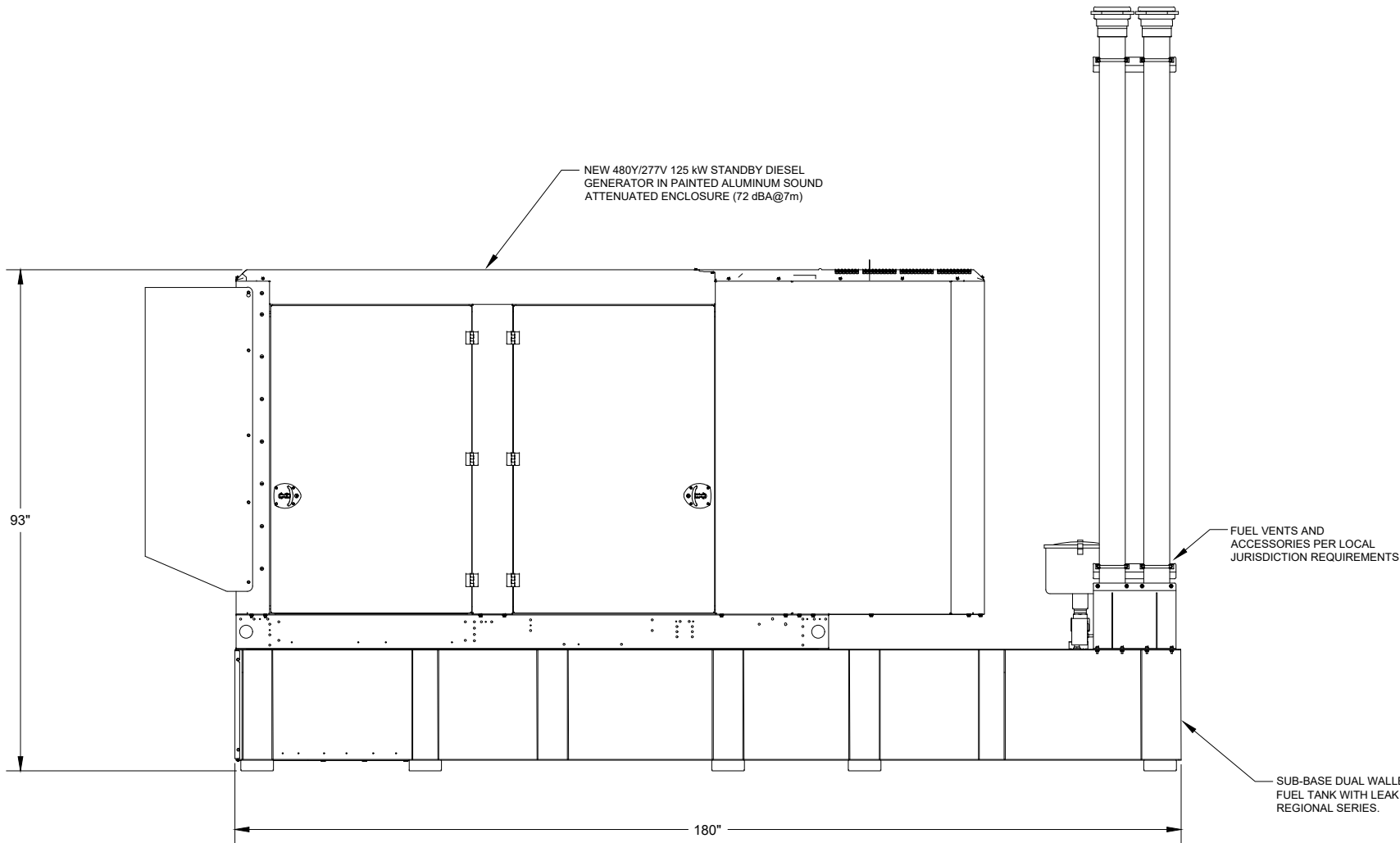


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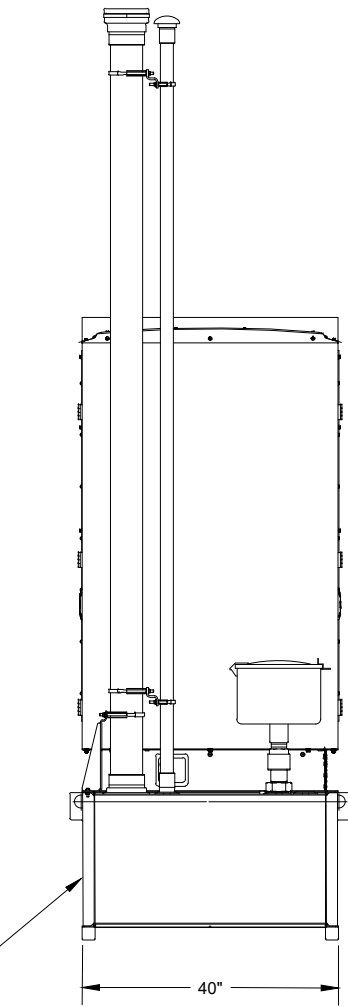
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GENERATOR PLAN VIEW
NOT TO SCALE



GENERATOR ELEVATION - SIDE
NOT TO SCALE



GENERATOR ELEVATION - END
NOT TO SCALE

GENERAL NOTES:

1. BASIS OF DESIGN: CUMMINS MODEL C125D6D 60 Hz DIESEL, STANDBY DUTY, EPA TIER 3 EMISSIONS. 125 kW / 156 kVA STANDBY RATING, 480Y/277V THREE PHASE.
2. FUEL TANK: REGIONAL SERIES WITH PRIMARY AND SECONDARY EMERGENCY VENTS, NORMAL VENT, SPILL BOX WITH OVERFILL PREVENTION VALVE, HIGH FUEL ALARM PANEL, LOW FUEL SWITCH, LEAK DETECTION, FUEL GAUGE. 351 GALLON USABLE TANK VOLUME WITH 10.1 GAL/HR PROVIDES 35 HOURS RUNTIME AT FULL LOAD, UP TO 48 HOURS UNDER NORMAL OPERATIONAL LOADS.
3. SEE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.



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WASTEWATER TREATMENT PLANT UPGRADE - PHASE I

STANDBY DIESEL GENERATOR ELEVATIONS

DATE	09-12-2022
SCALE	AS SHOWN
JOB NUMBER	2020-070

SHEET
E10.3

PAGE
96 OF 96



BID SET